

Goalkeepers

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Player counts

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In this project, there are a total of 60 goalkeepers considered, with

- 15 goalkeepers in 12 teams in the English FA Women's Super League
- 14 goalkeepers in 12 teams in the German Frauen Bundesliga
- 19 goalkeepers in 16 teams in the Spanish Liga F
- 12 goalkeepers in 10 teams in the Italian Serie A Women

Results

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Results for 2 clusters

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Breakdown of results by league

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league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_2_cluster				
0	10	9	15	8
1	5	5	4	4

The split for two clusters is around 2 for 1 for all leagues, although Liga F is much more lopsided.

Breakdown of results by features

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- Cluster 0 players on average had more share of their passes on the ground and from keepers' arms. Eventually this can align to a more modern style of ball playing goalkeepers
- On the other hand, cluster 1 players on average attempted longer passes including under pressure and had more share of passes from drop kicks. This can align to more traditional goalkeepers.

Eventually goalkeeper qualities such as saving or sweeping do not contribute to clustering results as much as passing qualities.

Results for 3 clusters

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Breakdown of results by league

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league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_3_cluster				
0	3	3	3	3
1	6	7	10	6
2	6	4	6	3

The split is more concentrated towards cluster 1 for all leagues.

Comparing results for 2 clusters against 3 clusters

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kmeans_3_cluster	0	1	2
kmeans_2_cluster			
0	12	24	6
1	0	5	13

Most of k = 2 cluster 1 is retained in k = 3 cluster 2, while k = 3 is where both clusters break into, with more coming from k = 2 cluster 0.

Breakdown of results by features

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- Cluster 0 goalkeepers stand out for high share of passes on the ground, passes from keepers' arms, and lower sweeping positions. These keepers can be read as ball playing keepers.
- While cluster 1 goalkeepers, in contrast, sweep at higher positions, while passing share stays in the middle of the other two clusters. These keepers can be interpreted as sweeper keepers.
- Cluster 2 goalkeepers are characterized by higher average pass lengths including under pressure. Similar to k = 2 cluster 1, these keepers are more traditional than the other two types.

Once again, goalkeeper positioning and saving by body parts do not contribute to clustering, although sweeping positioning does.

However, these results are not robust as number of components are added, so results are not highly confident.

Breakdown of results by players

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Zecira Musovic - Chelsea

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For both $k = 2$ and $k = 3$, Musovic safely got clustered into traditional keepers. What stand out for her is longer pass lengths, especially under pressure. While attempting a higher than average share of passes on the ground than an average traditional keeper, when this metric is adjusted according to Chelsea's dominance, it becomes rather low.

Hannah Hampton - Chelsea

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Similar to her teammate at Chelsea, Hampton was also clustered into traditional keepers quite safely, although the margin was slightly less than Musovic. Similar to Musovic, Hampton also attempted longer passes, although lengths of passes under pressure do not differ by much than overall average. In addition, her share of passes on the ground was quite similar to Musovic, and when it is adjusted according to Chelsea's dominance, it also becomes low. However, what made Hampton different to Musovic is that she tended to receipt in more advanced positions, which made her look more modern than traditional.

Rebecca Leigh Spencer - Tottenham Hotspur

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Despite Tottenham being a mid table team, Spencer was quite safely clustered into ball playing keepers cluster. What made her stand out are her passing lengths on average shorter than average for keepers in the same cluster while her high ground pass percentage stayed in line with other players in this cluster, although her receiving and sweeping positions were lower than average for players of her cluster.

Maria Korenciova - Como

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Korenciova was clustered into $k = 2$ Ball Playing cluster and $k = 3$ Sweeper cluster, in spite of Como finishing only 7th out of 10 in Serie A Women that season. This can be explained by her sweeping and positioning pointing towards the average of players in sweeper cluster, even though the figures were still lower by quite a margin, while not relying not as much on ground passes.

Mary Earps - Manchester United

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Earps was clustered into ball playing keepers' cluster for both $k = 2$ and $k = 3$. However, her assignments showed quite parity among all clusters in both scenarios, as her $k = 2$ margin reflects a genuine, if modest, lean toward Ball-Playing, while her $k = 3$ margin is where the true near-three-way-tie lives. Most of her passing styles, including passing lengths and percentage of passes on the ground, were similar to ball playing keepers' cluster, although the trends were not as clear as Spencer. She also stood out for much more advanced sweeping positions, which make her look close to Sweeper Keeper cluster.

Appendix

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Radar plots for 2 clusters goalkeepers' averages

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All_Features — kmeans_2_cluster comparison (raw)

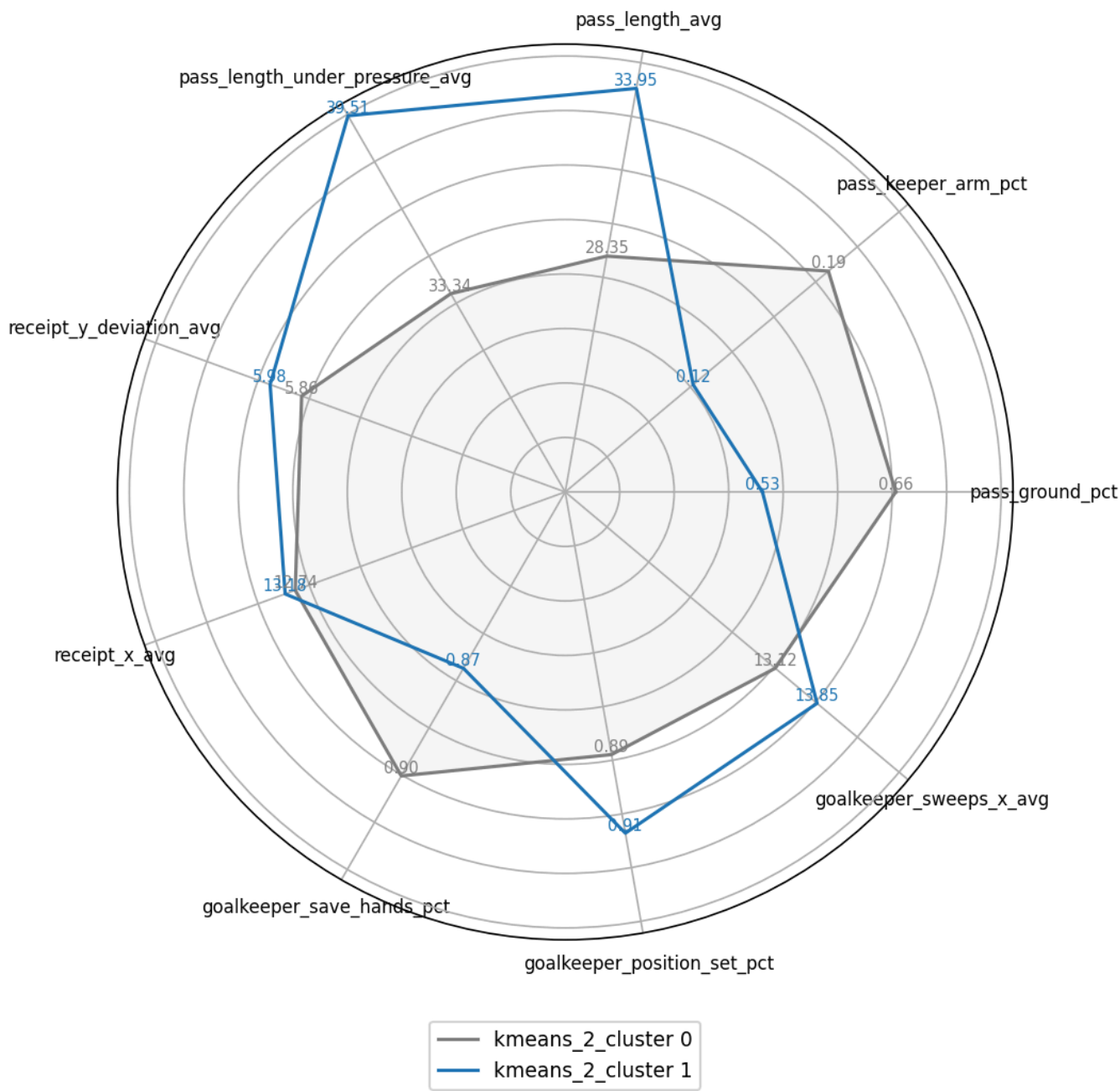


Figure 1: All Goalkeepers - Raw Statistics Comparison

All_Features — kmeans_2_cluster comparison (adj)

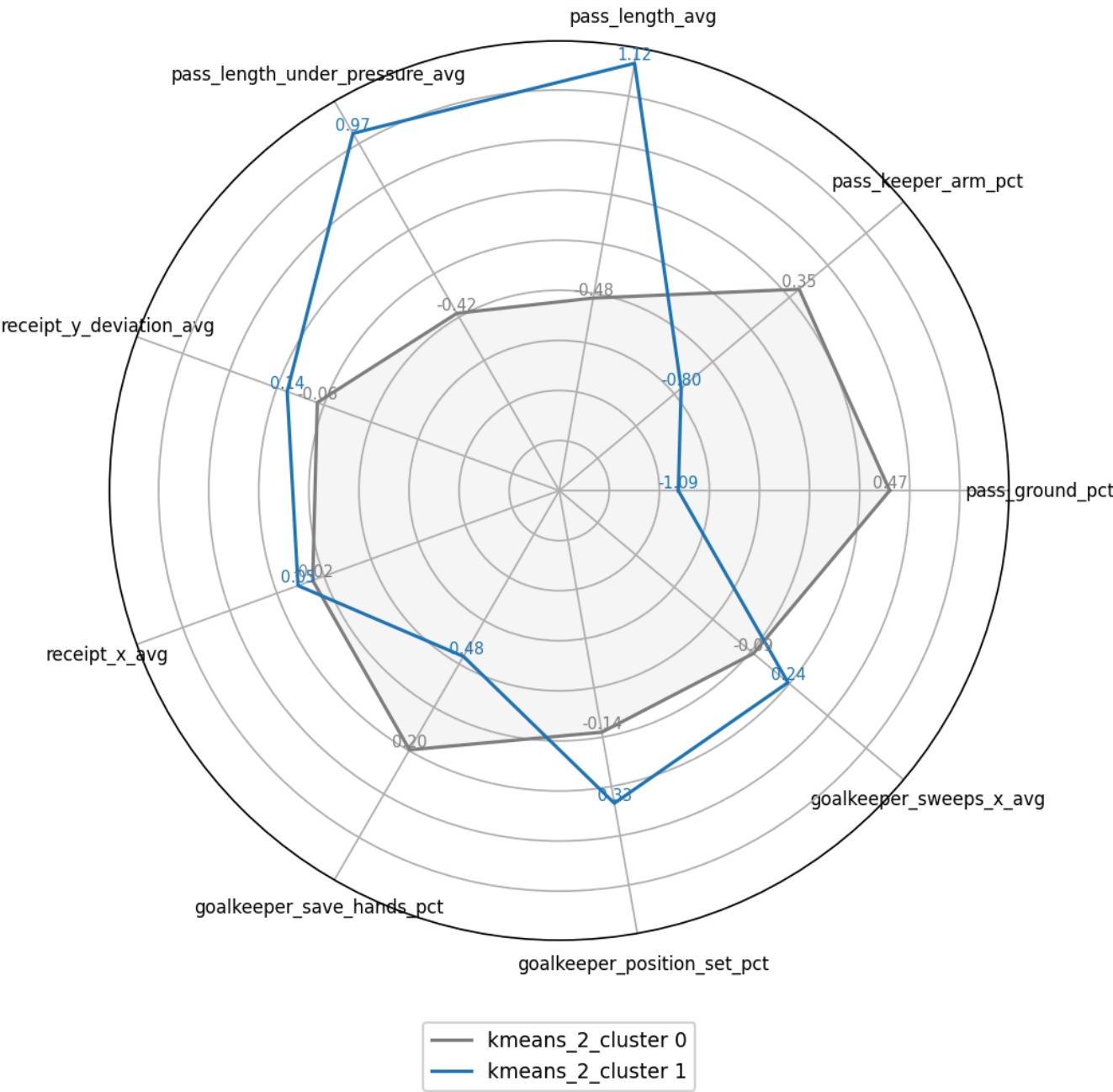


Figure 2: All Goalkeepers - Adjusted Statistics Comparison

All_Features — kmeans_2_cluster comparison (FA Women's Super League, raw)

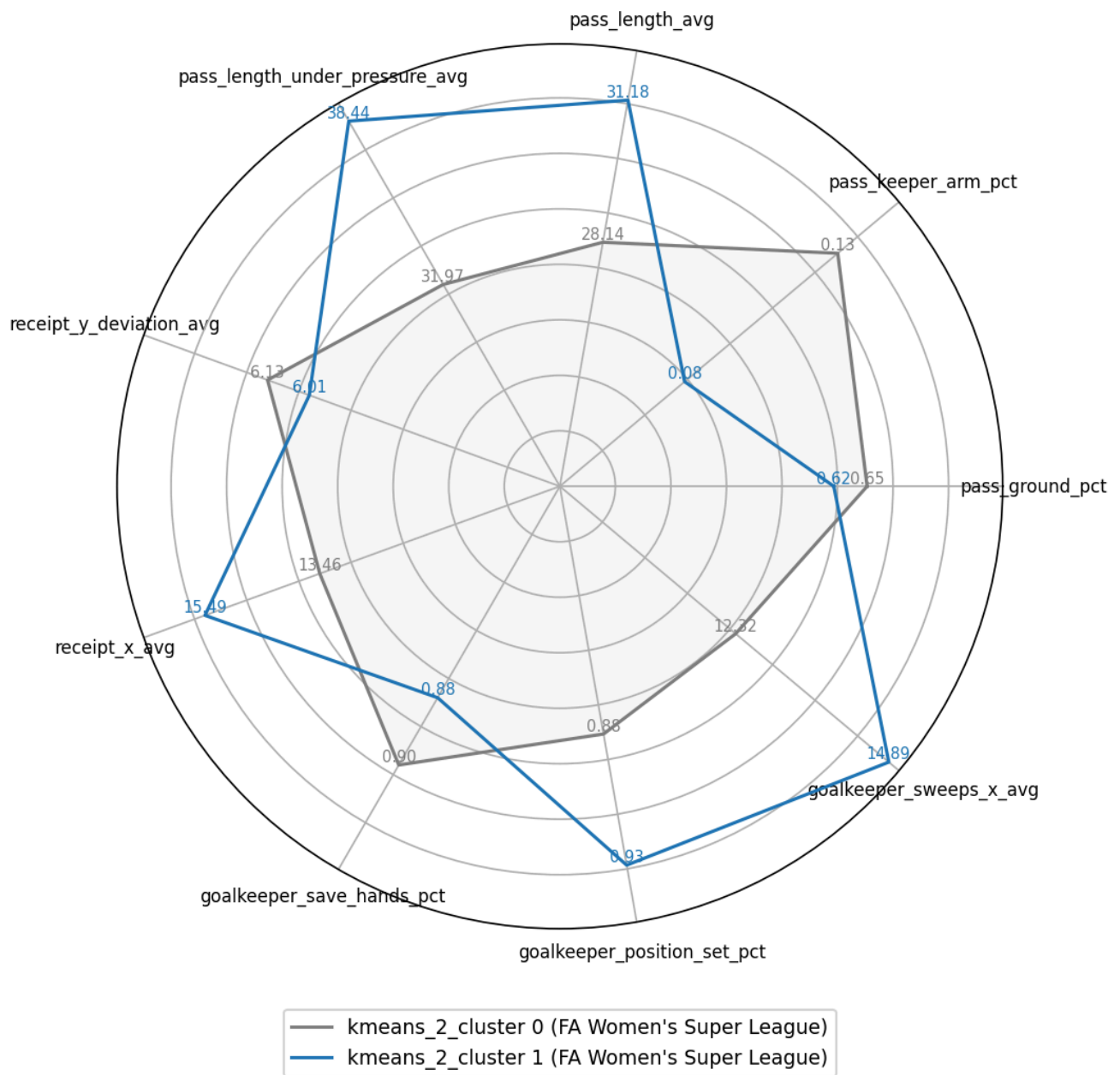


Figure 3: FA Women's Super League Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_2_cluster comparison (FA Women's Super League, adj)

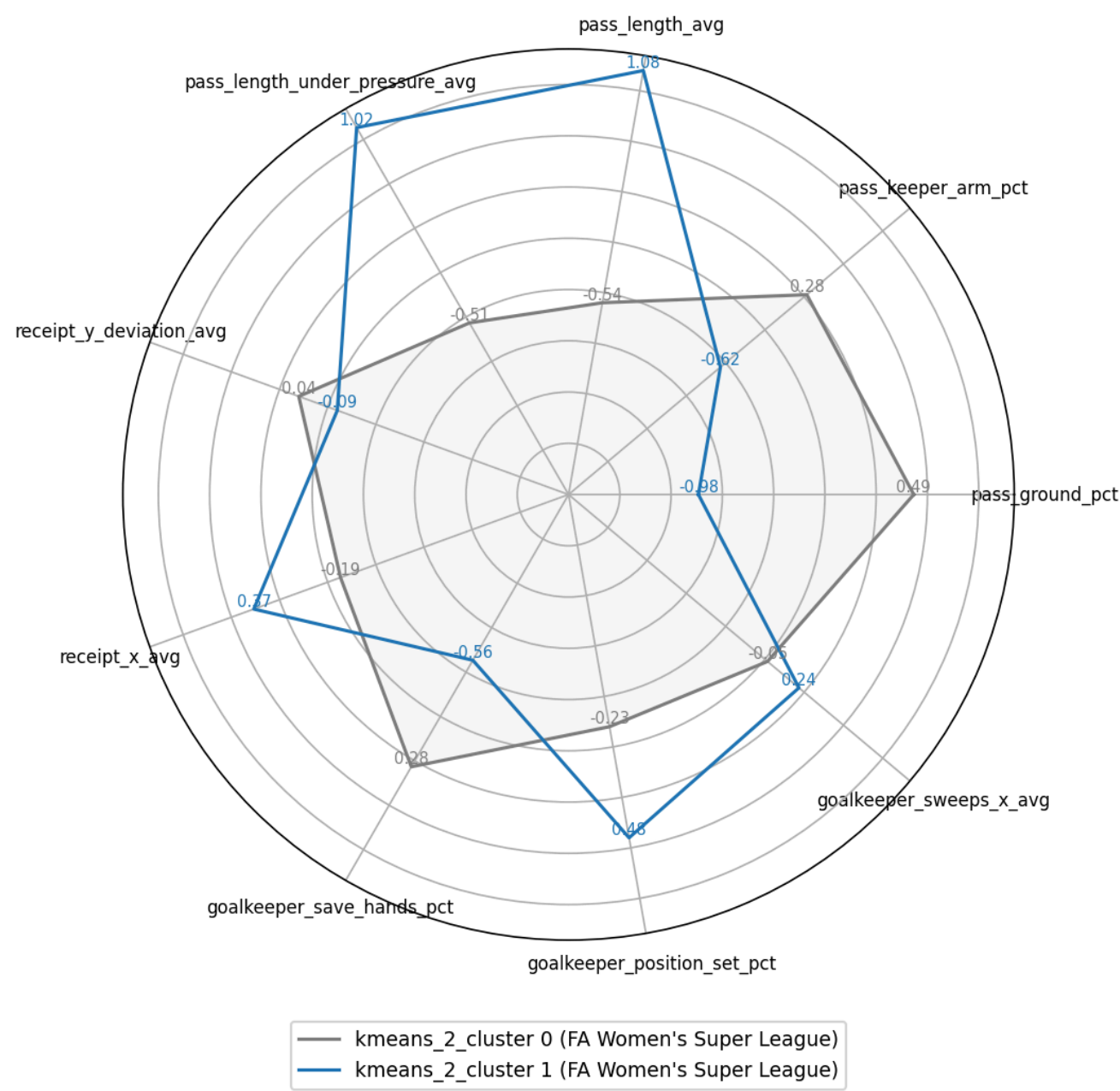


Figure 4: FA Women's Super League Goalkeepers — Adjusted Statistics Comparison

Frauen Bundesliga

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All_Features — kmeans_2_cluster comparison (Frauen Bundesliga, raw)

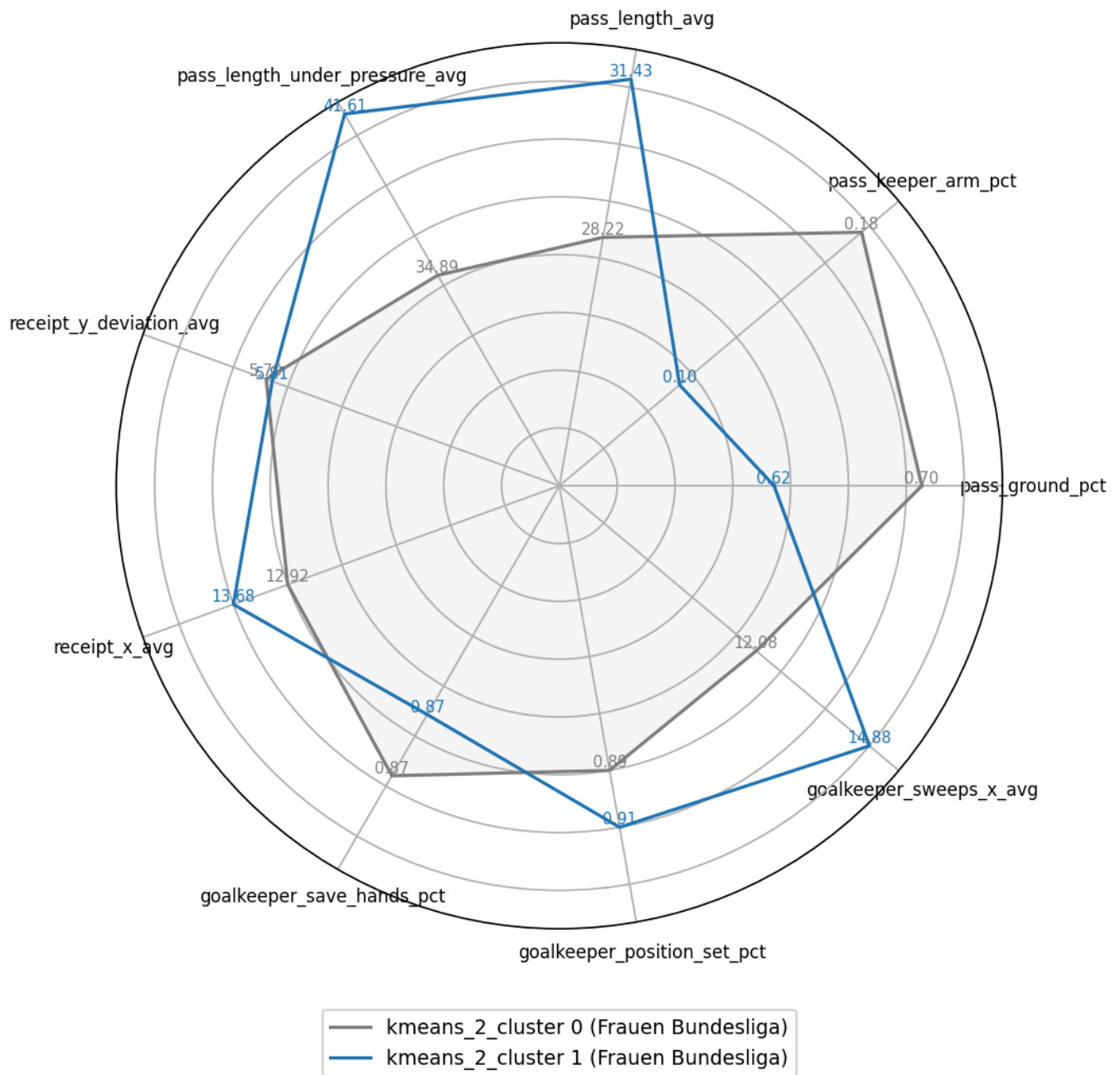


Figure 5: Frauen Bundesliga Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_2_cluster comparison (Frauen Bundesliga, adj)

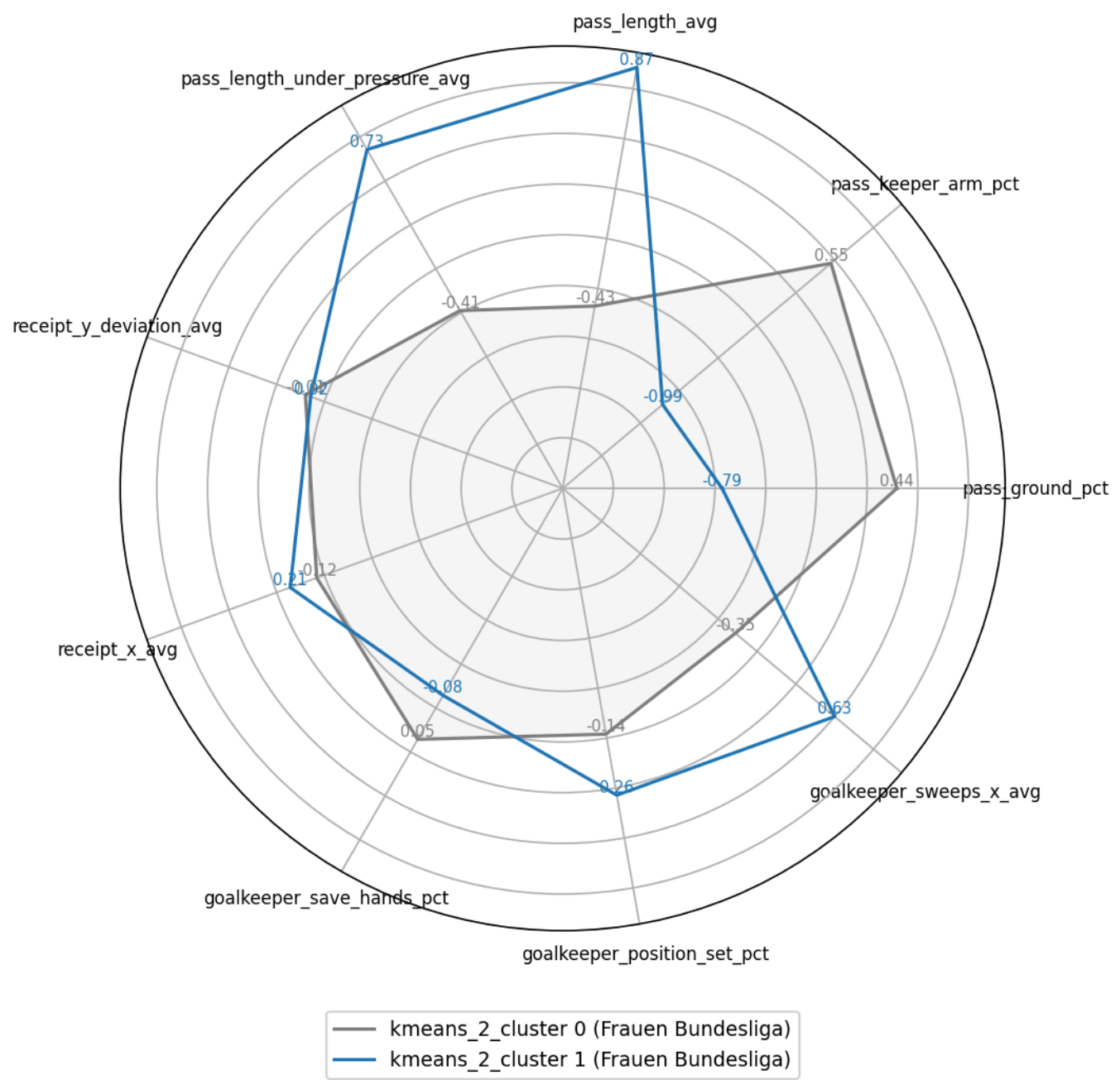


Figure 6: Frauen Bundesliga Goalkeepers — Adjusted Statistics Comparison

Liga F

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All_Features — kmeans_2_cluster comparison (Liga F, raw)

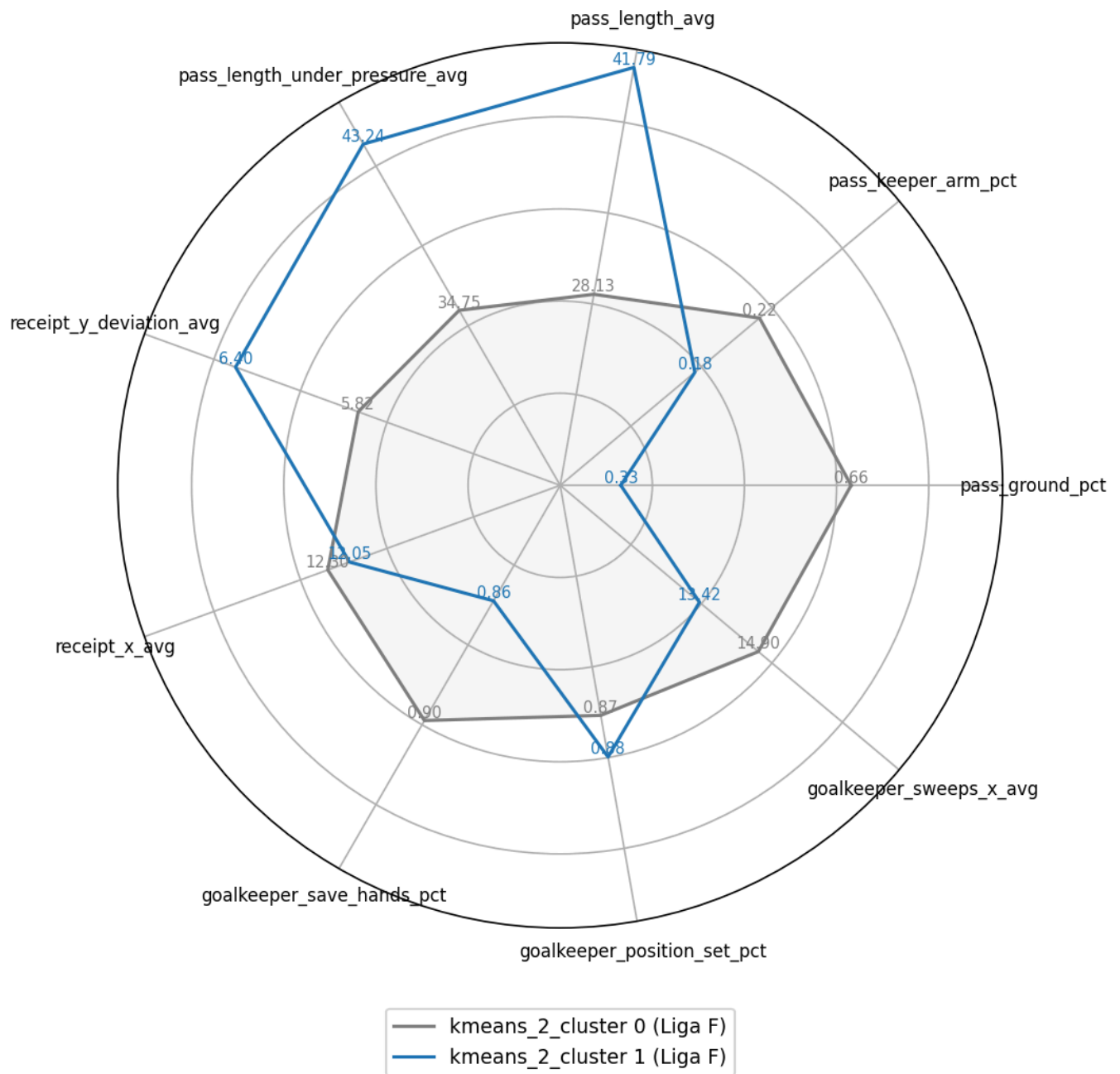


Figure 7: Liga F Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_2_cluster comparison (Liga F, adj)

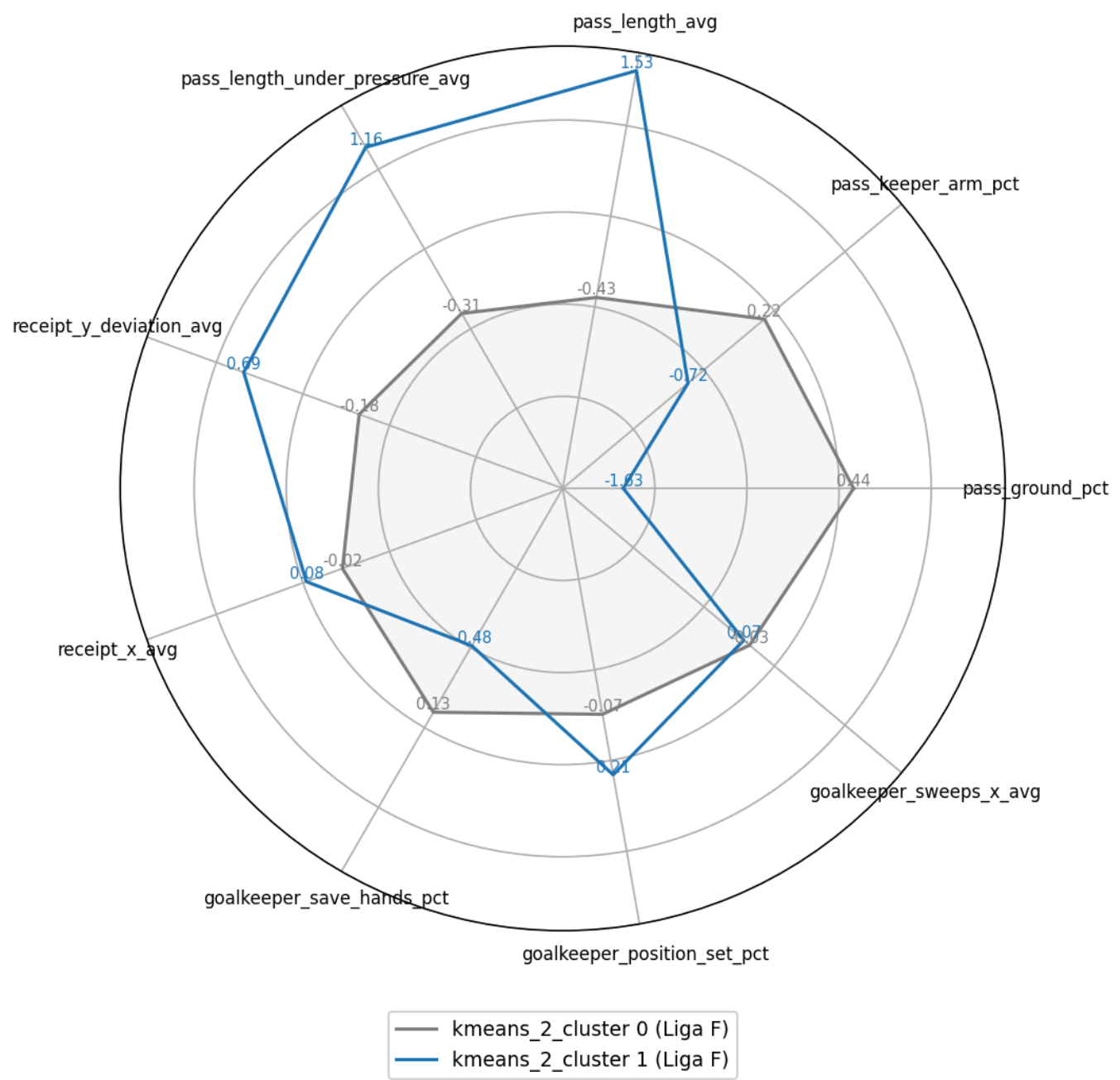


Figure 8: Liga F Goalkeepers — Adjusted Statistics Comparison

Serie A Women

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All_Features — kmeans_2_cluster comparison (Serie A Women, raw)

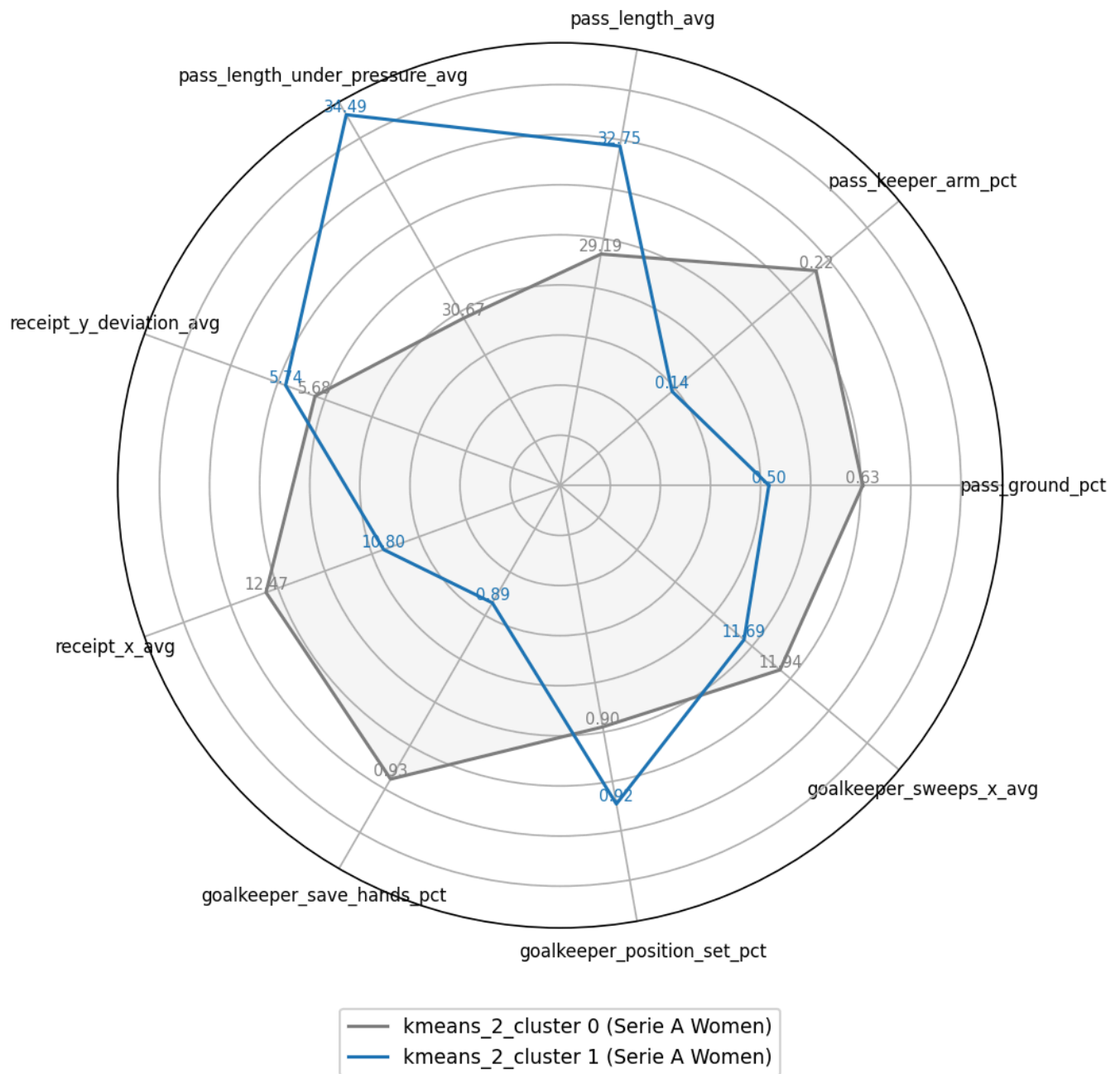


Figure 9: Serie A Women Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_2_cluster comparison (Serie A Women, adj)

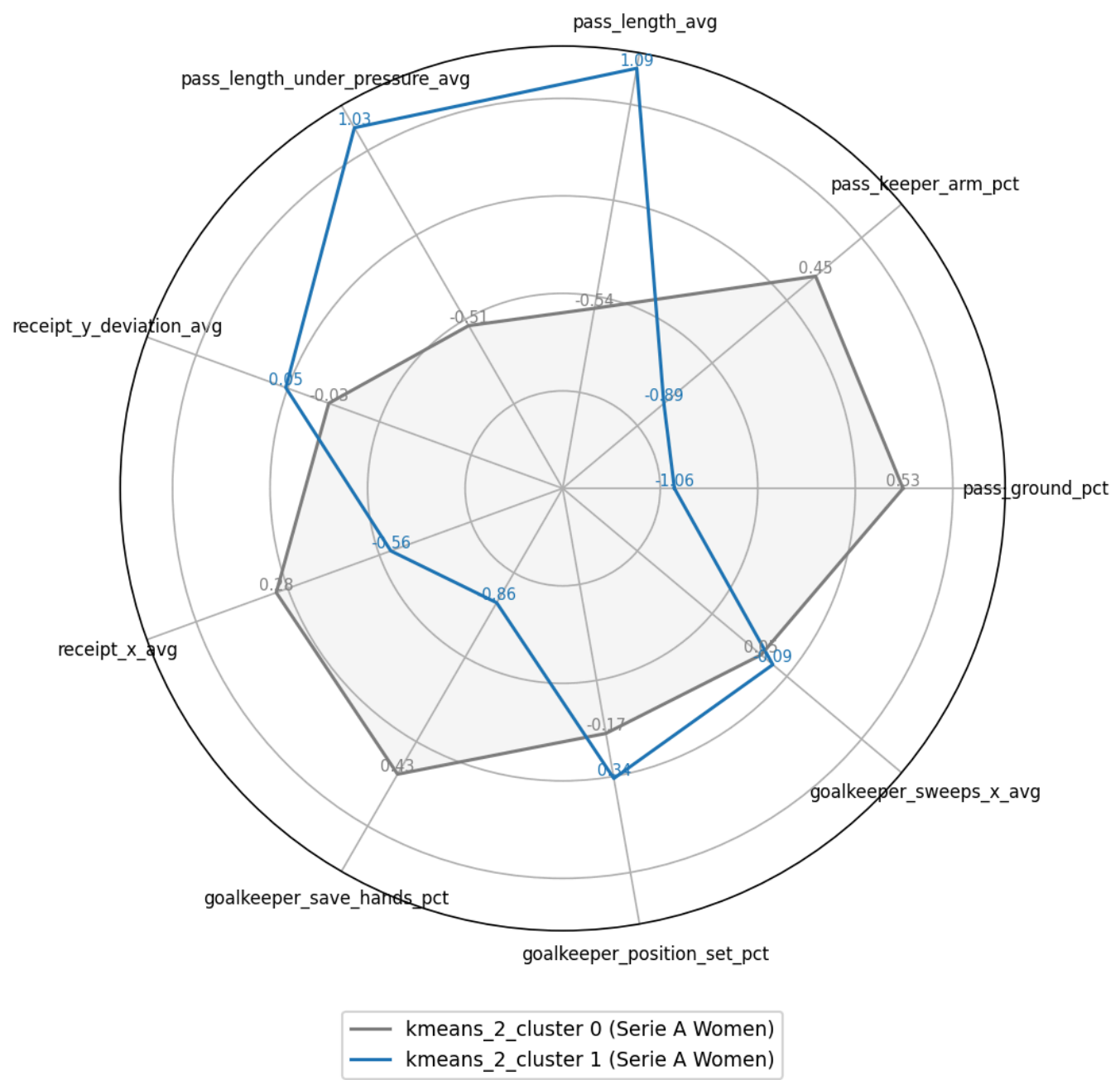


Figure 10: Serie A Women Goalkeepers — Adjusted Statistics Comparison

Radar plots for 3 clusters goalkeepers' averages

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All_Features — kmeans_3_cluster comparison (raw)

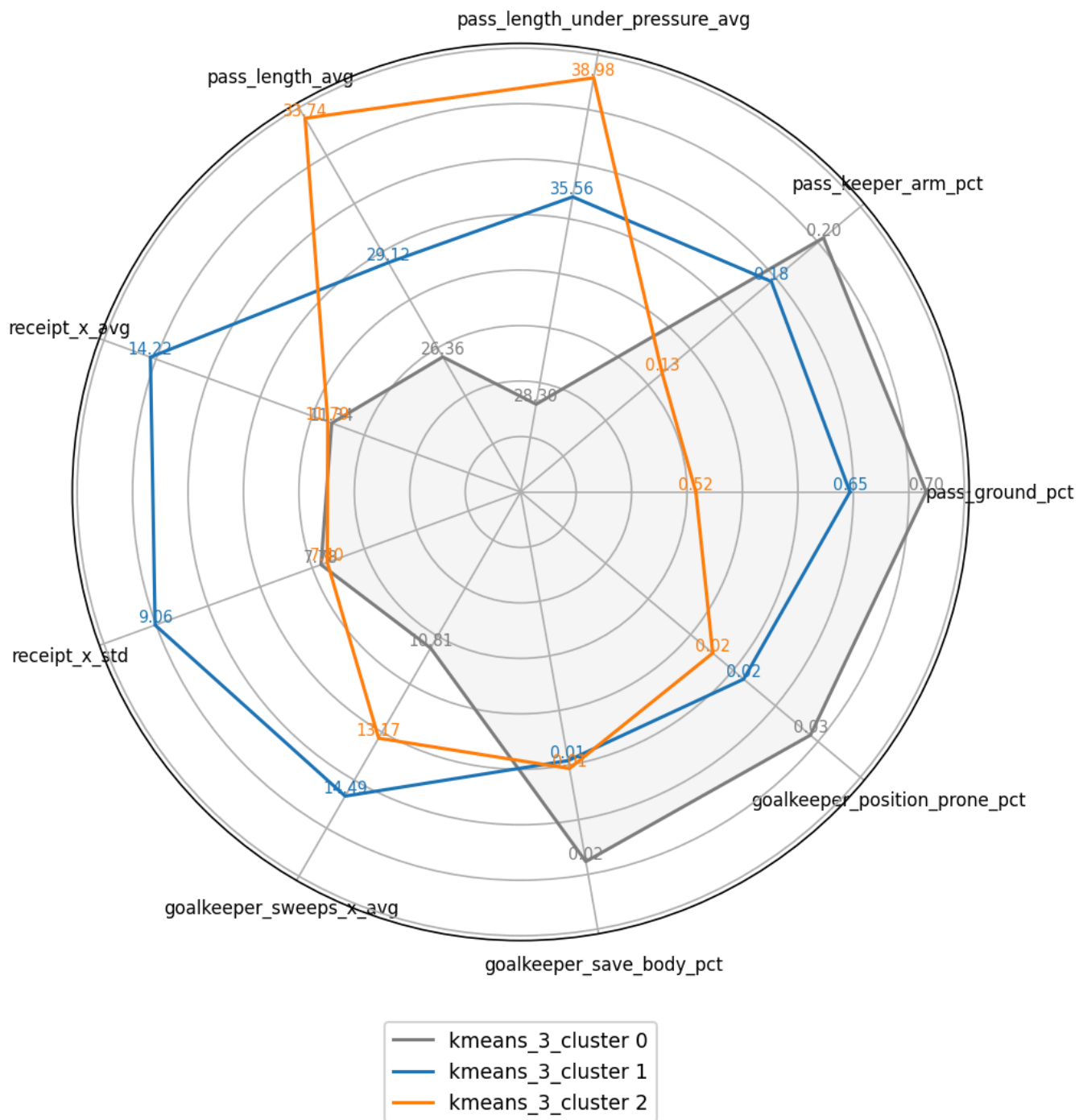


Figure 11: Raw Statistics Comparison

All_Features — kmeans_3_cluster comparison (adj)

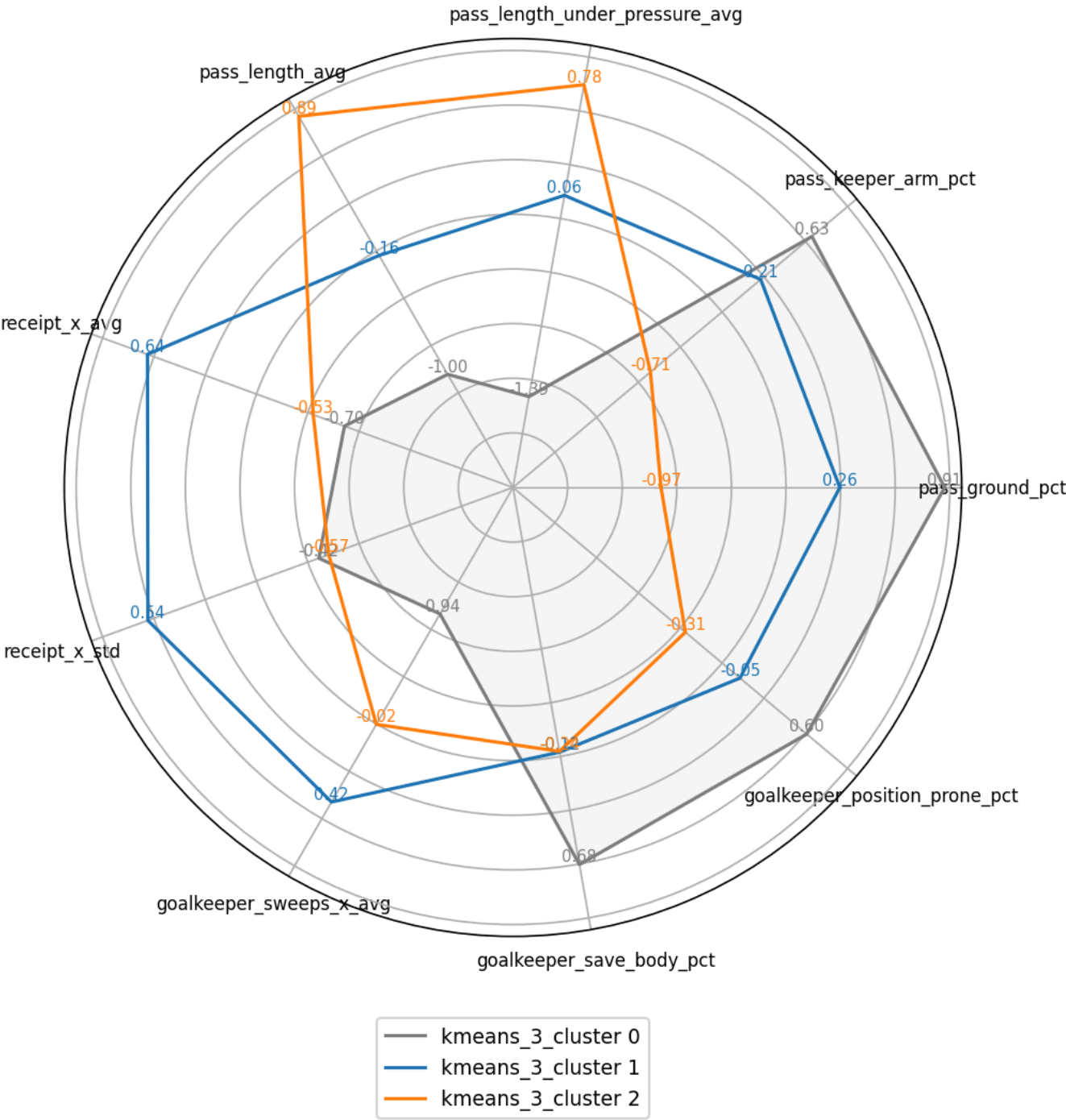


Figure 12: Adjusted Statistics Comparison

FA Women's Super League

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All_Features — kmeans_3_cluster comparison (FA Women's Super League, raw)

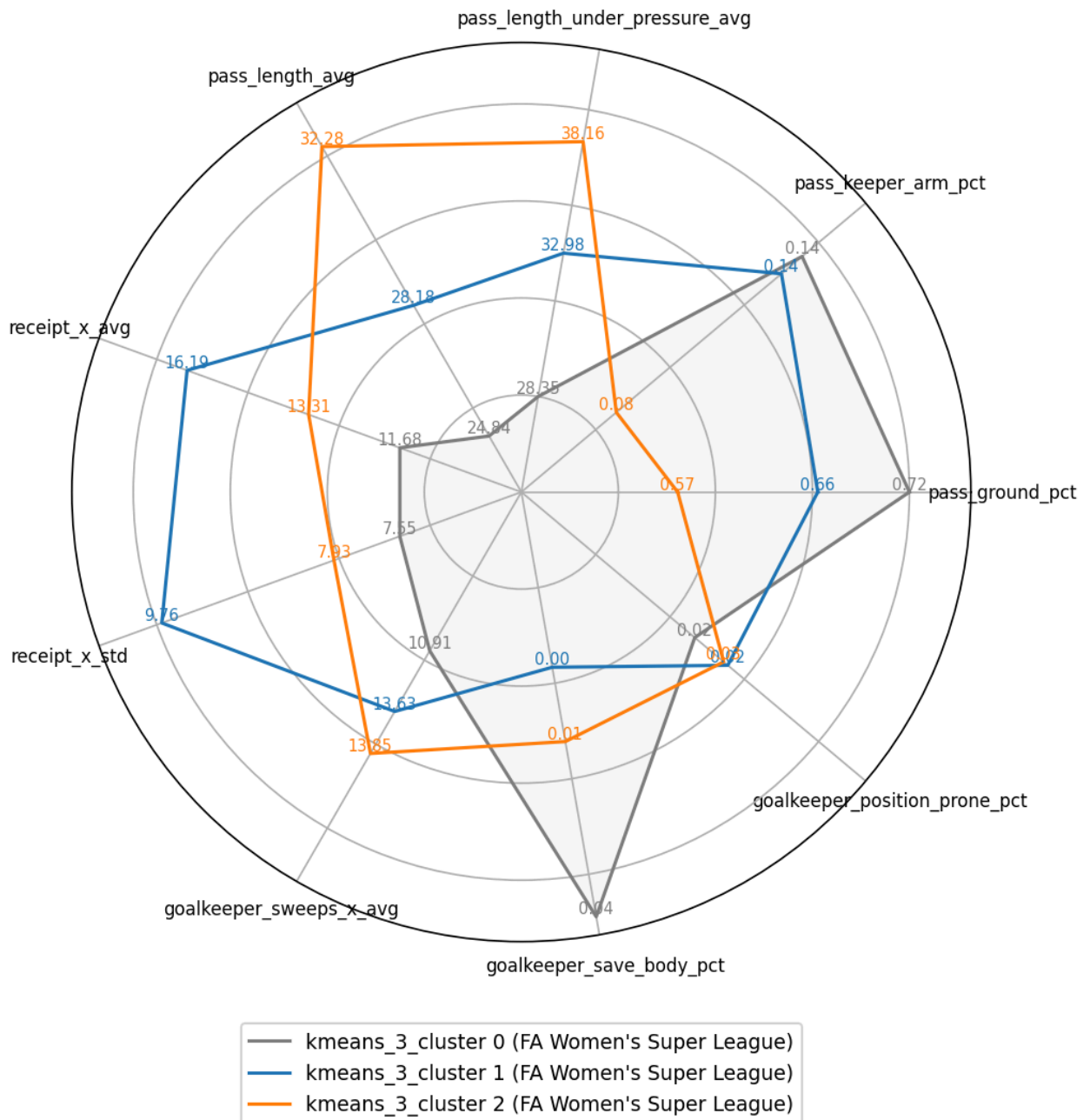


Figure 13: FA Women's Super League Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_3_cluster comparison (FA Women's Super League, adj)

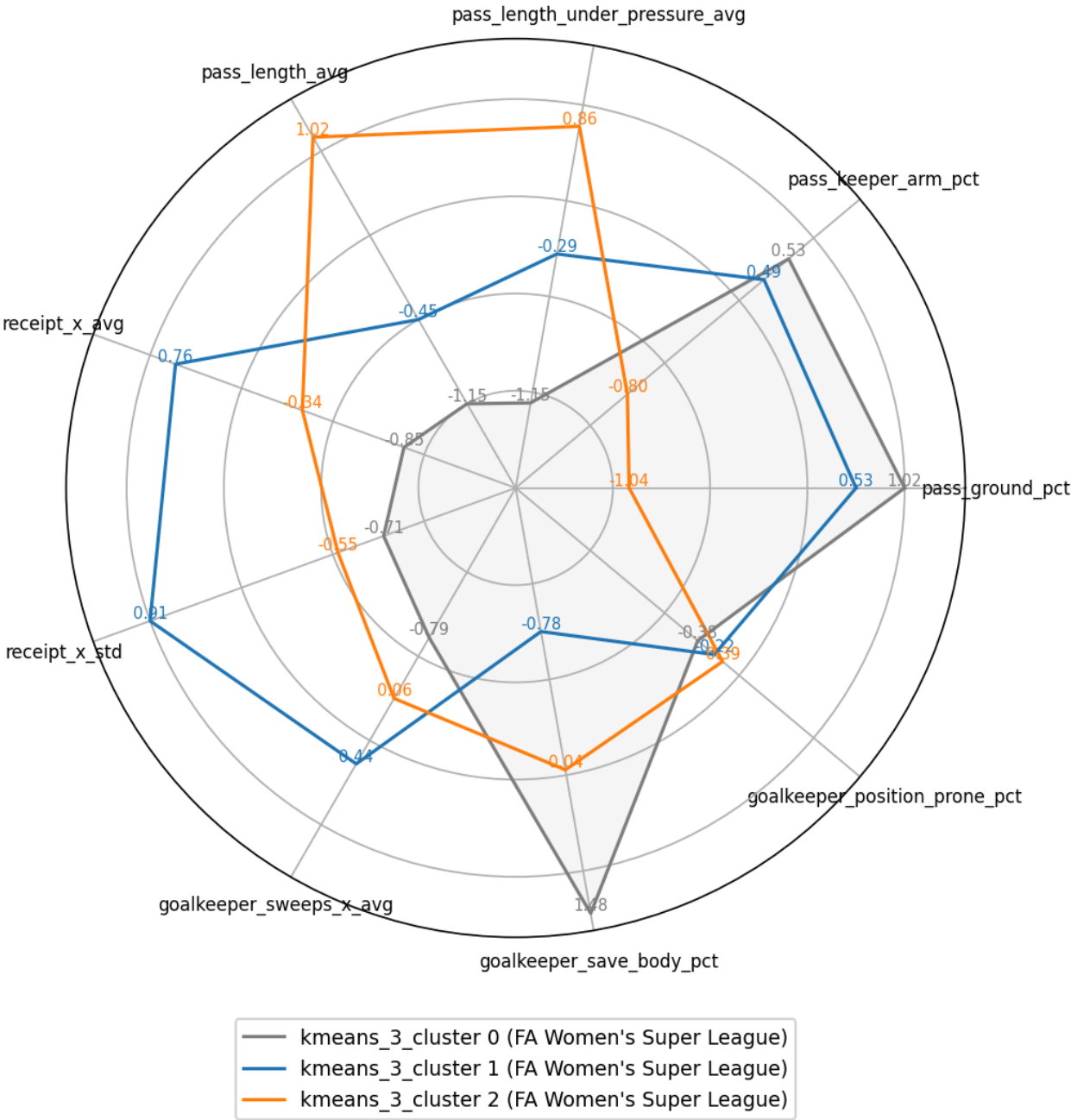


Figure 14: FA Women's Super League Goalkeepers — Adjusted Statistics Comparison

Frauen Bundesliga

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All_Features — kmeans_3_cluster comparison (Frauen Bundesliga, raw)

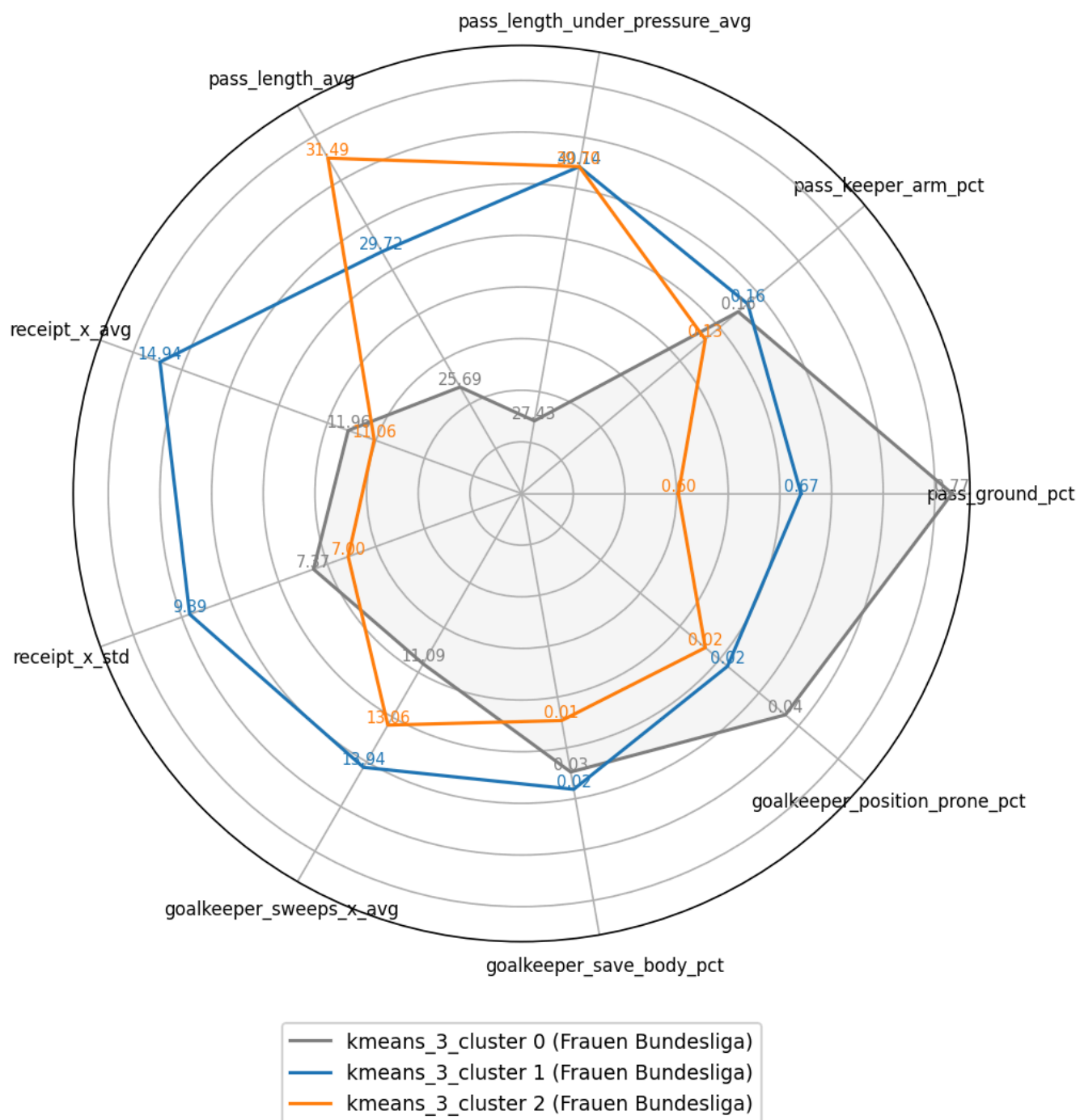


Figure 15: Frauen Bundesliga Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_3_cluster comparison (Frauen Bundesliga, adj)

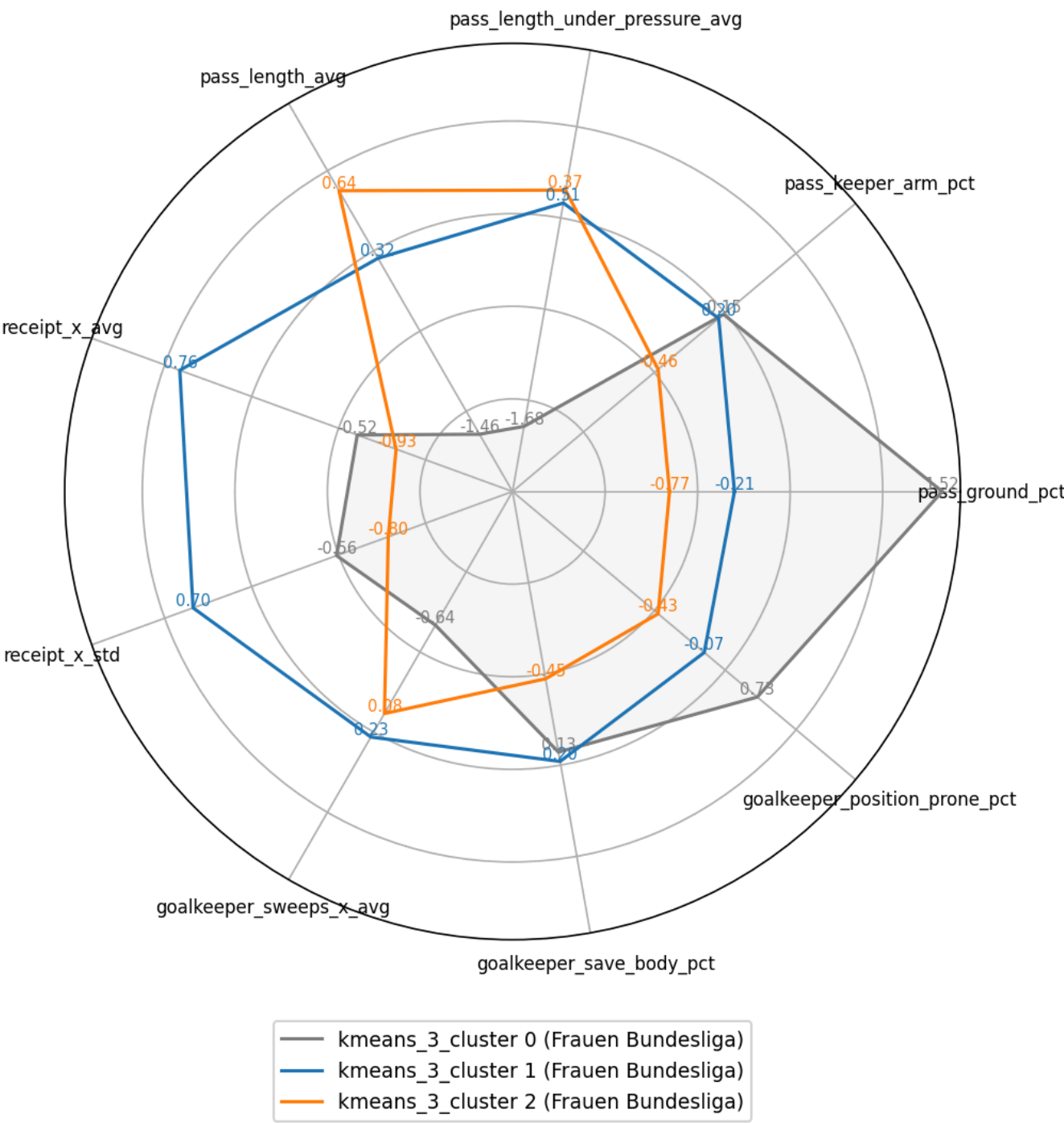


Figure 16: Frauen Bundesliga Goalkeepers — Adjusted Statistics Comparison

Liga F

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All_Features — kmeans_3_cluster comparison (Liga F, raw)

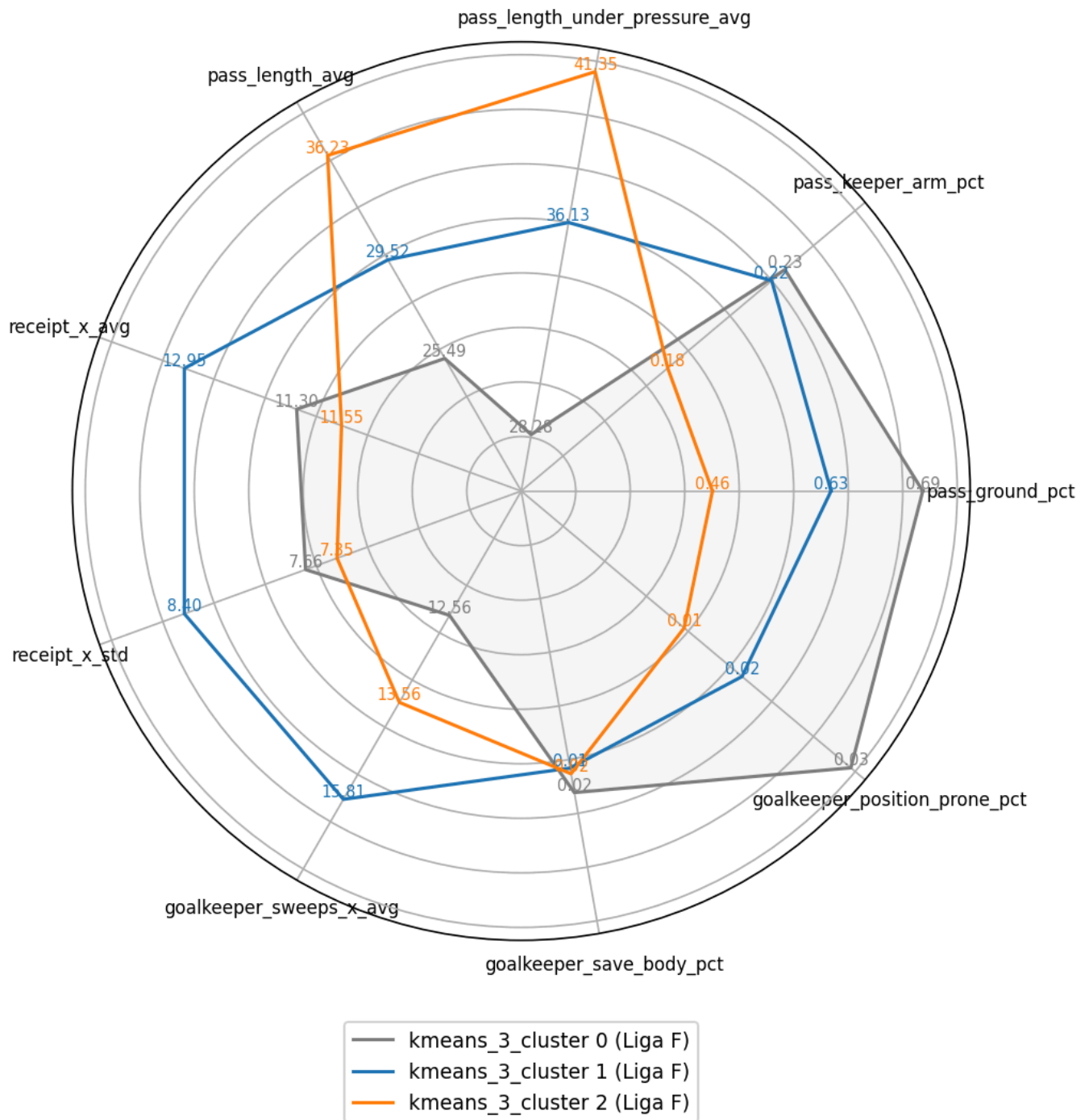


Figure 17: Liga F Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_3_cluster comparison (Liga F, adj)

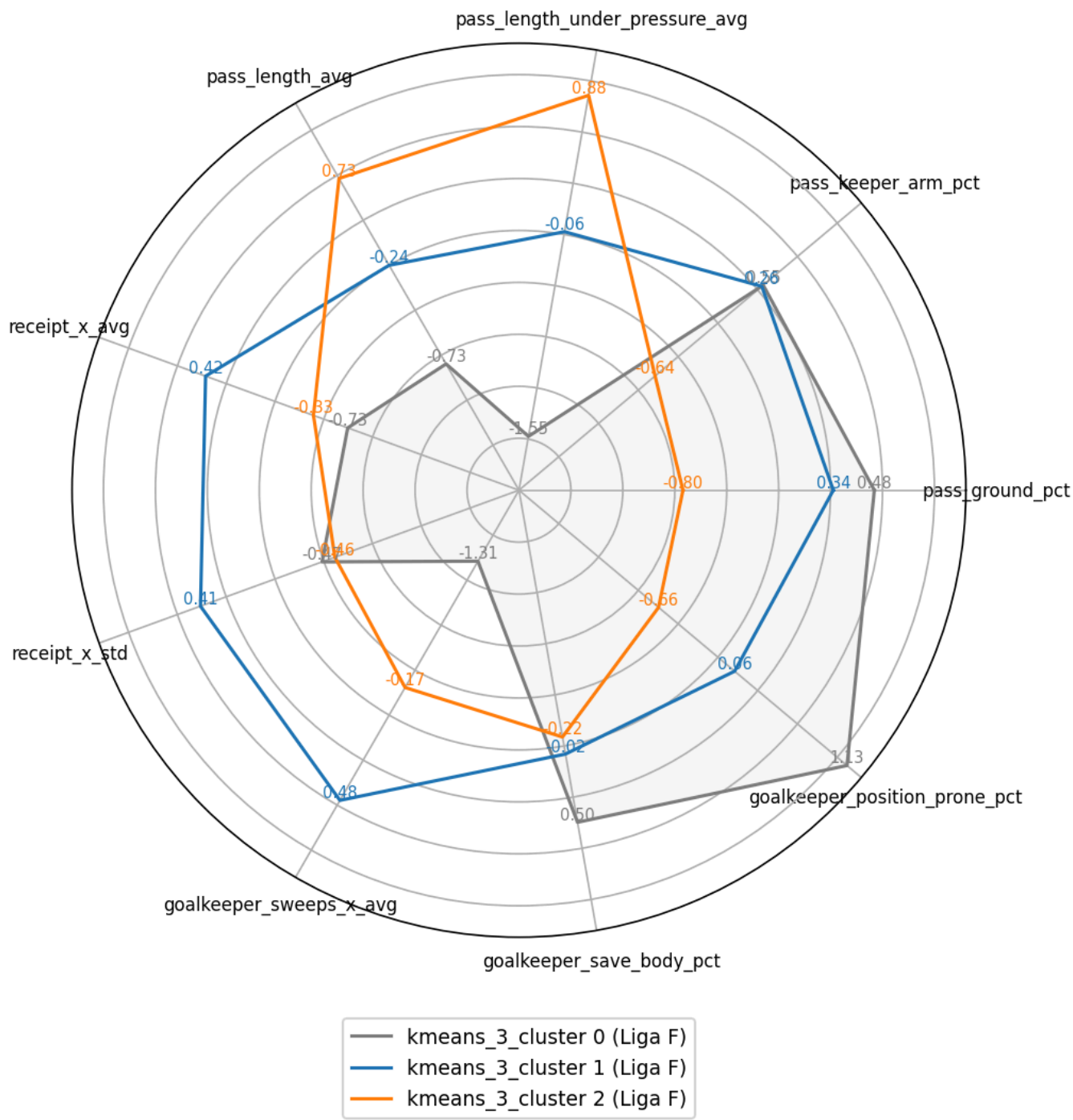


Figure 18: Liga F Goalkeepers — Adjusted Statistics Comparison

Serie A Women

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All_Features — kmeans_3_cluster comparison (Serie A Women, raw)

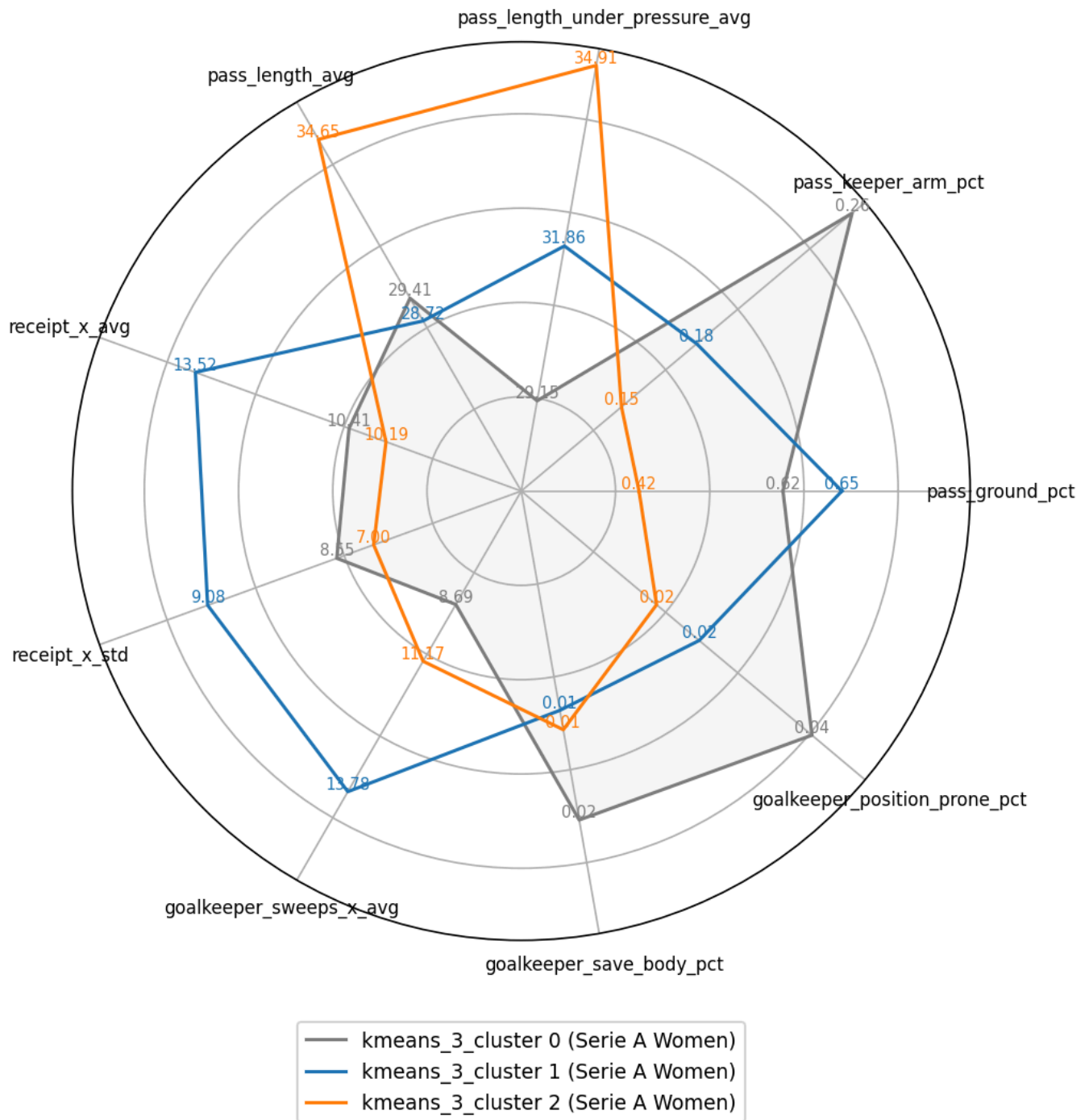


Figure 19: Serie A Women Goalkeepers — Raw Statistics Comparison

All_Features — kmeans_3_cluster comparison (Serie A Women, adj)

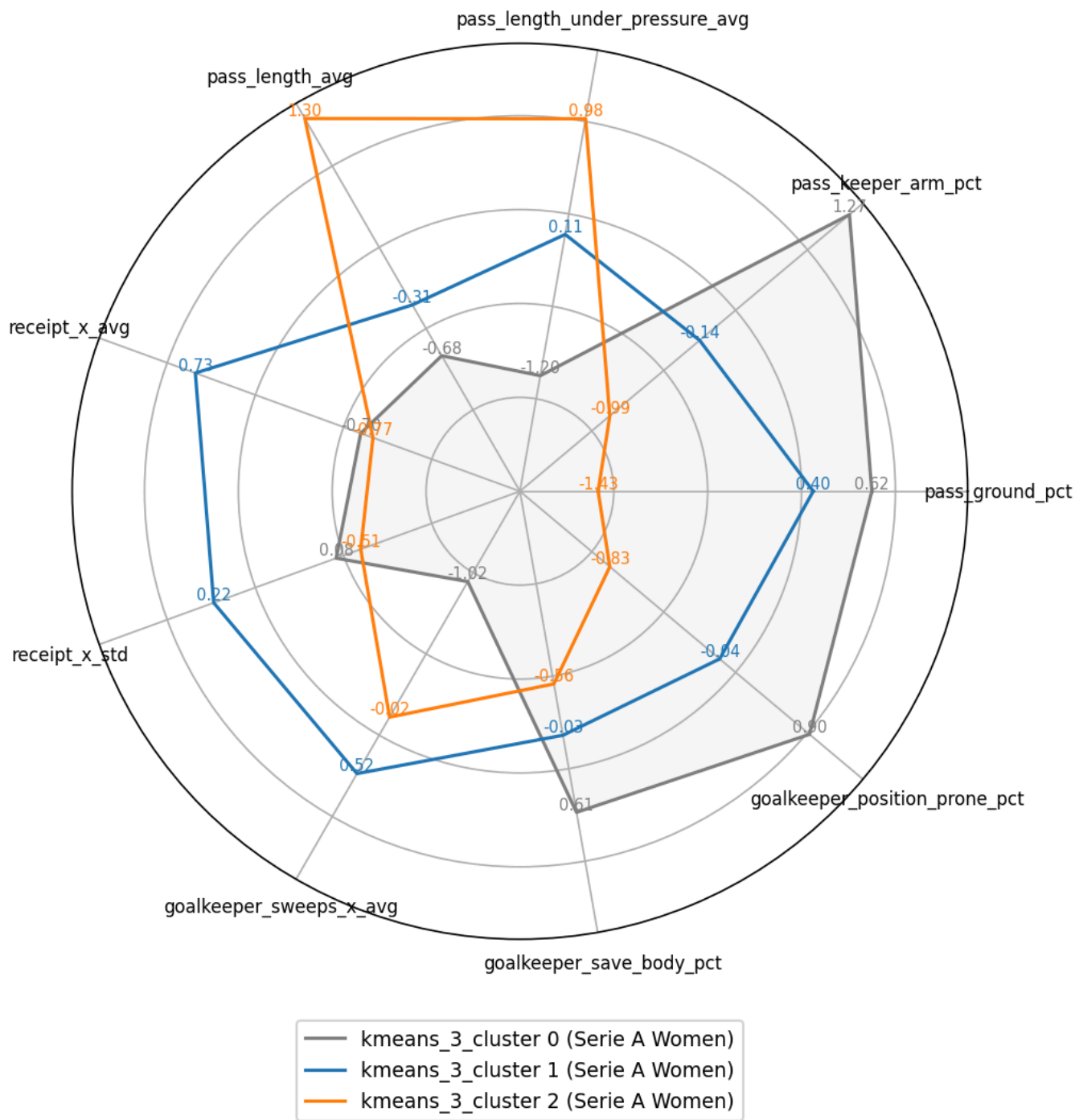


Figure 20: Serie A Women Goalkeepers — Adjusted Statistics Comparison

Radar plots for players

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Zecira Musovic - Chelsea

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2 cluster plots

All_Features — Zecira Musovic (Chelsea FCW) (raw)

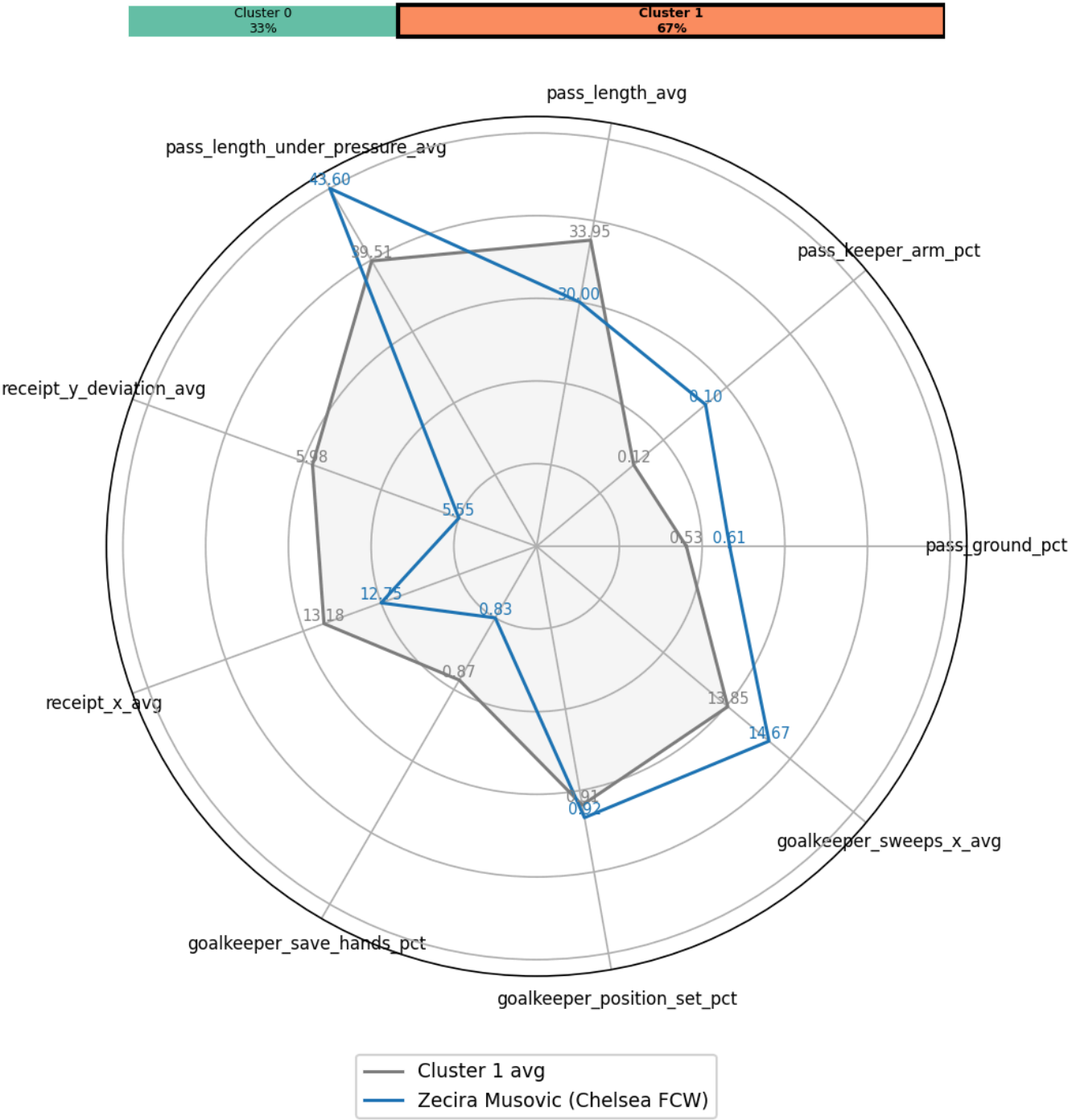


Figure 21: Zecira Musovic — Raw Statistics Comparison, 2 Clusters

All_Features — Zecira Musovic (Chelsea FCW) vs FA Women's Super League peers (raw)

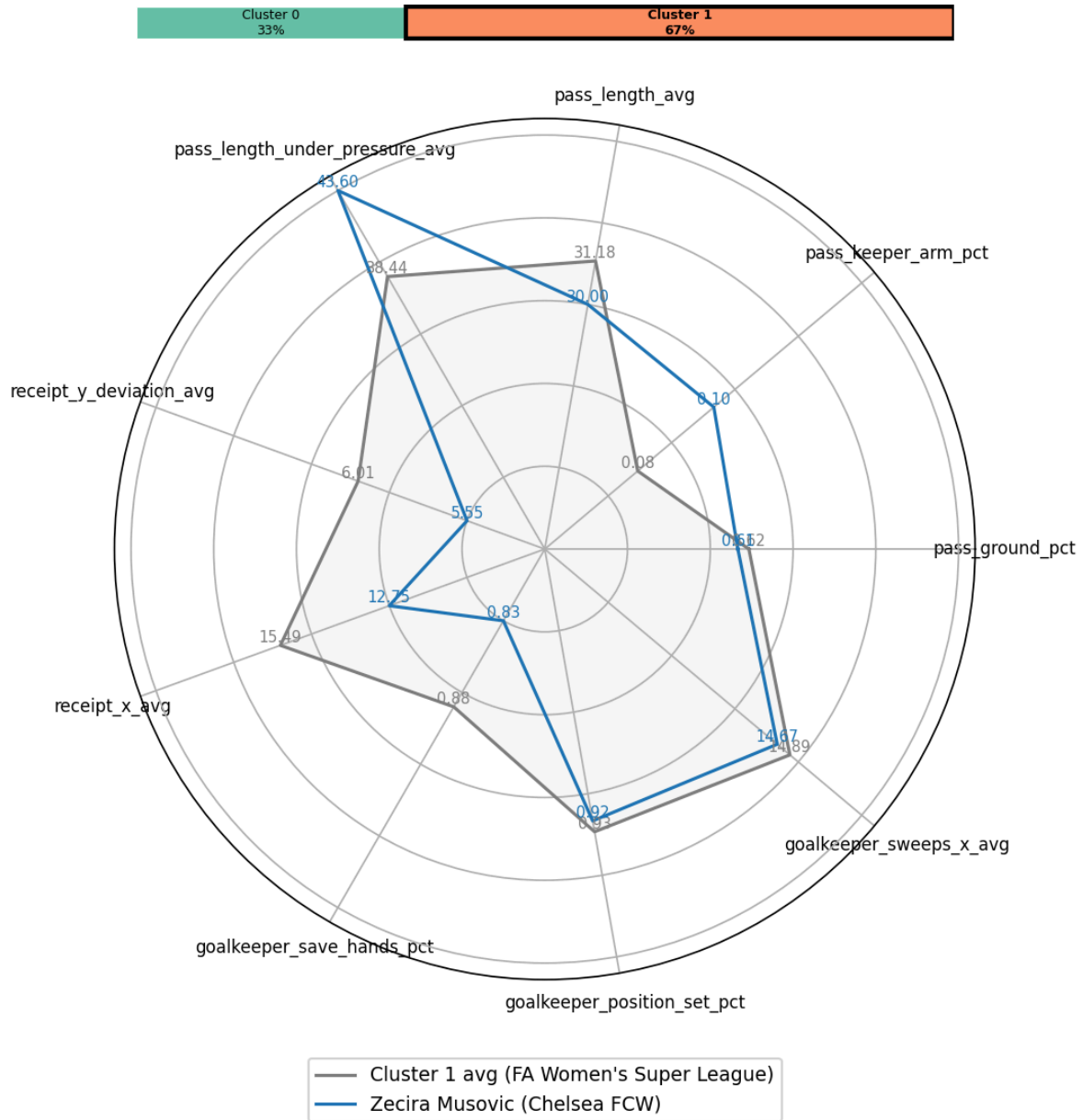


Figure 22: Zecira Musovic — League-Aware Raw Statistics Comparison, 2 Clusters

All_Features — Zecira Musovic (Chelsea FCW) (adj)

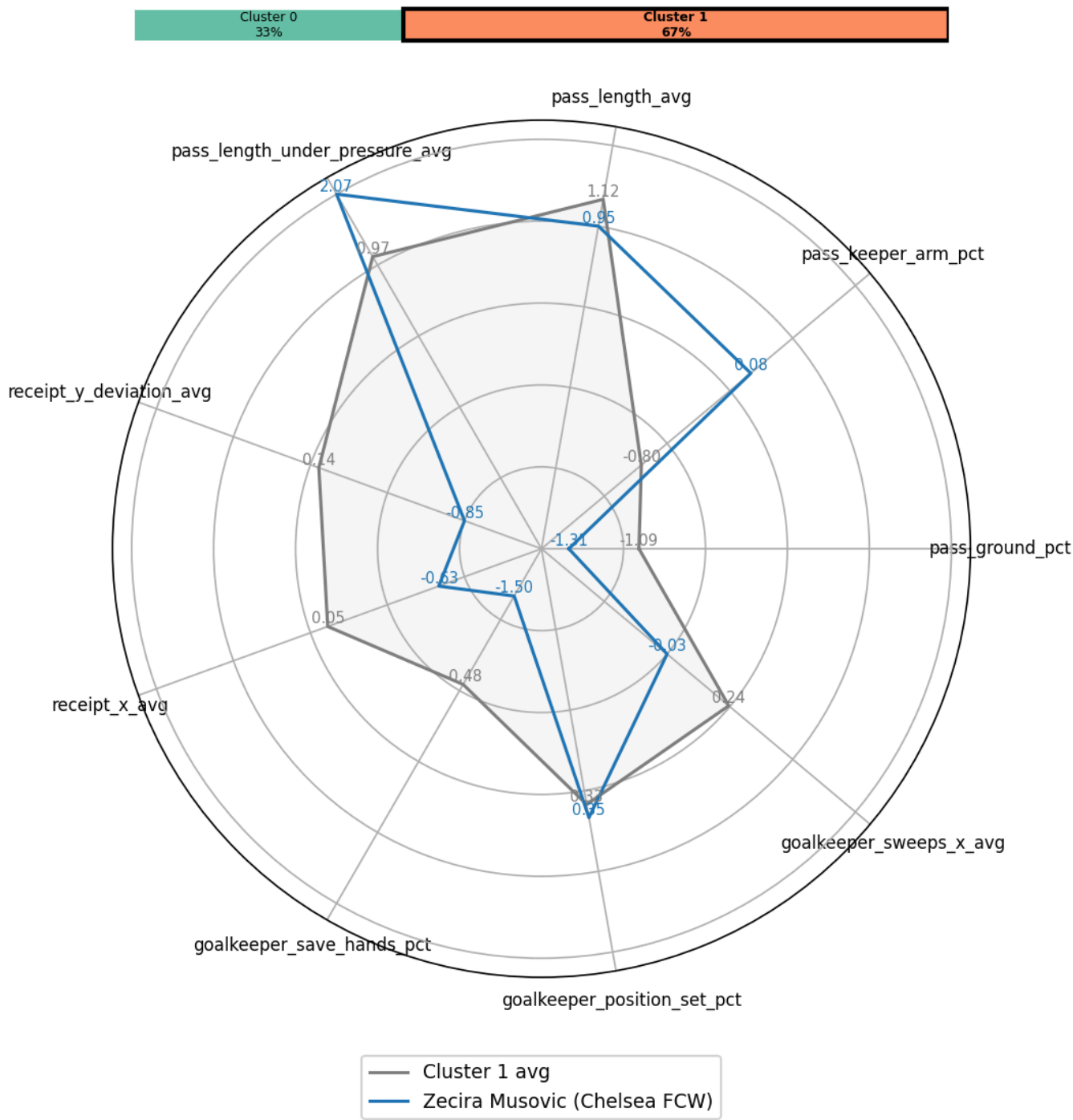


Figure 23: Zecira Musovic — Adjusted Statistics Comparison, 2 Clusters

All_Features — Zecira Musovic (Chelsea FCW) vs FA Women's Super League peers (adj)

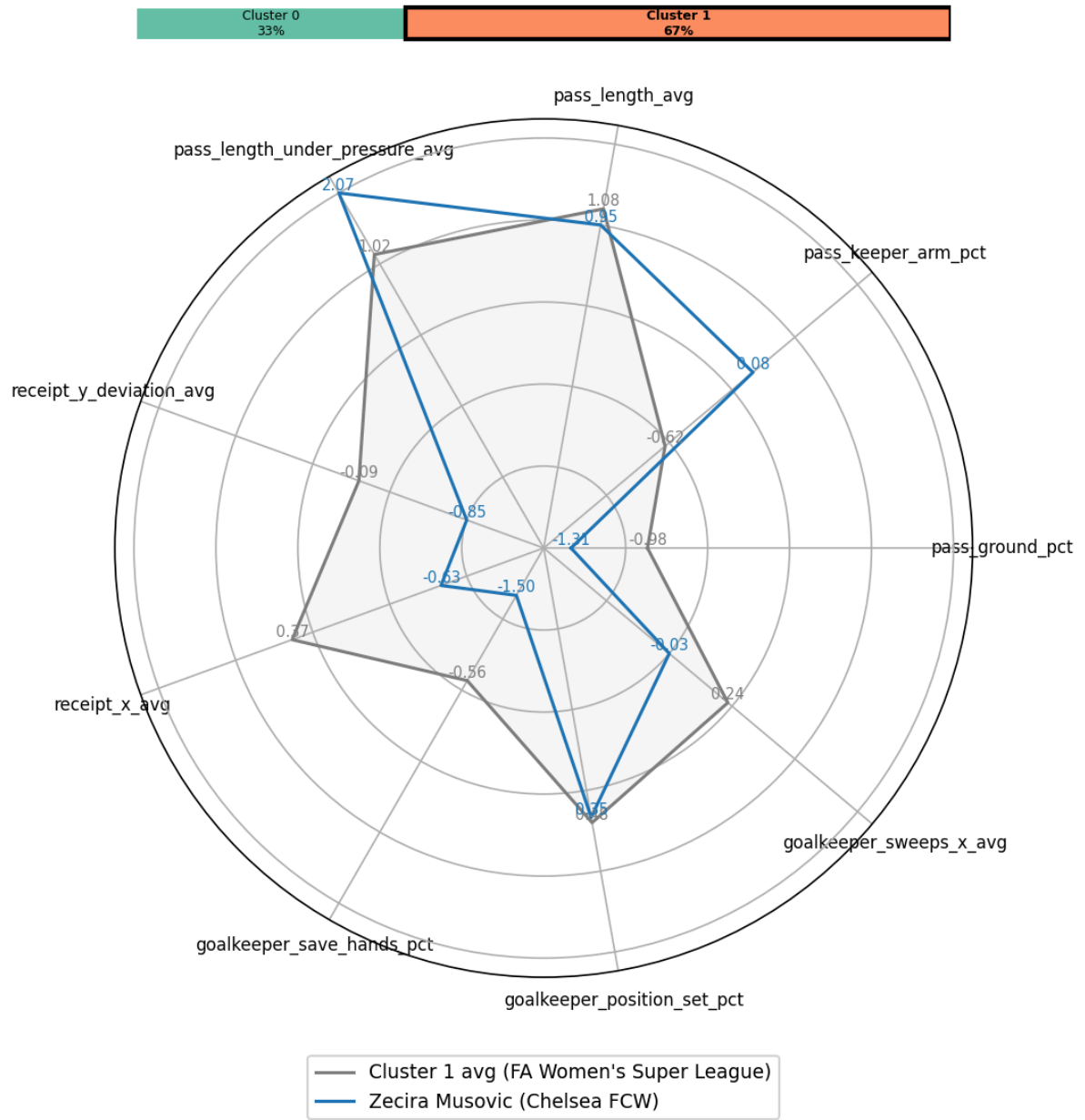


Figure 24: Zecira Musovic — League-Aware Adjusted Statistics Comparison, 2 Clusters

3 cluster plots

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All_Features — Zecira Musovic (Chelsea FCW) (raw)

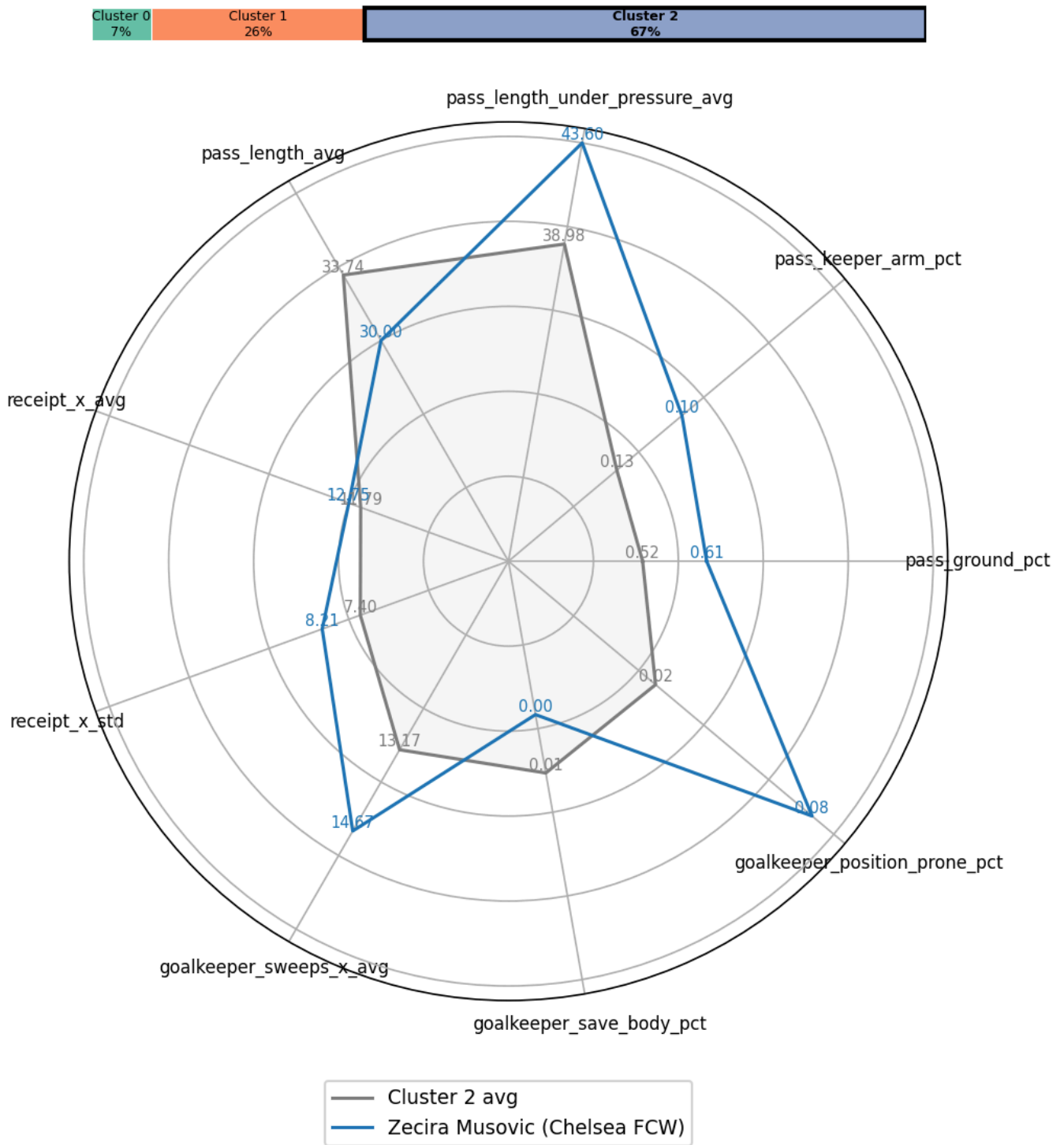


Figure 25: Zecira Musovic — Raw Statistics Comparison, 3 Clusters

All_Features — Zecira Musovic (Chelsea FCW) vs FA Women's Super League peers (raw)

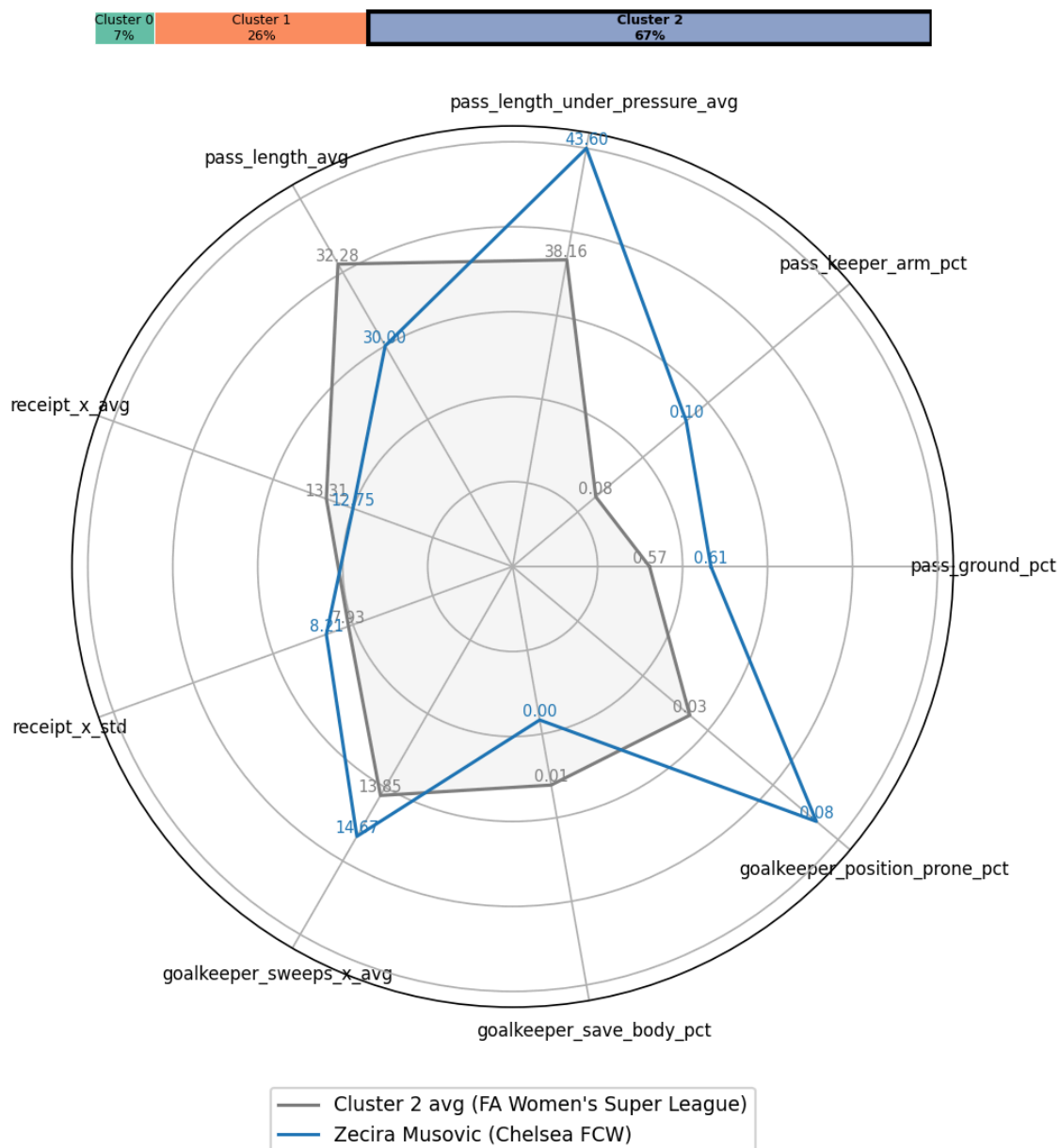


Figure 26: Zecira Musovic — League-Aware Raw Statistics Comparison, 3 Clusters

All_Features — Zecira Musovic (Chelsea FCW) (adj)

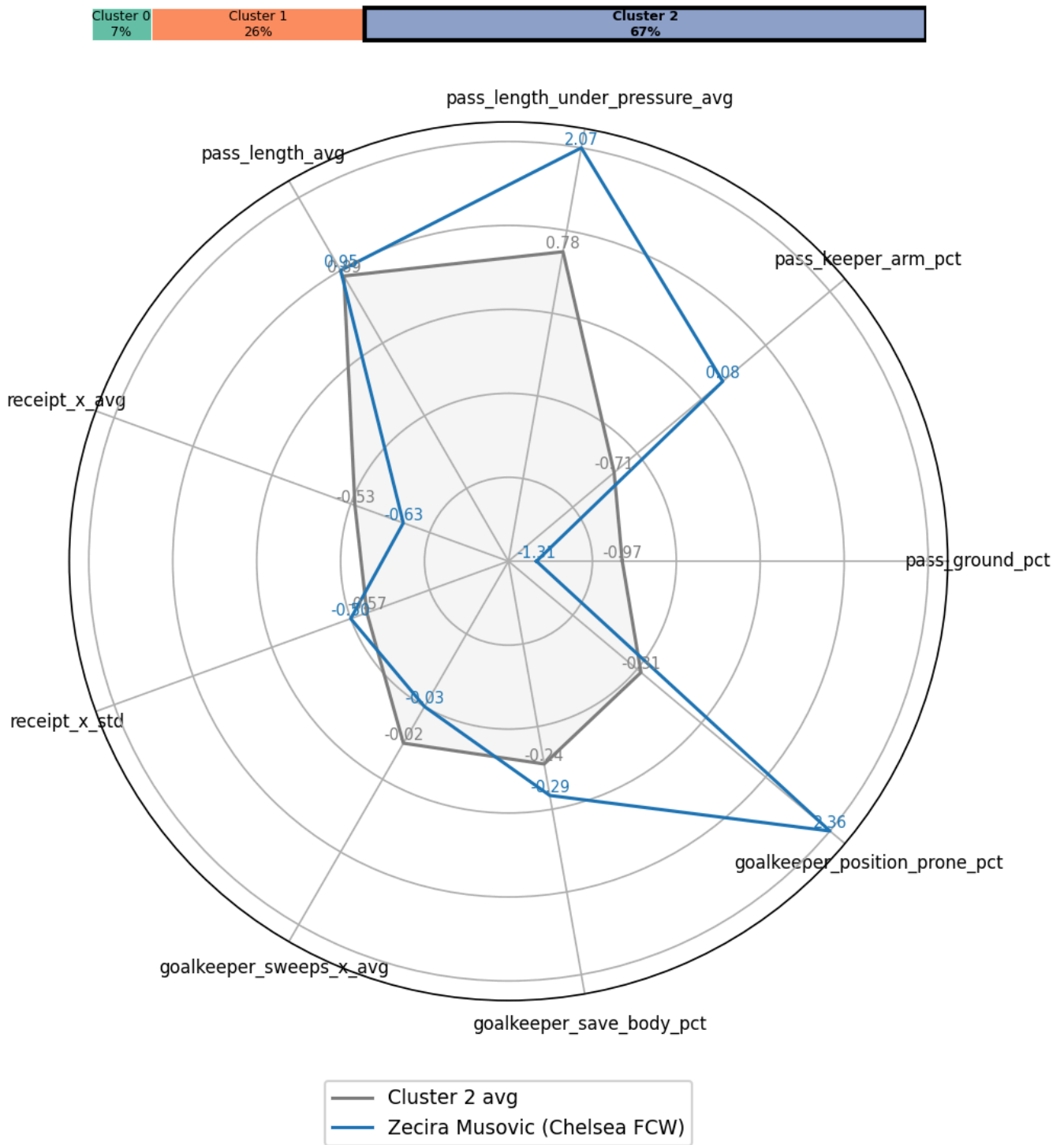


Figure 27: Zecira Musovic — Adjusted Statistics Comparison, 3 Clusters

All_Features — Zecira Musovic (Chelsea FCW) vs FA Women's Super League peers (adj)

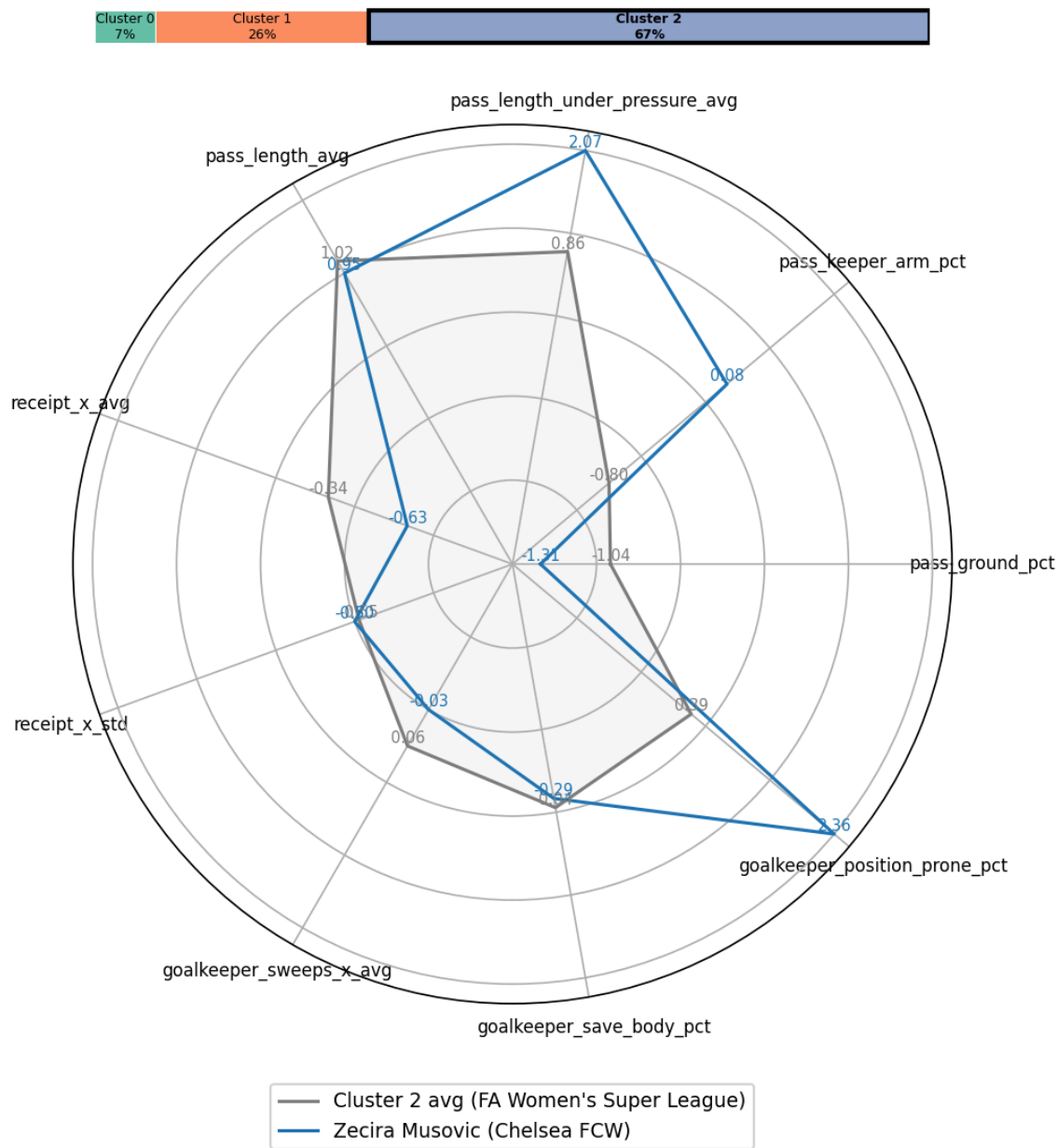


Figure 28: Zecira Musovic — League-Aware Adjusted Statistics Comparison, 3 Clusters

Hannah Hampton - Chelsea

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2 cluster plots

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All_Features — Hannah Hampton (Chelsea FCW) (raw)

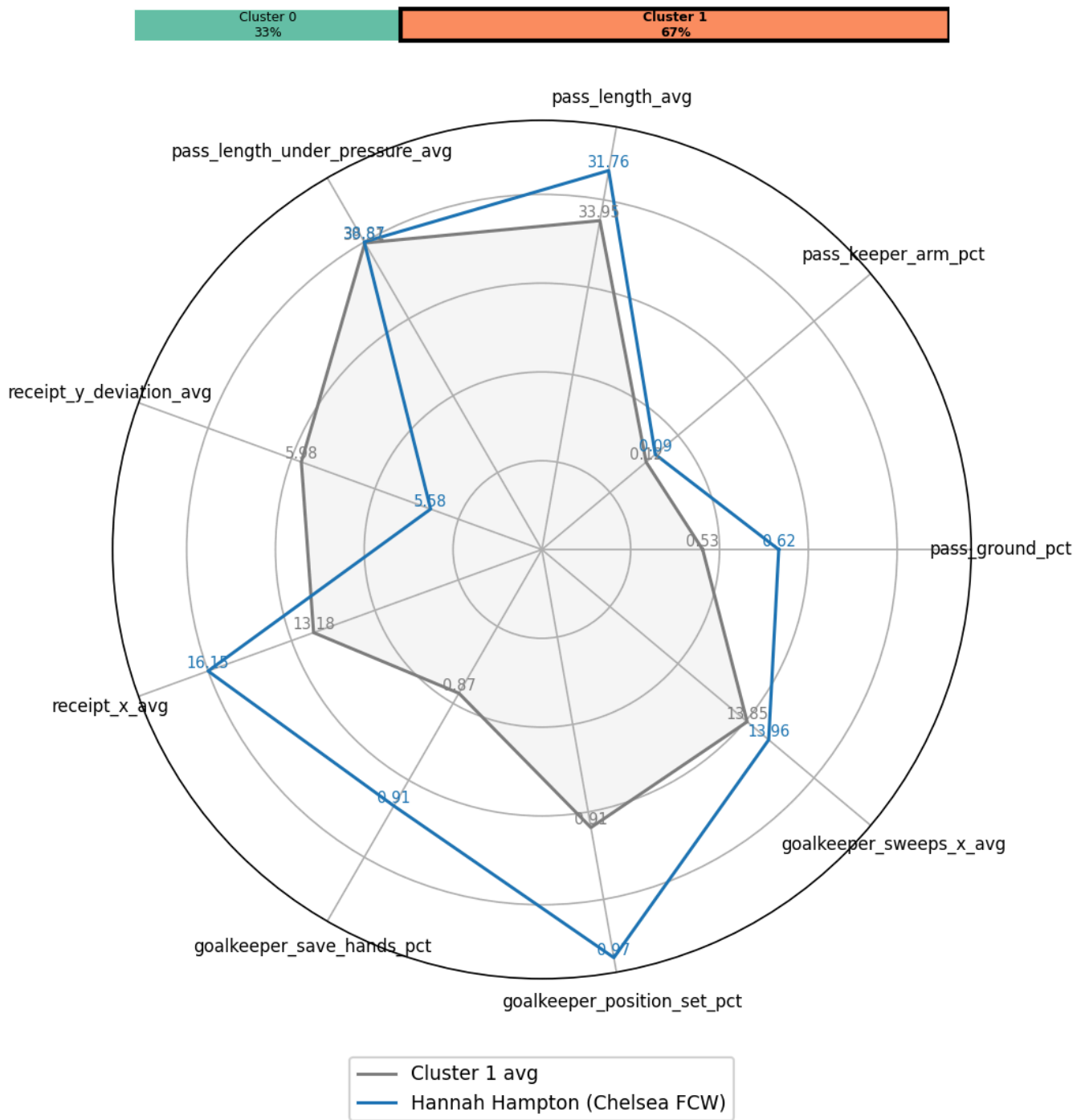


Figure 29: Hannah Hampton — Raw Statistics Comparison, 2 Clusters

All_Features — Hannah Hampton (Chelsea FCW) vs FA Women's Super League peers (raw)

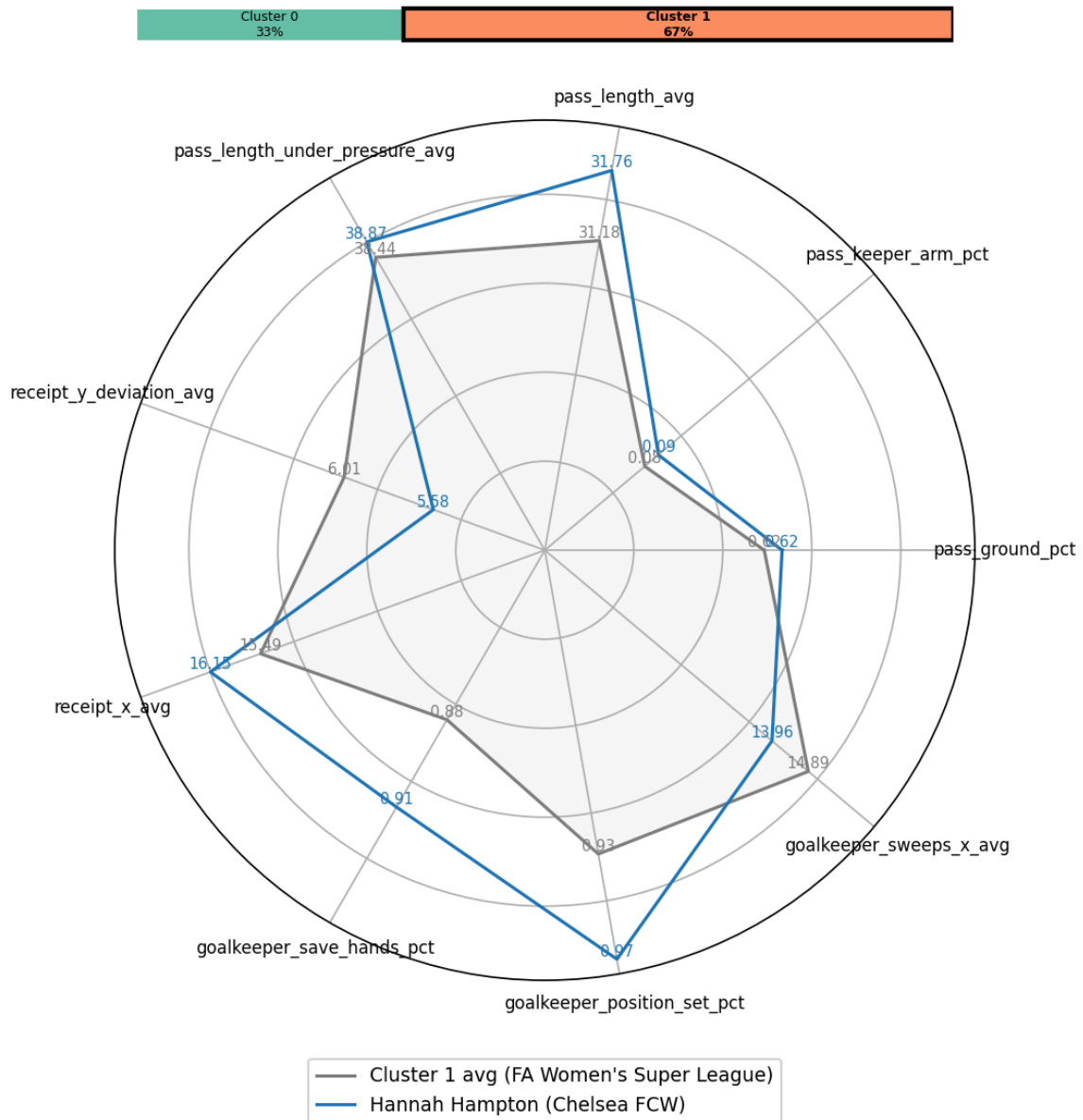


Figure 30: Hannah Hampton — League-Aware Raw Statistics Comparison, 2 Clusters

All_Features — Hannah Hampton (Chelsea FCW) (adj)

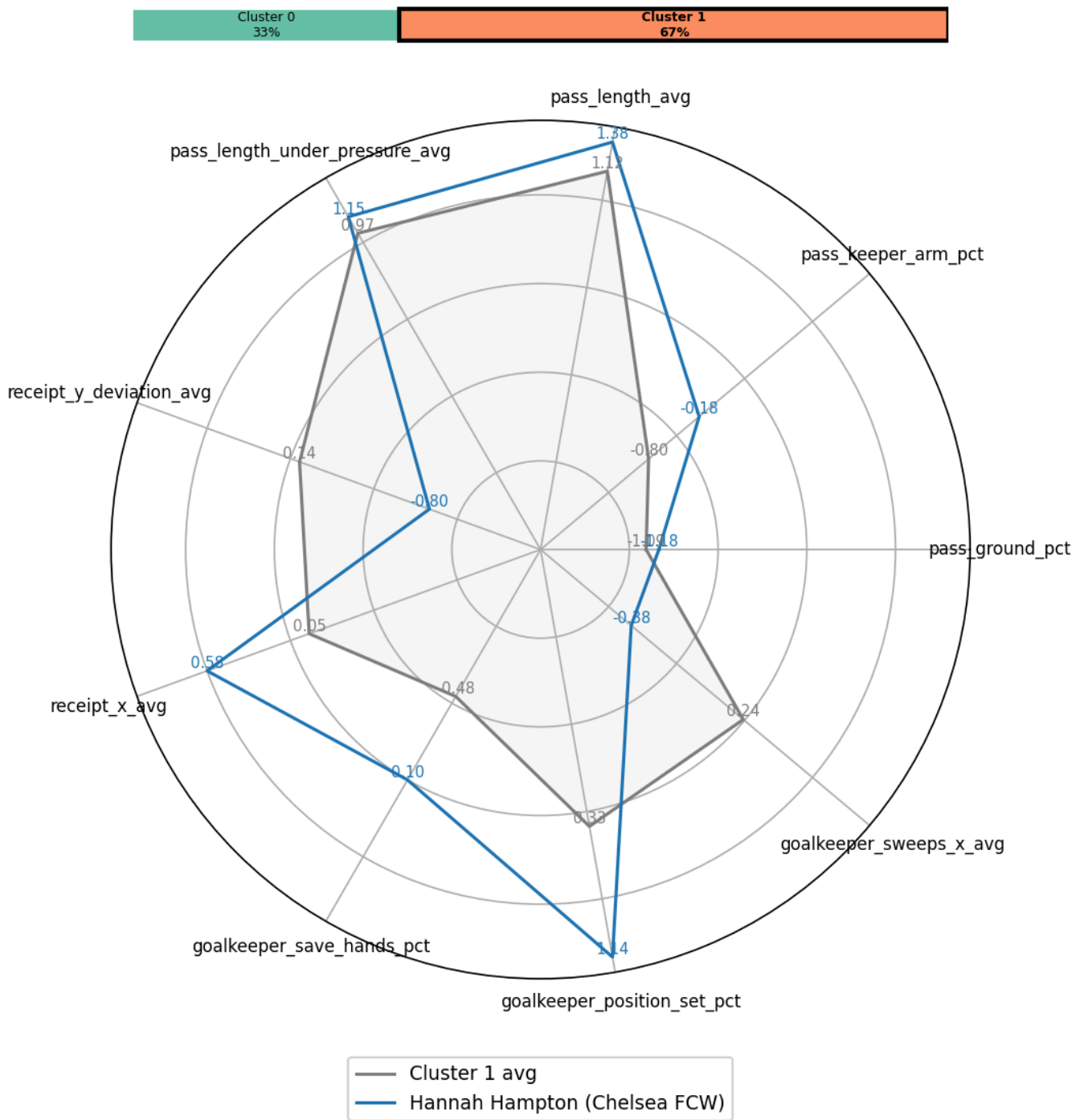


Figure 31: Hannah Hampton — Adjusted Statistics Comparison, 2 Clusters

All_Features — Hannah Hampton (Chelsea FCW) vs FA Women's Super League peers (adj)

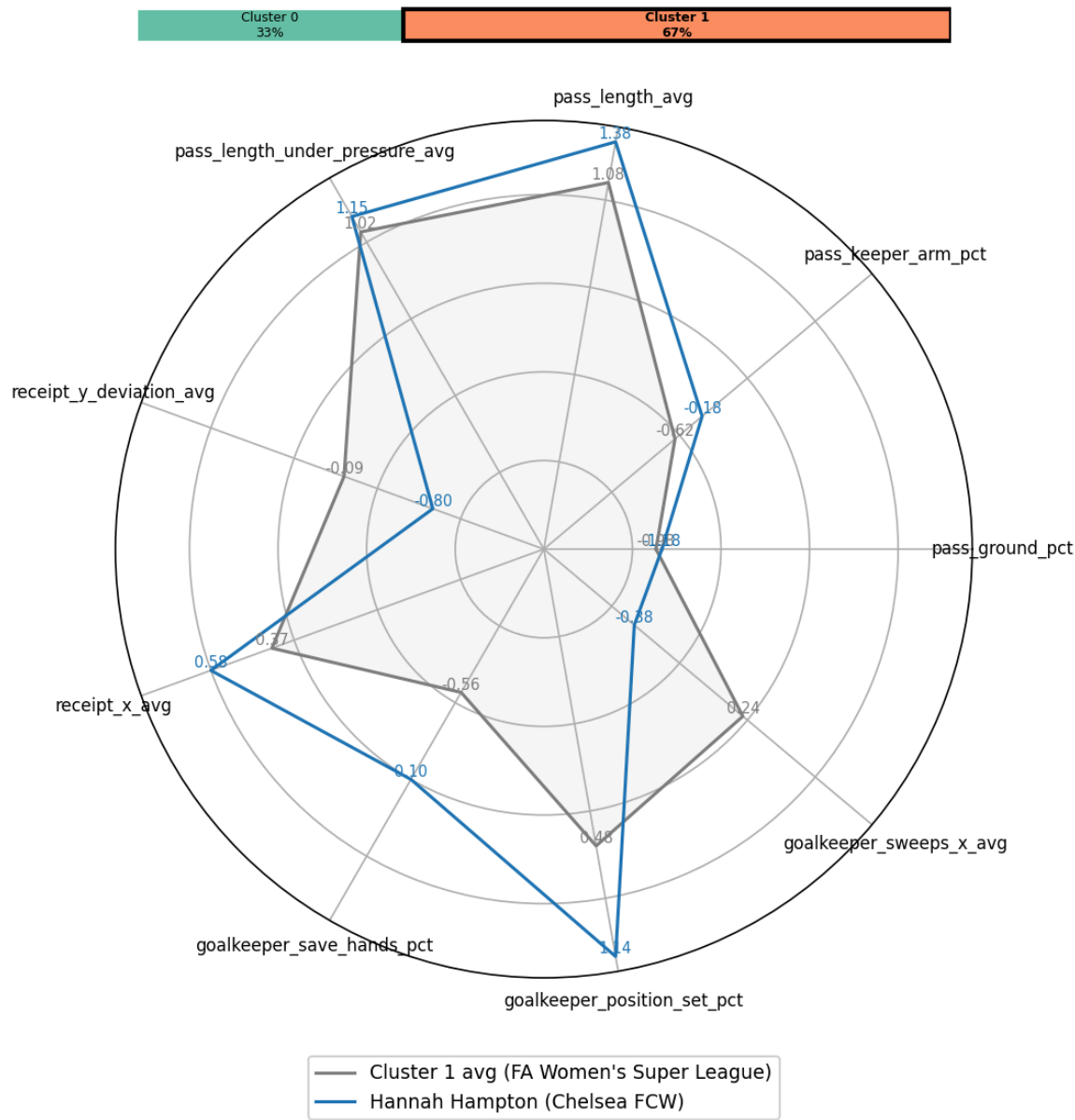


Figure 32: Hannah Hampton — League-Aware Adjusted Statistics Comparison, 2 Clusters

3 cluster plots

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All_Features — Hannah Hampton (Chelsea FCW) (raw)

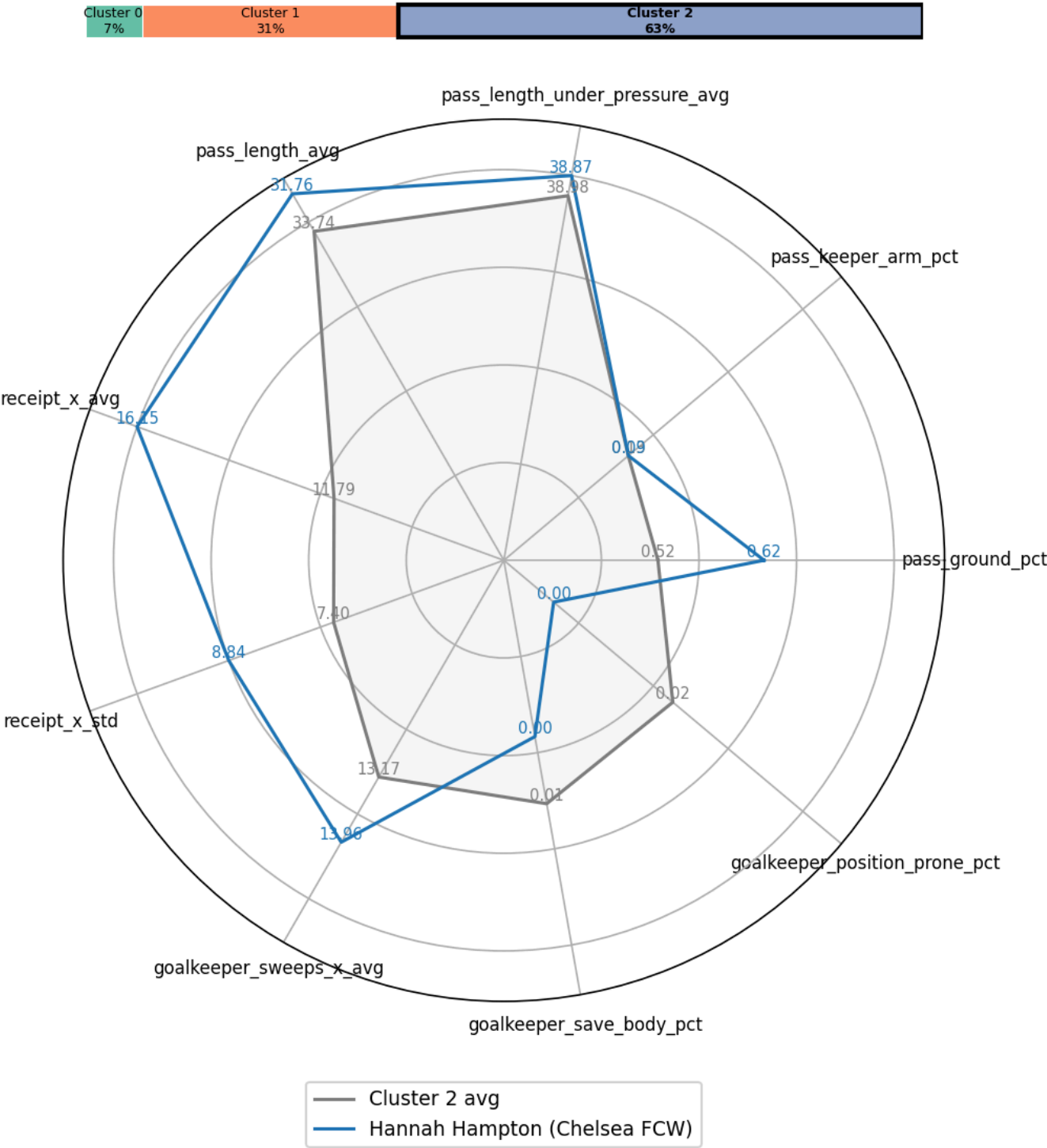


Figure 33: Hannah Hampton — Raw Statistics Comparison, 3 Clusters

All_Features — Hannah Hampton (Chelsea FCW) vs FA Women's Super League peers (raw)

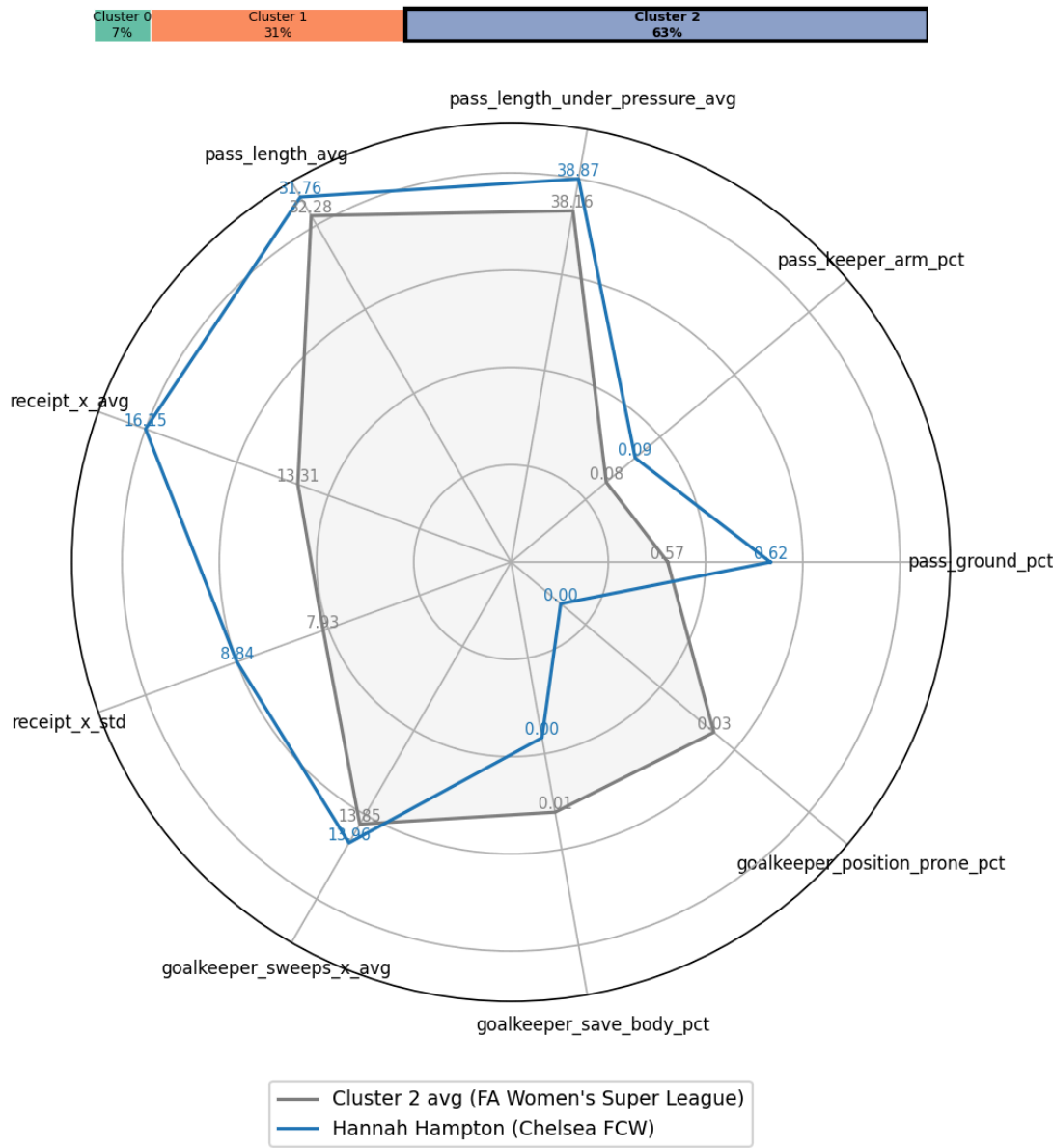


Figure 34: Hannah Hampton — League-Aware Raw Statistics Comparison, 3 Clusters

All_Features — Hannah Hampton (Chelsea FCW) (adj)

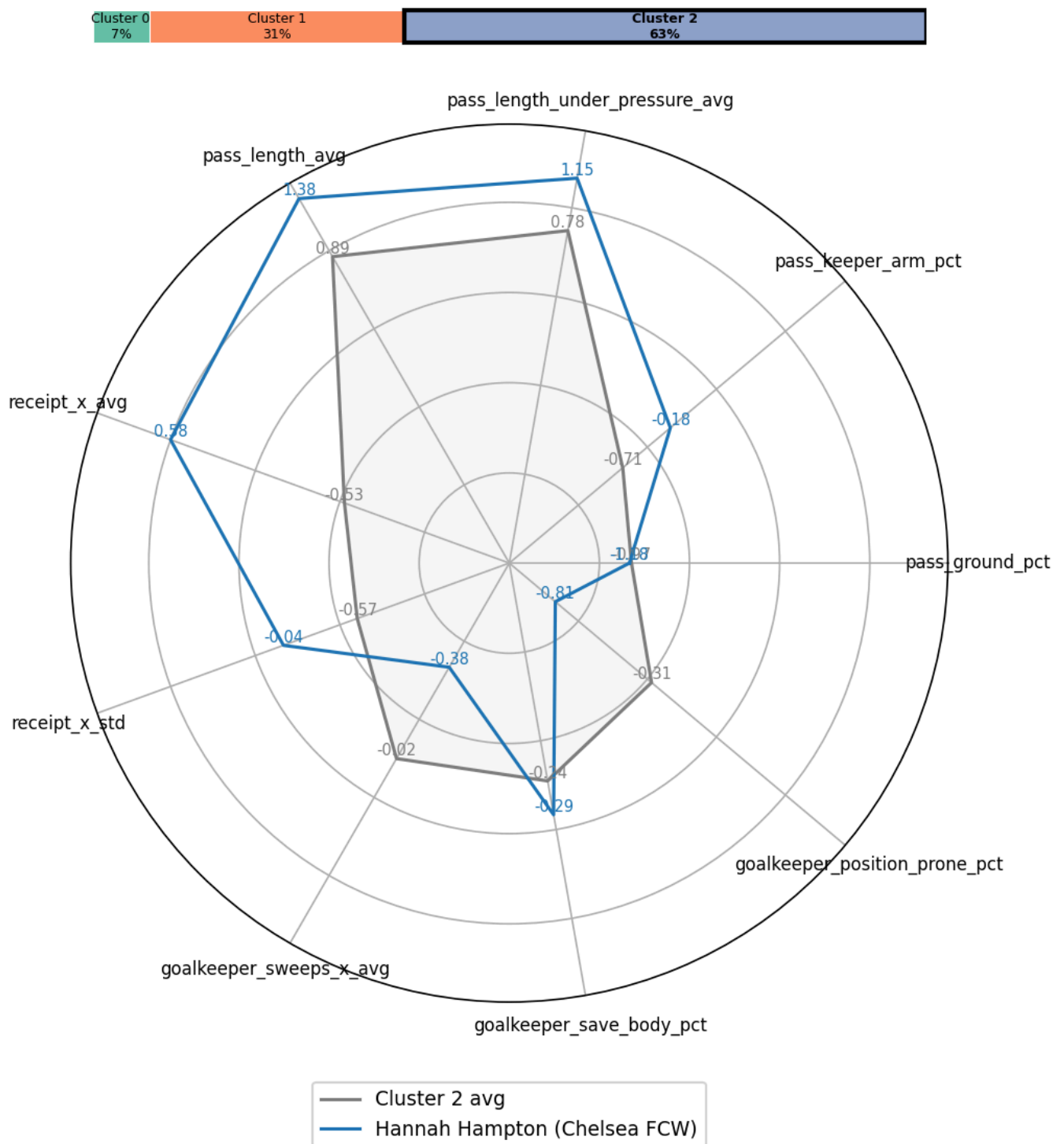


Figure 35: Hannah Hampton — Adjusted Statistics Comparison, 3 Clusters

All_Features — Hannah Hampton (Chelsea FCW) vs FA Women's Super League peers (adj)

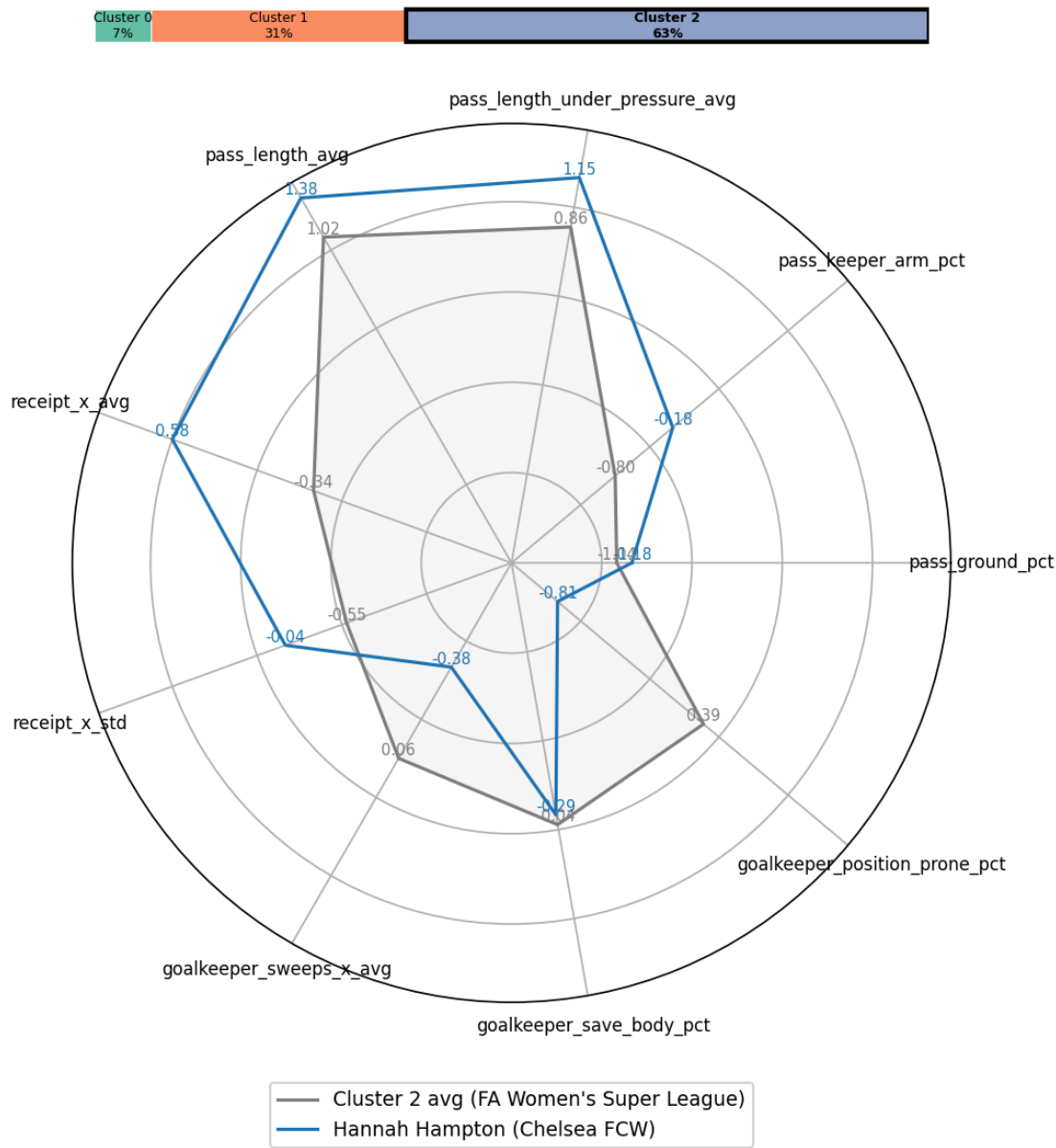


Figure 36: Hannah Hampton — League-Aware Adjusted Statistics Comparison, 3 Clusters

Rebecca Leigh Spencer - Tottenham

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2 cluster plots

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All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) (raw)

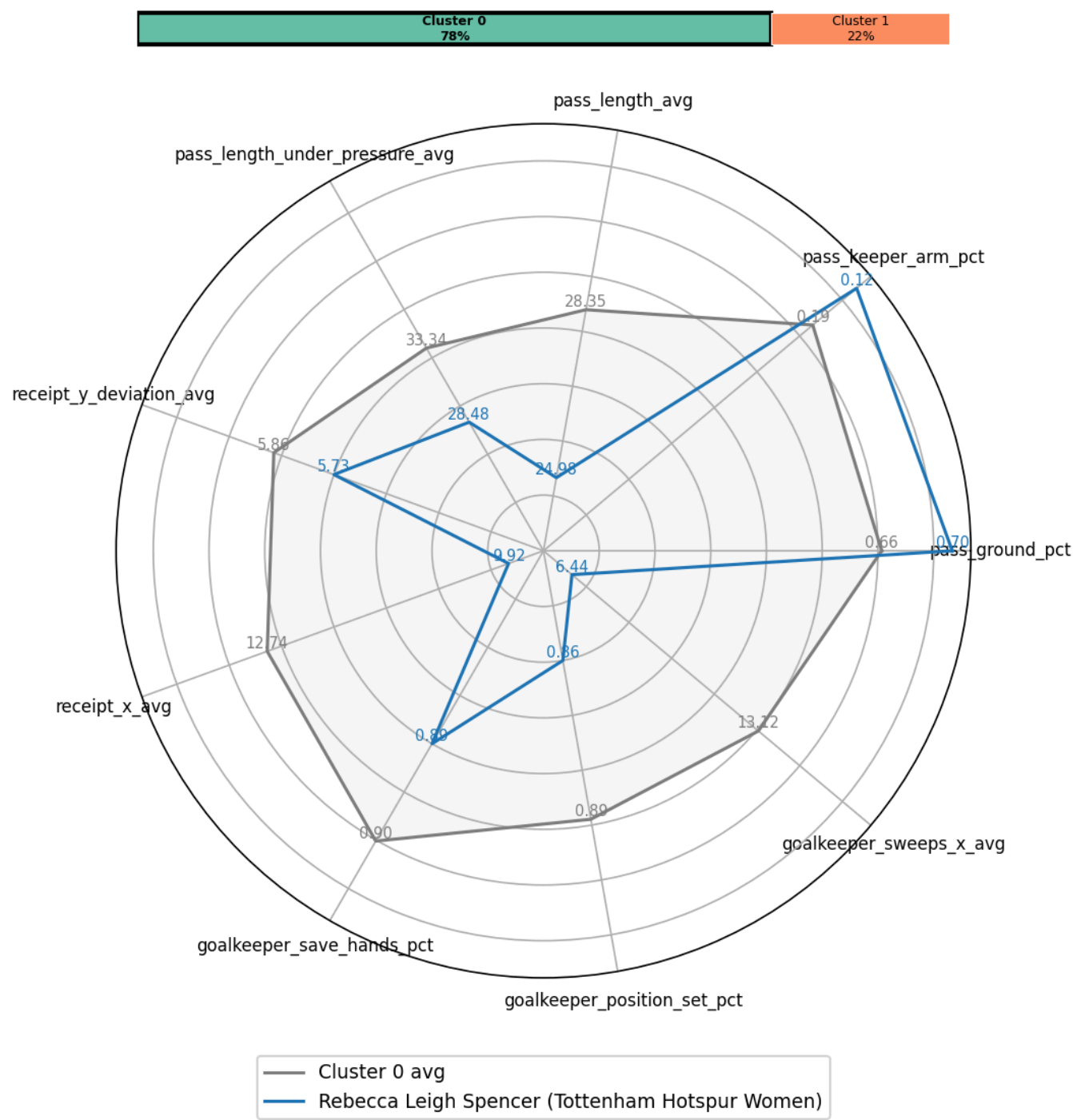


Figure 37: Rebecca Leigh Spencer — Raw Statistics Comparison, 2 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) vs FA Women's Super League peers (raw)

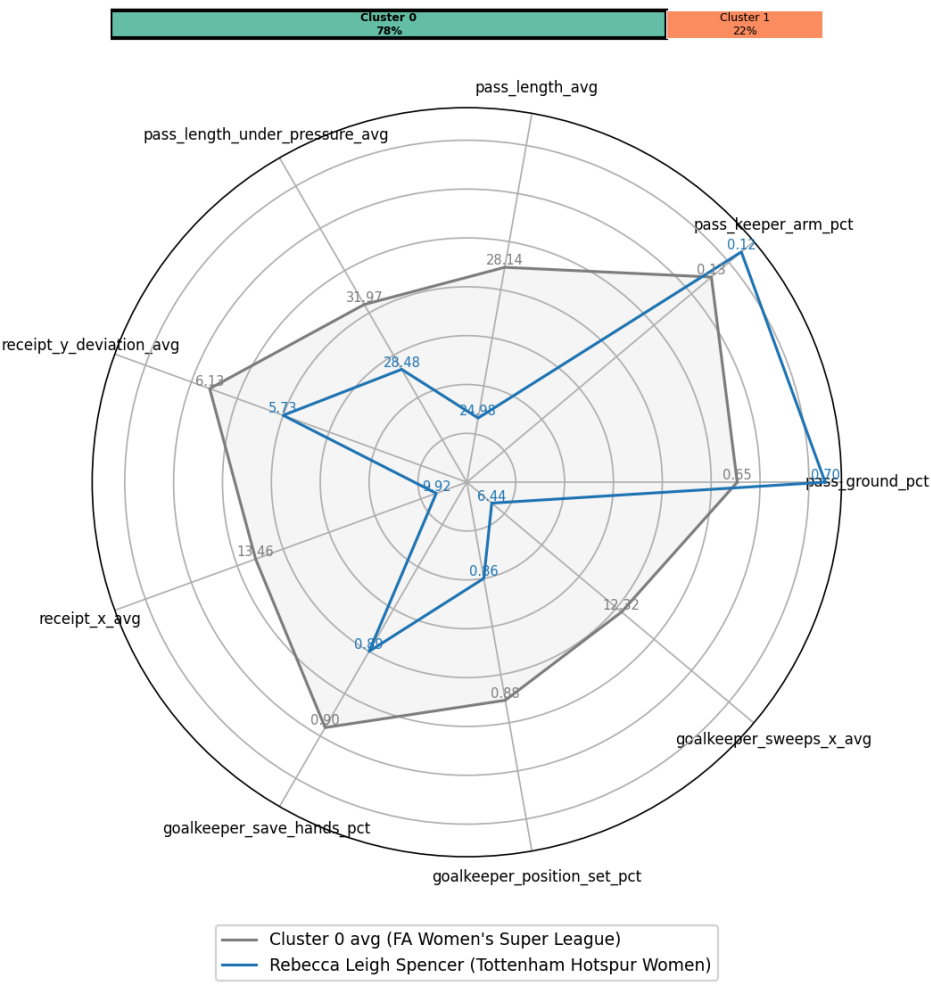


Figure 38: Rebecca Leigh Spencer — League-Aware Raw Statistics Comparison, 2 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) (adj)

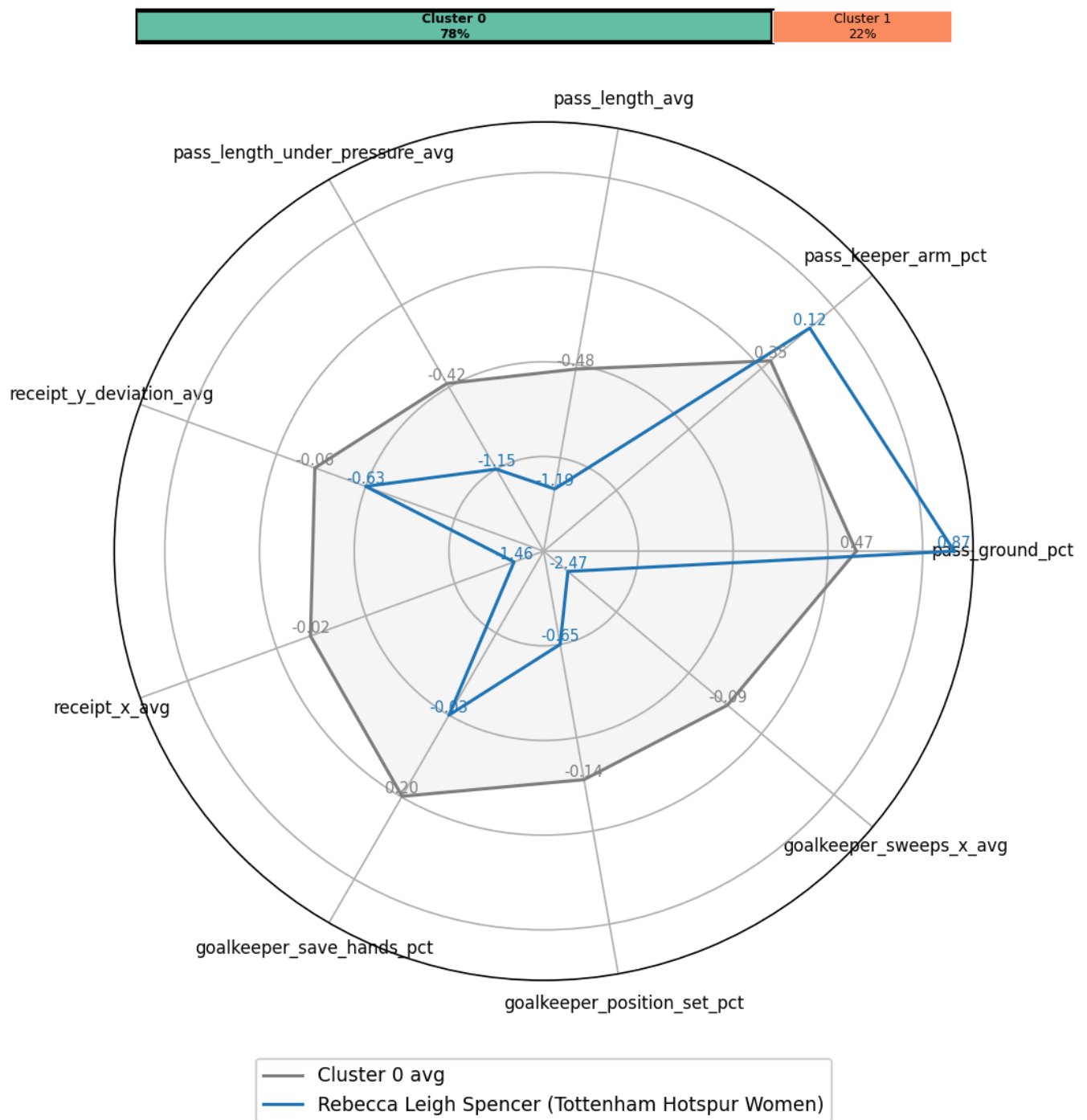


Figure 39: Rebecca Leigh Spencer — Adjusted Statistics Comparison, 2 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) vs FA Women's Super League peers (adj)

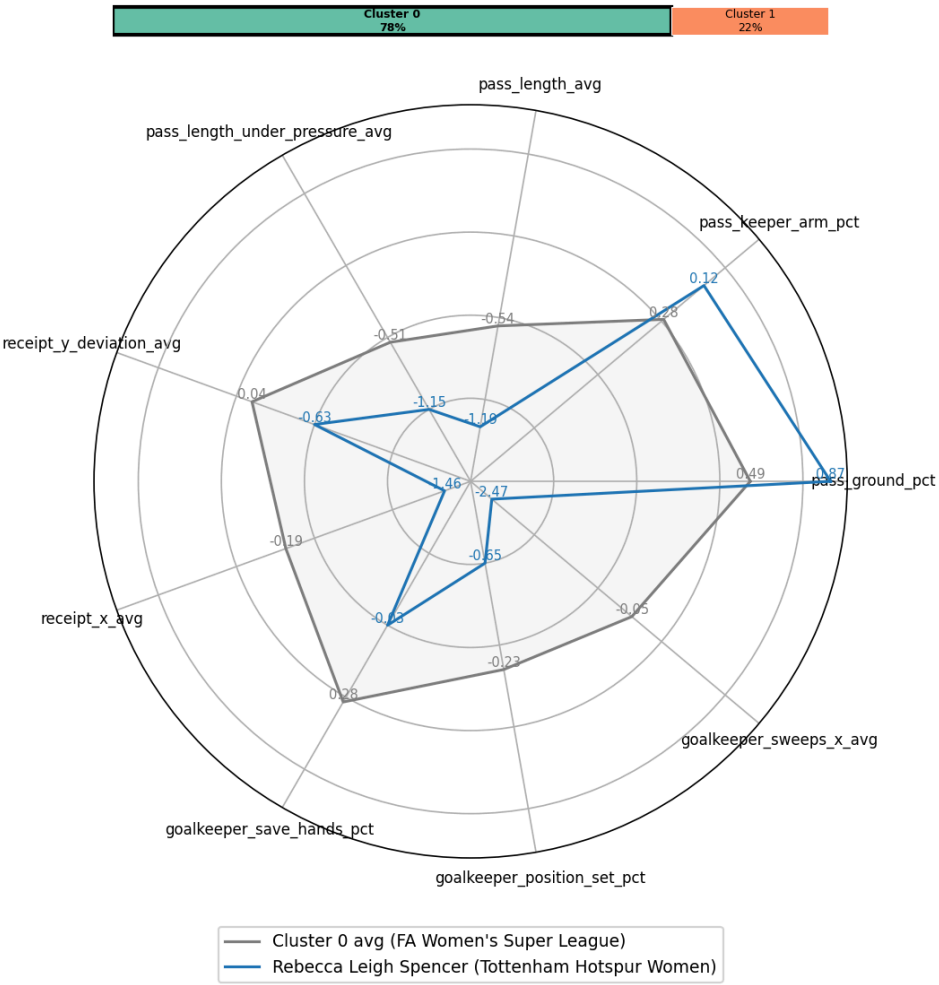


Figure 40: Rebecca Leigh Spencer — League-Aware Adjusted Statistics Comparison, 2 Clusters

3 cluster plots

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All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) (raw)

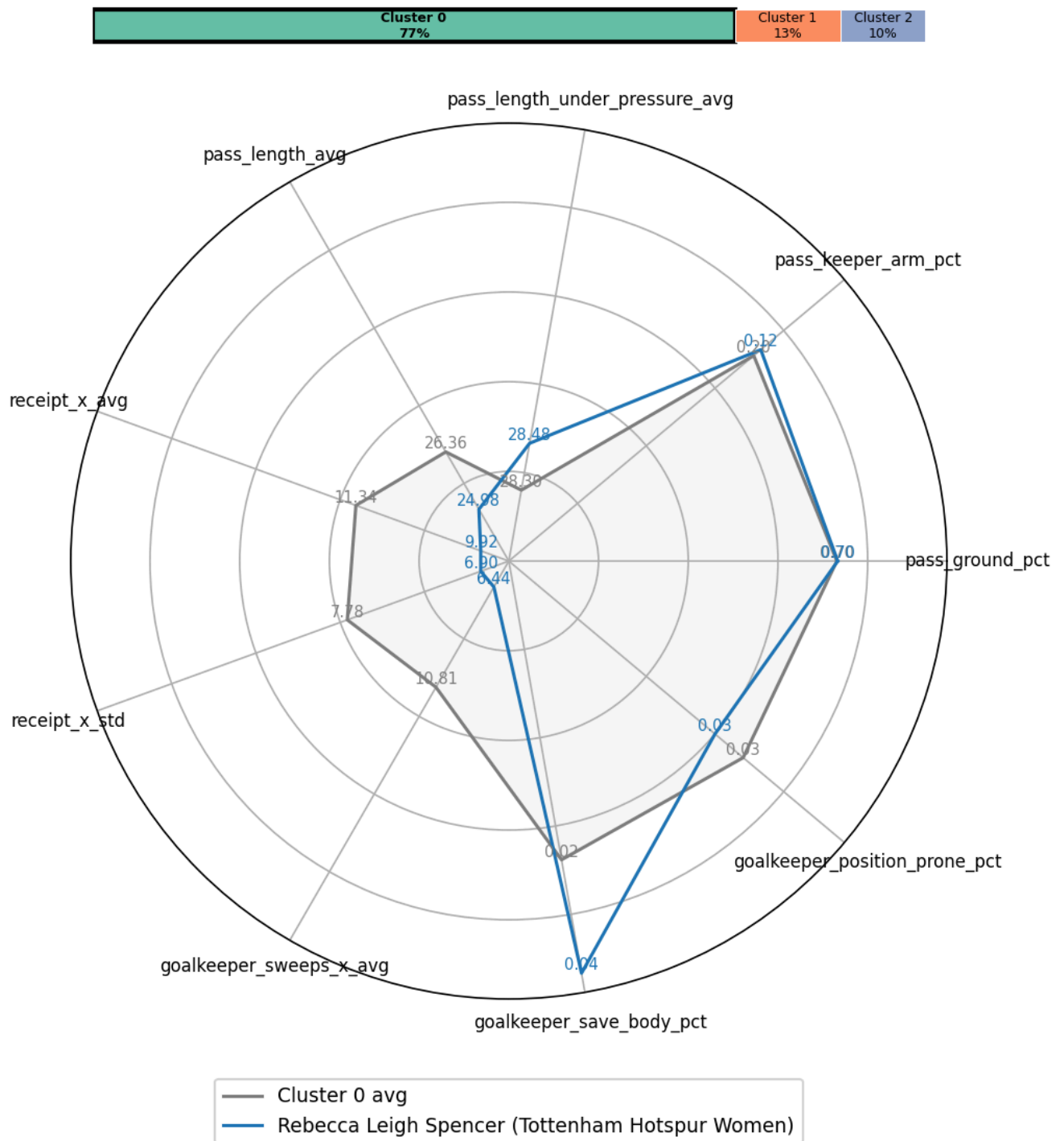


Figure 41: Rebecca Leigh Spencer — Raw Statistics Comparison, 3 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) vs FA Women's Super League peers (raw)

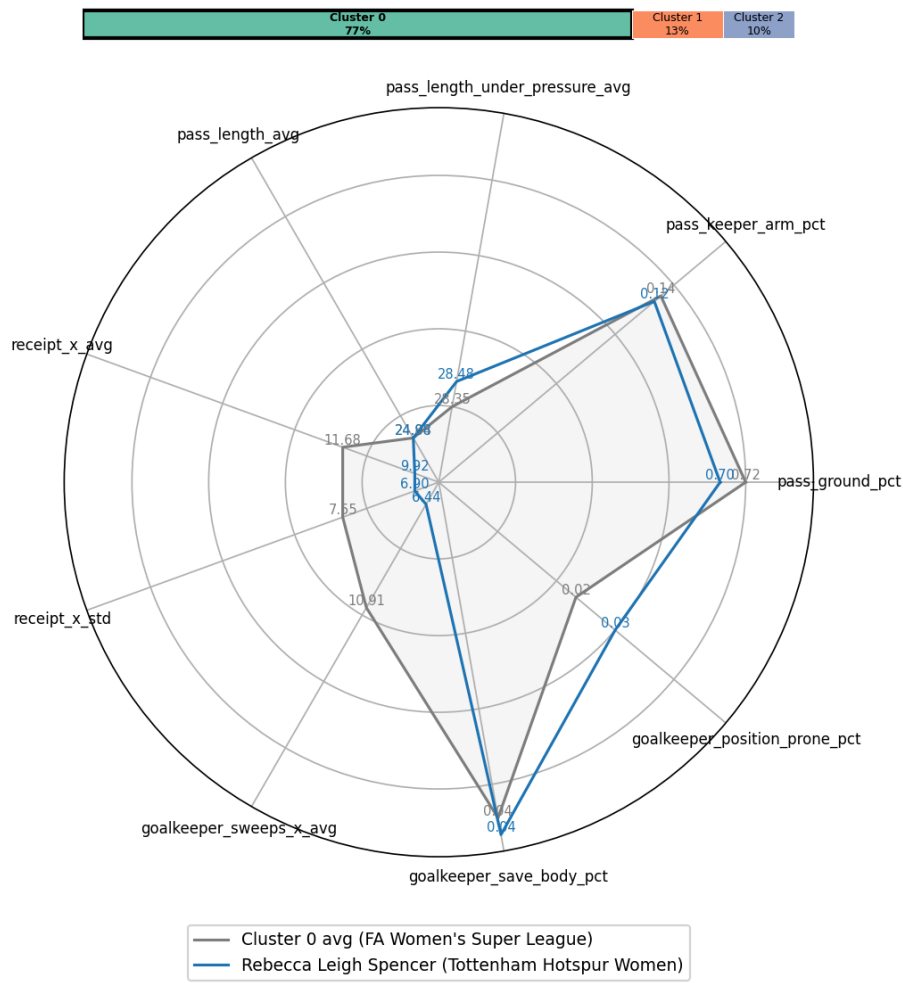


Figure 42: Rebecca Leigh Spencer — League-Aware Raw Statistics Comparison, 3 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) (adj)

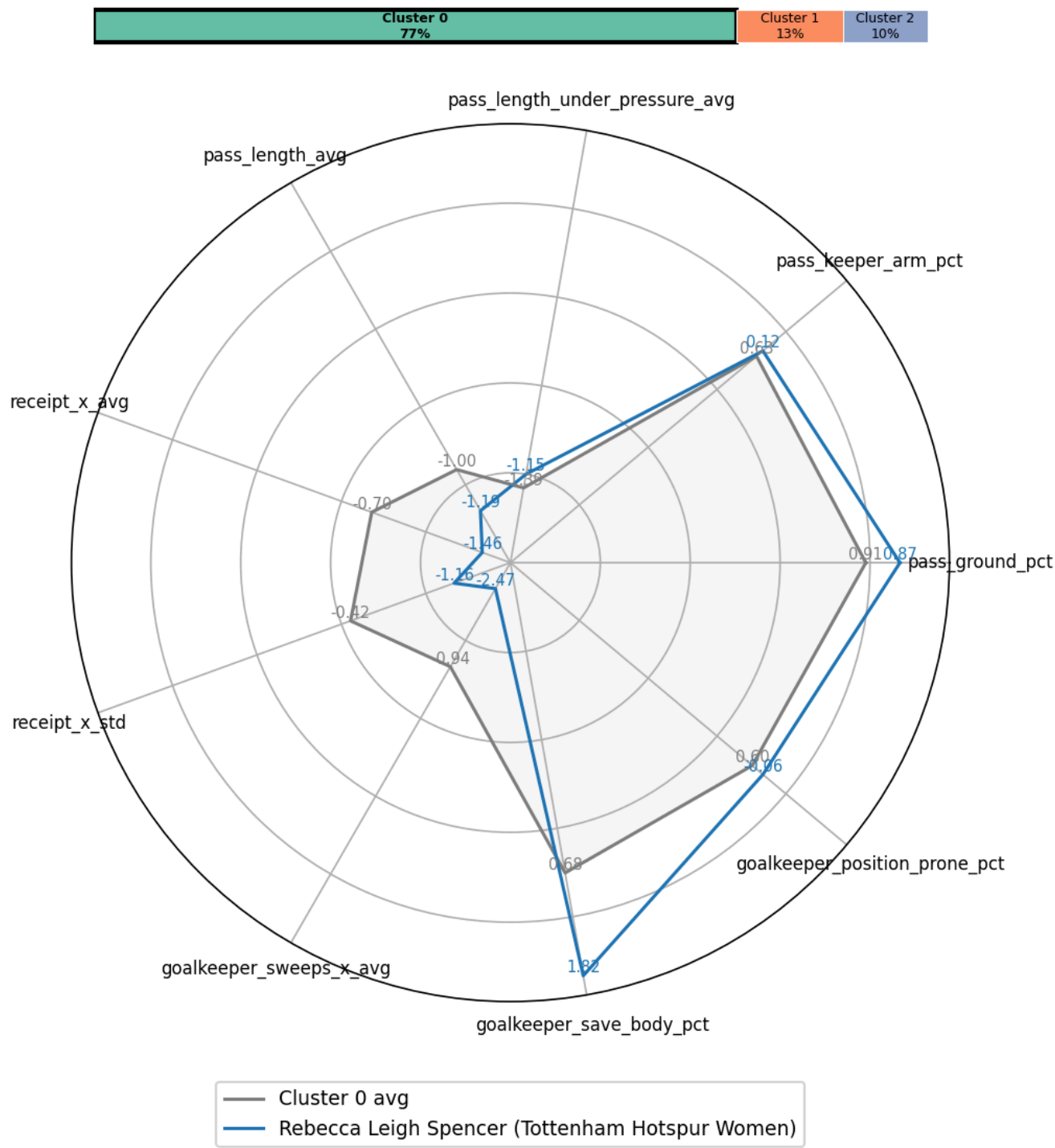


Figure 43: Rebecca Leigh Spencer — Adjusted Statistics Comparison, 3 Clusters

All_Features — Rebecca Leigh Spencer (Tottenham Hotspur Women) vs FA Women's Super League peers (adj)

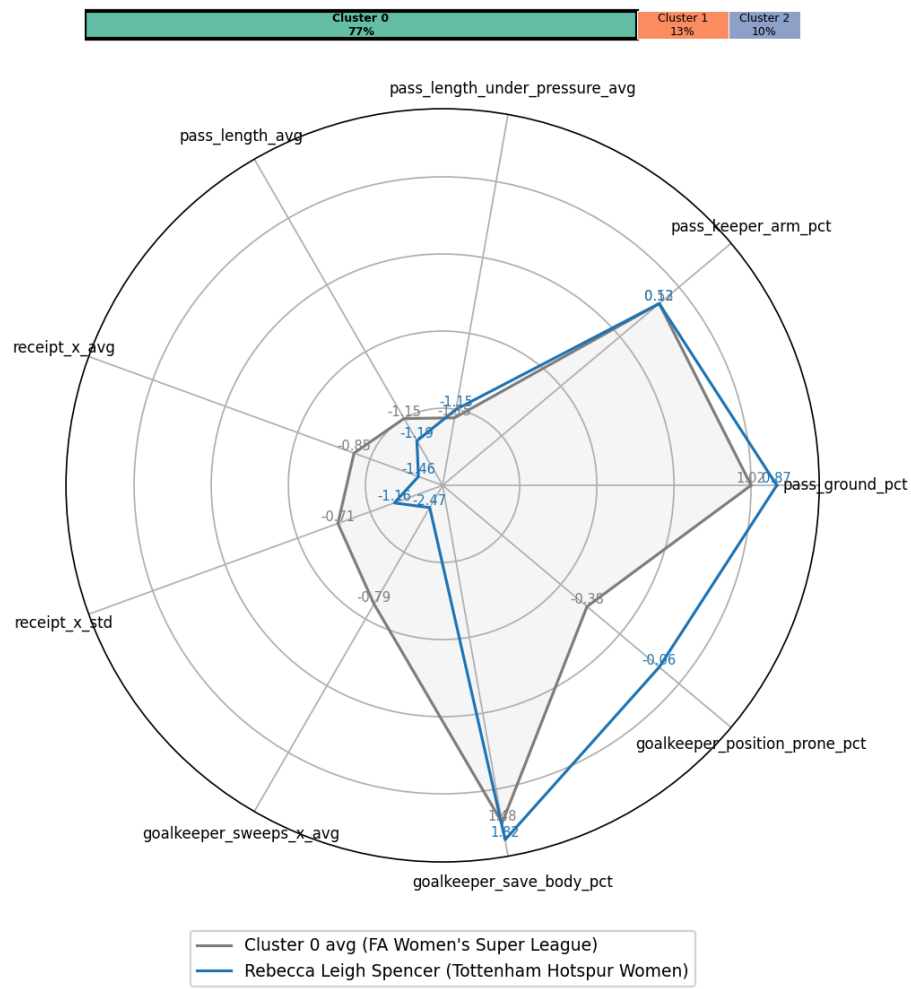


Figure 44: Rebecca Leigh Spencer — League-Aware Adjusted Statistics Comparison, 3 Clusters

Maria Korenciova - Como

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2 cluster plots

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All_Features — Mária Korenčiová (Como W) (raw)

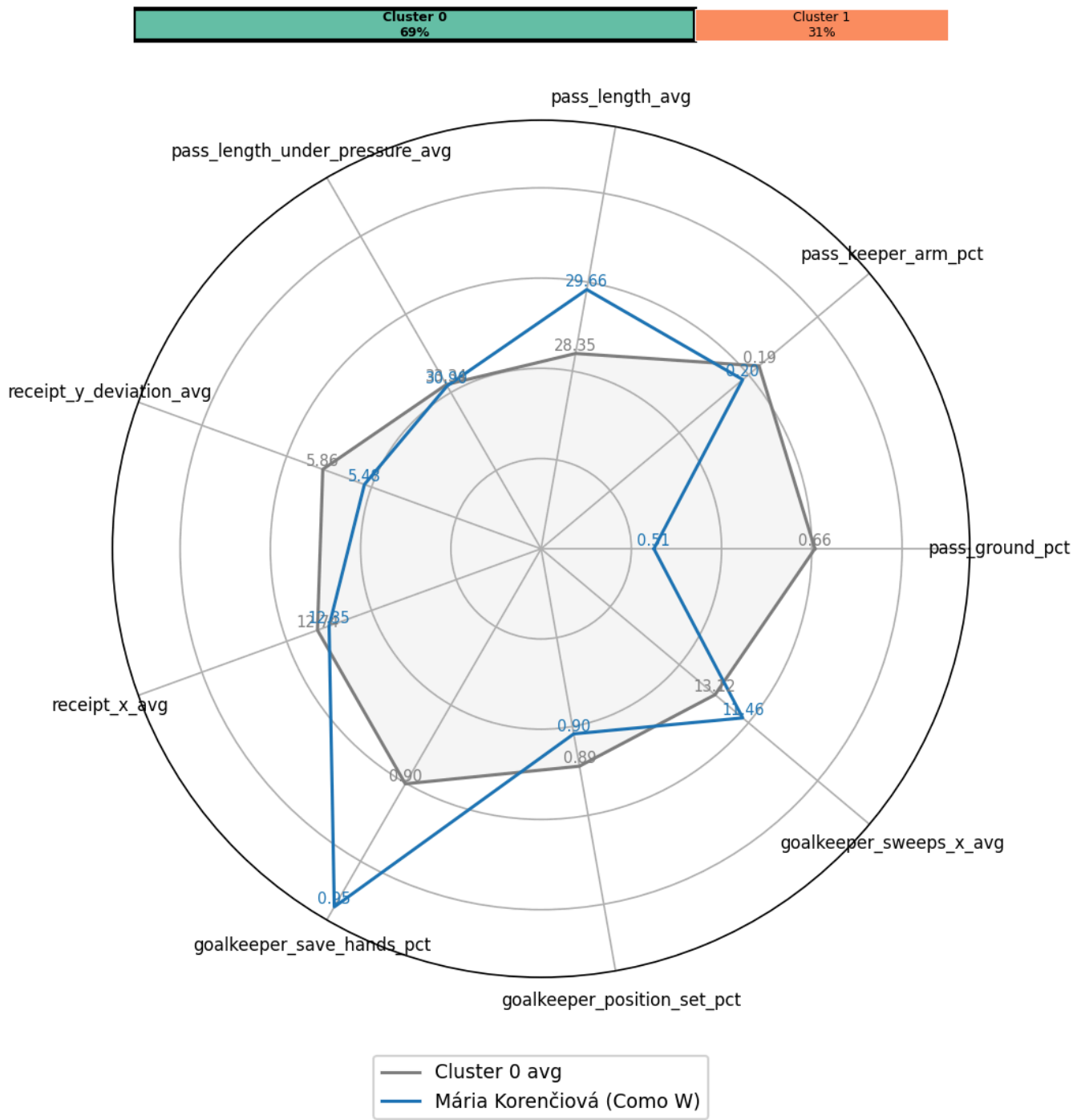


Figure 45: Maria Korenciova — Raw Statistics Comparison, 2 Clusters

All_Features — Mária Koreňčiová (Como W) vs Serie A Women peers (raw)

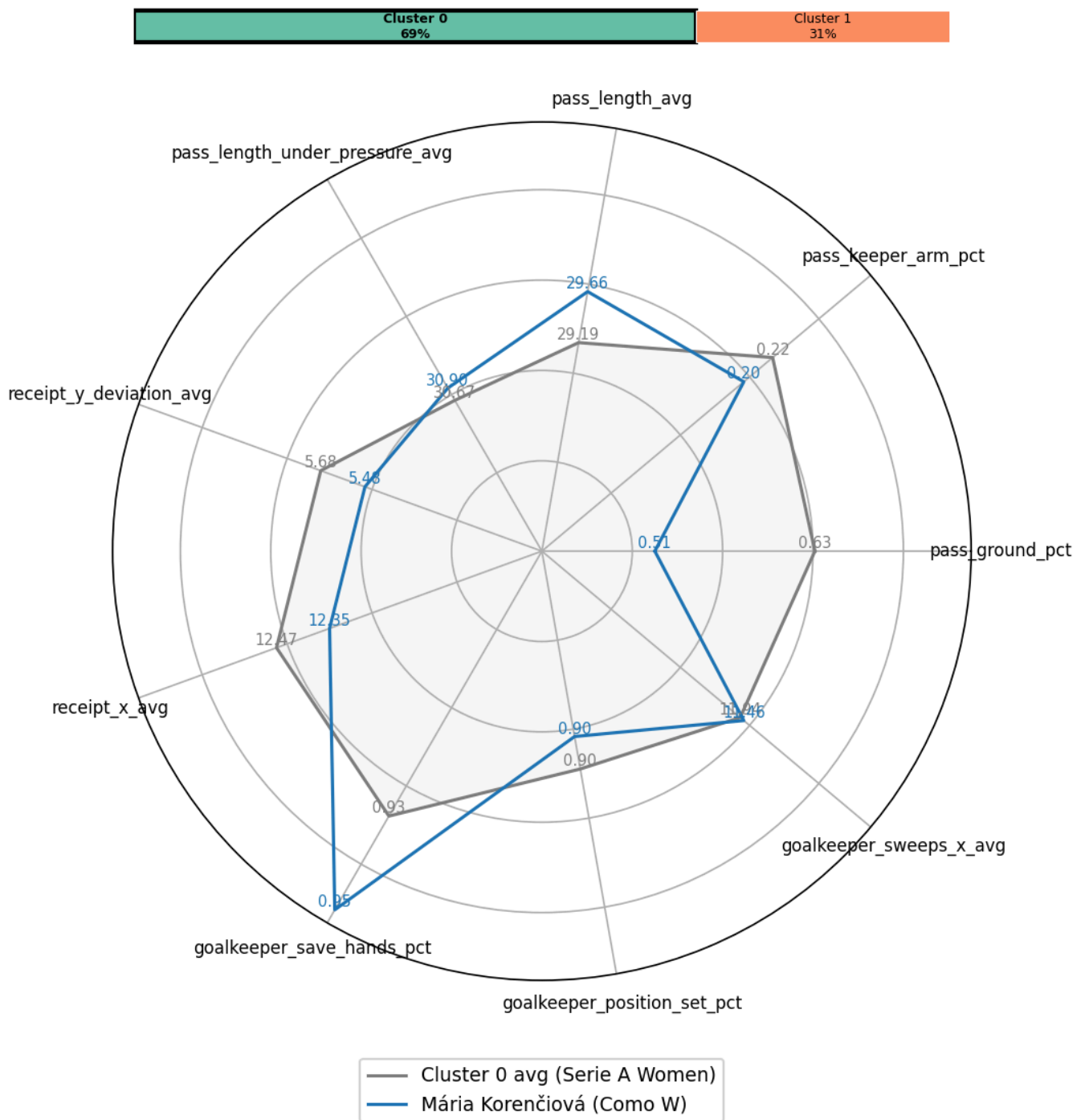


Figure 46: Maria Korenciova — League-Aware Raw Statistics Comparison, 2 Clusters

All_Features — Mária Koreňčiová (Como W) (adj)

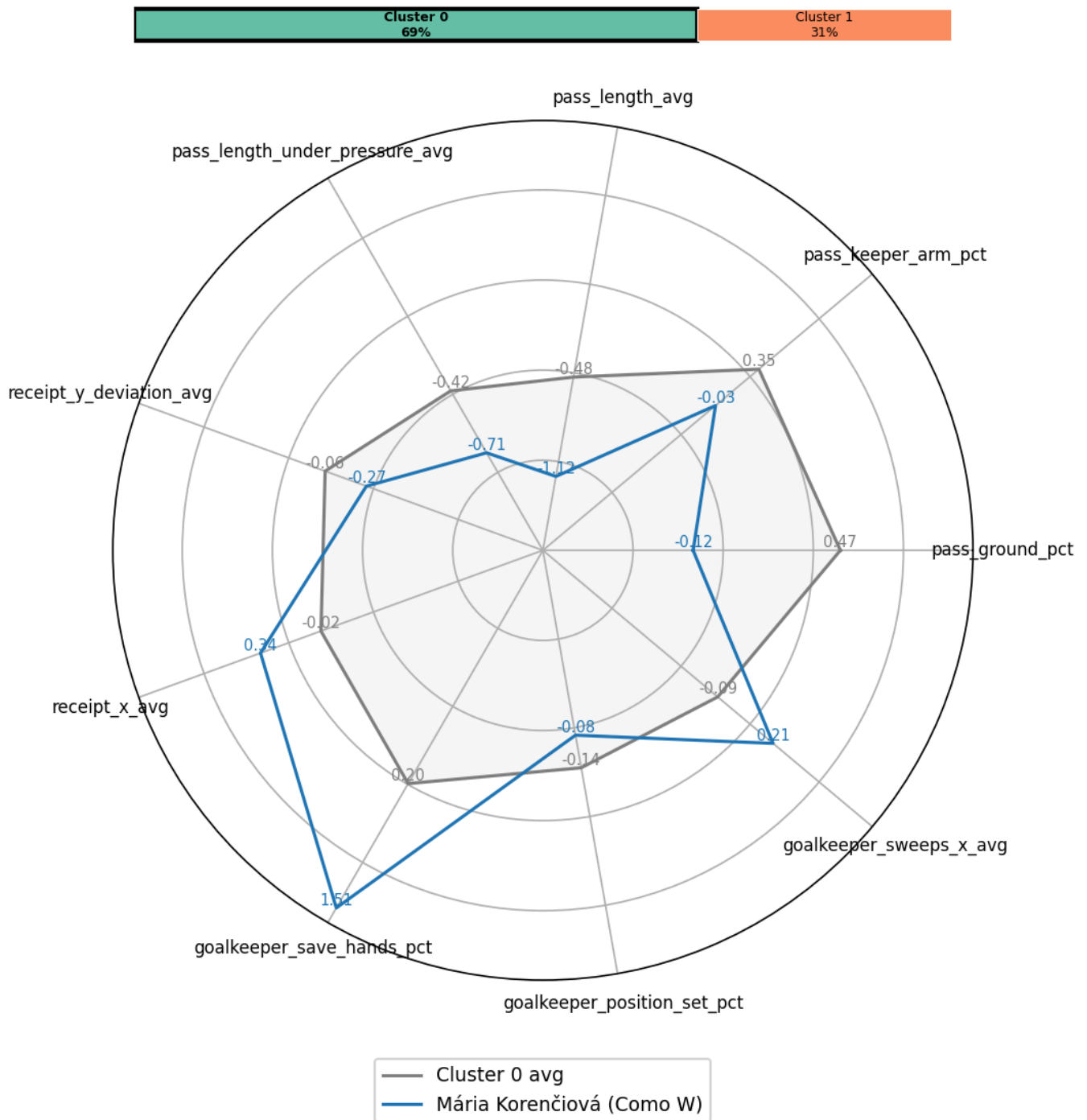


Figure 47: Maria Korenciova — Adjusted Statistics Comparison, 2 Clusters

All_Features — Mária Korenčiová (Como W) vs Serie A Women peers (adj)

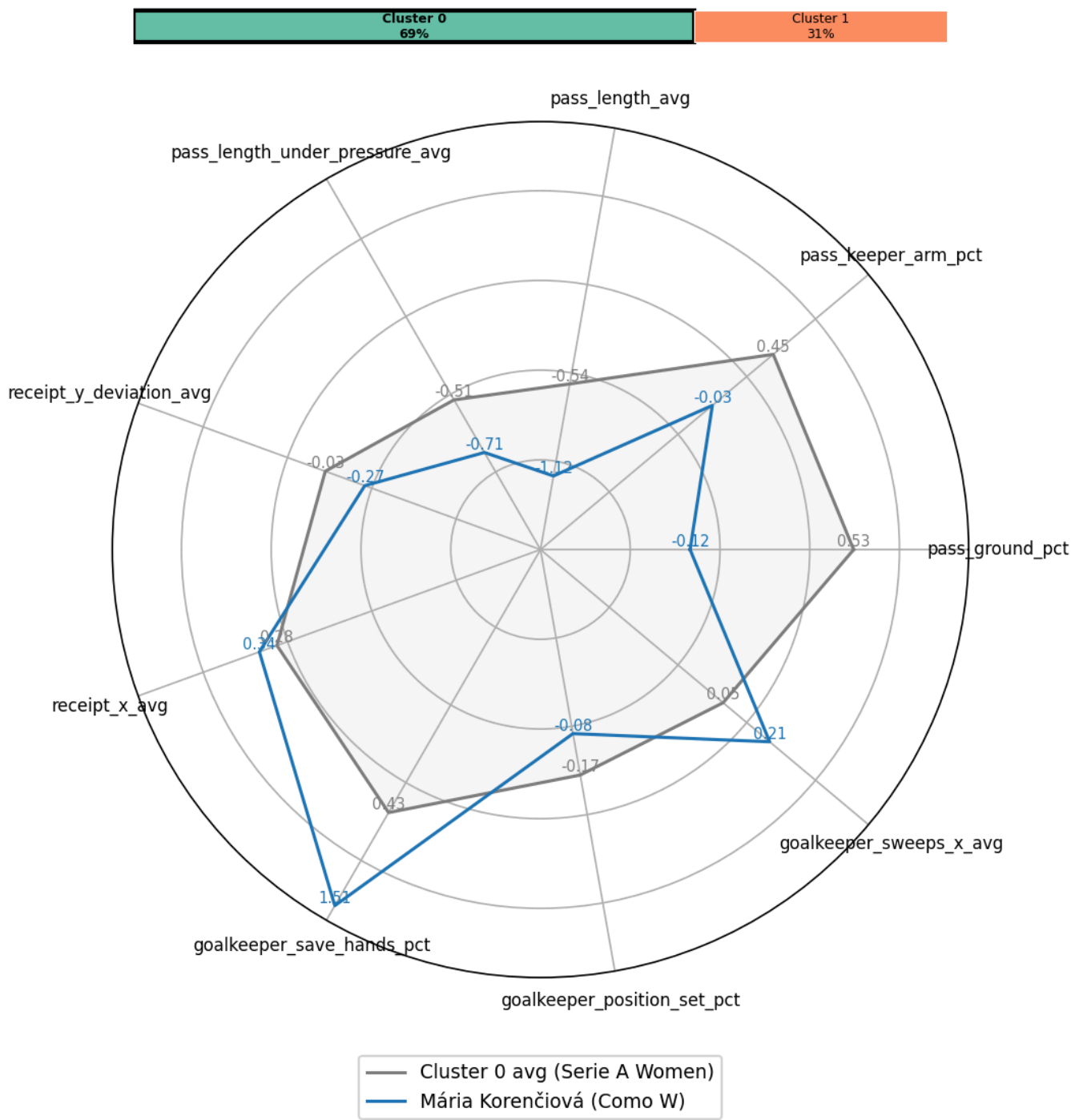


Figure 48: Maria Korenciova — League-Aware Adjusted Statistics Comparison, 2 Clusters

3 cluster plots

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All_Features — Mária Koreňčiová (Como W) (raw)

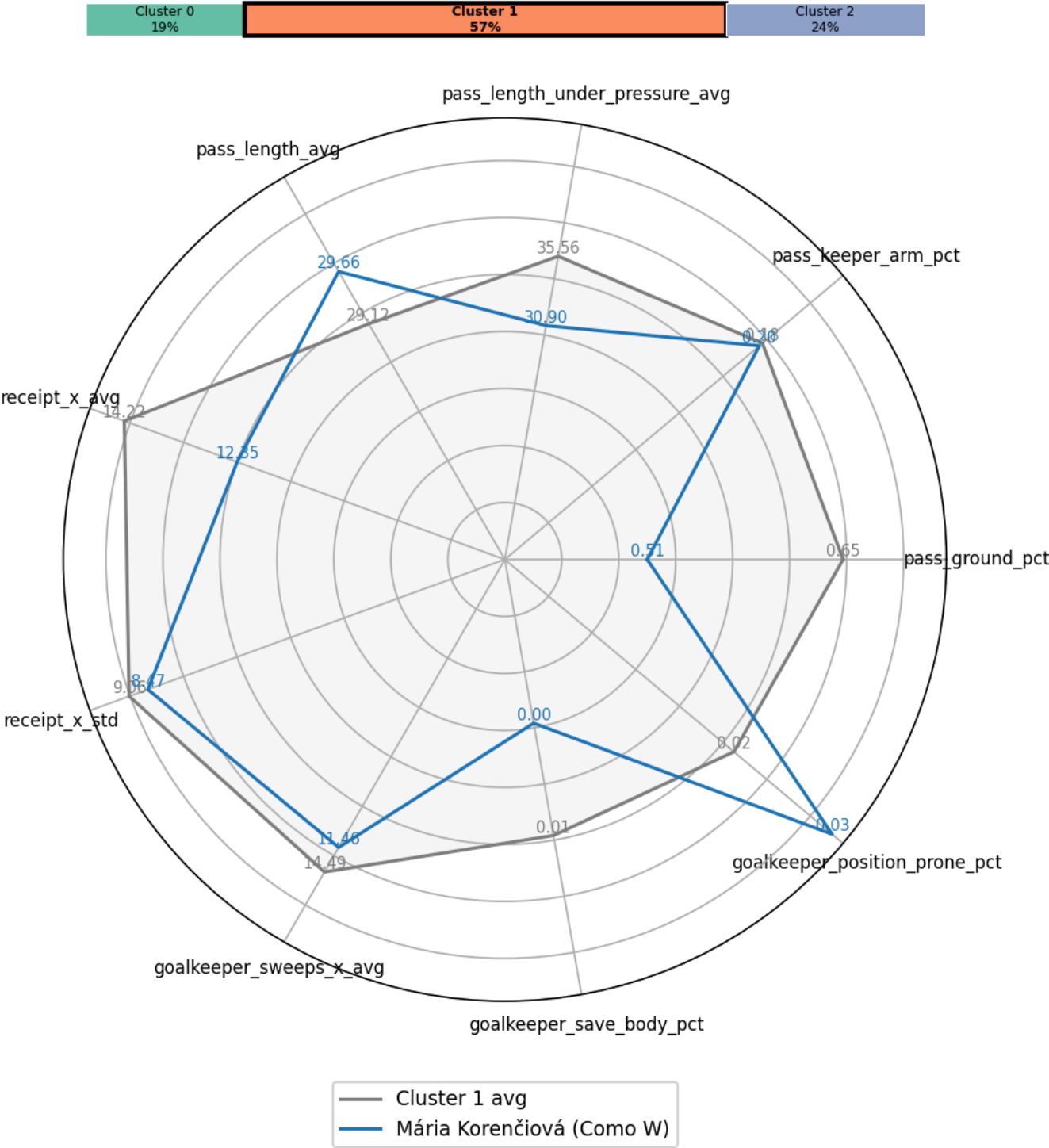


Figure 49: Maria Korenciova — Raw Statistics Comparison, 3 Clusters

All_Features — Mária Korenčiová (Como W) vs Serie A Women peers (raw)

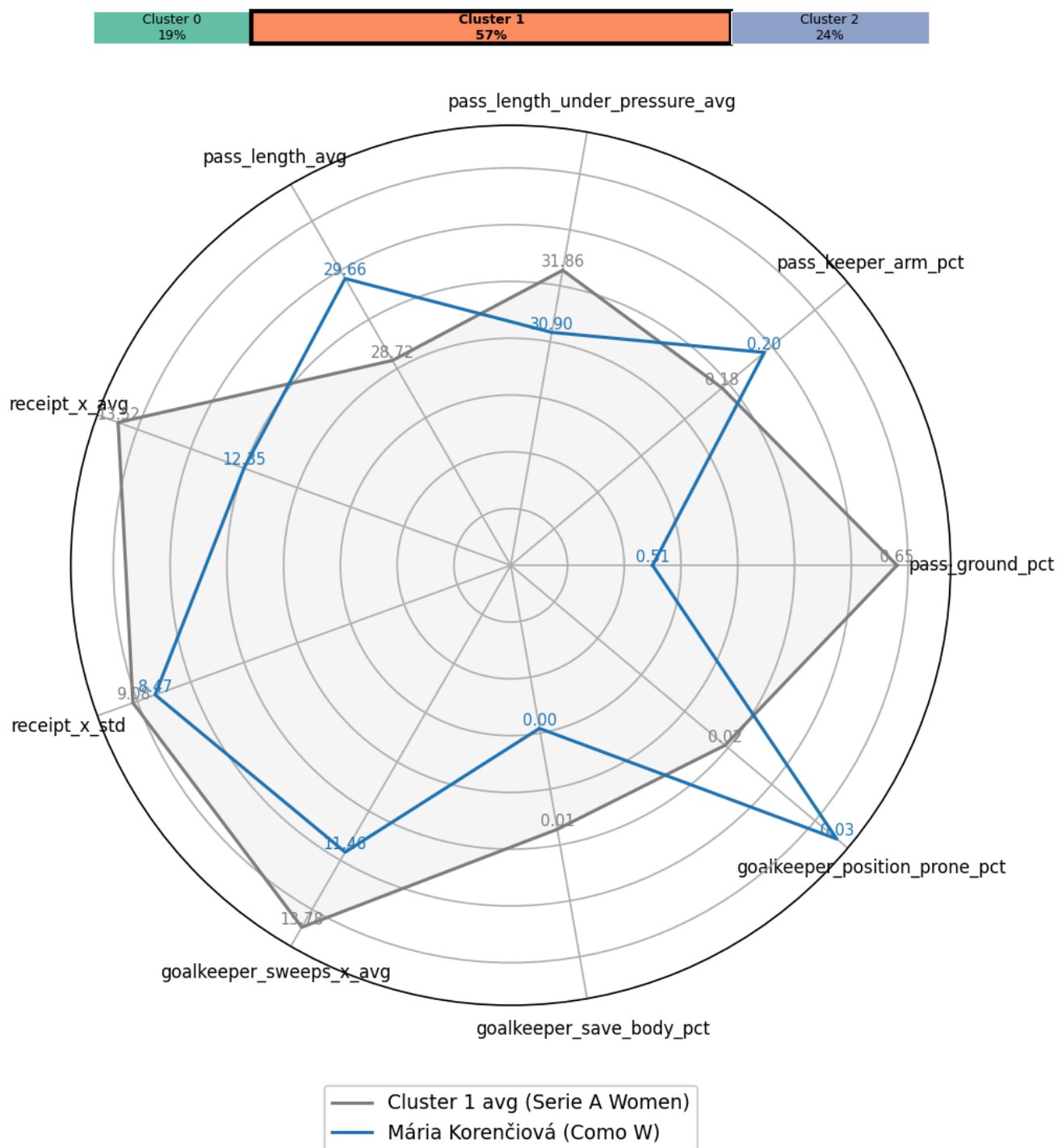


Figure 50: Maria Korenciova — League-Aware Raw Statistics Comparison, 3 Clusters

All_Features — Mária Korenčiová (Como W) (adj)

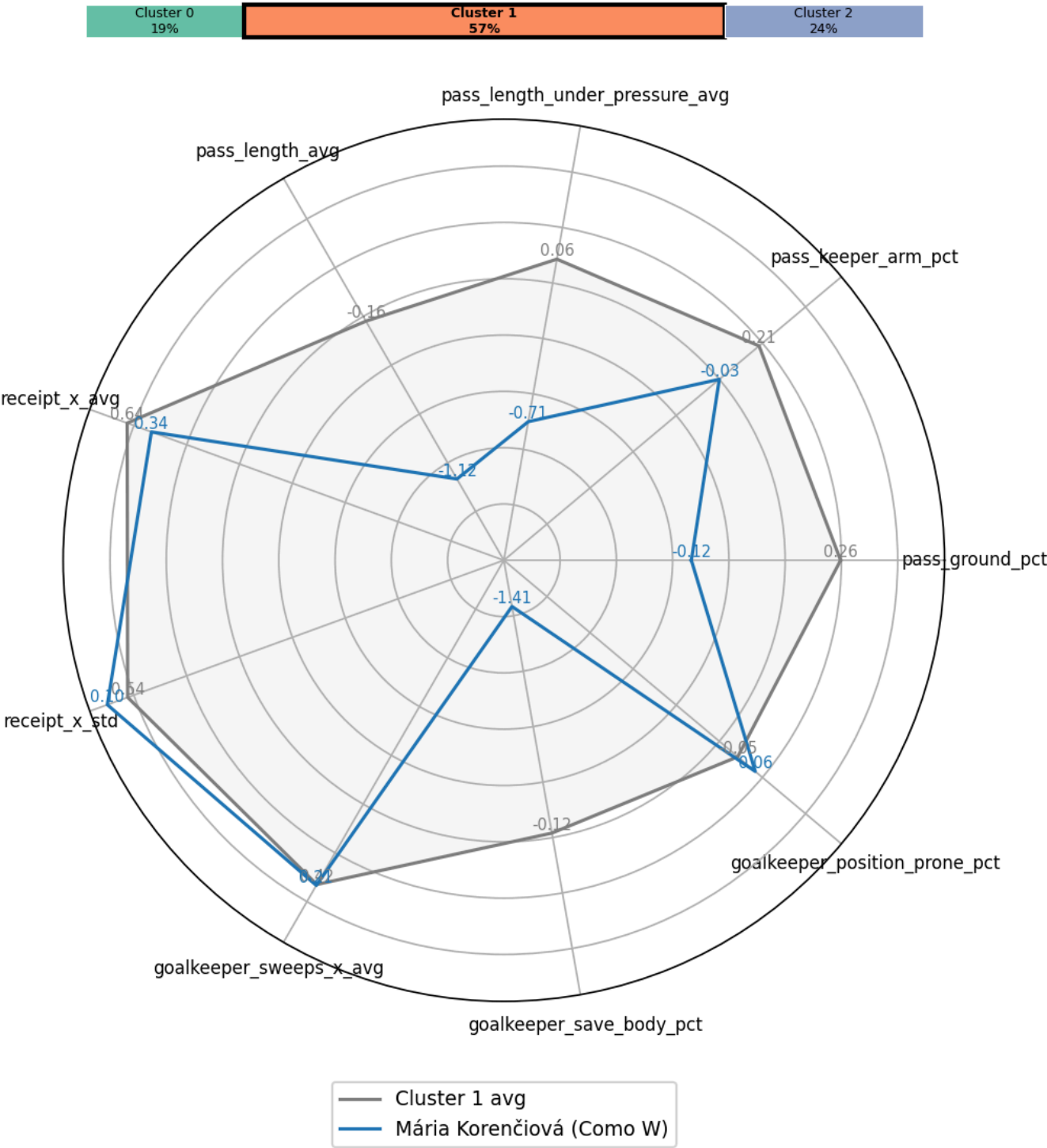


Figure 51: Maria Korenciova — Adjusted Statistics Comparison, 3 Clusters

All_Features — Mária Korenčiová (Como W) vs Serie A Women peers (adj)



Figure 52: Maria Korenciova — League-Aware Adjusted Statistics Comparison, 3 Clusters

Mary Earps - Manchester United

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2 cluster plots

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All_Features — Mary Alexandra Earps (Manchester United W) (raw)

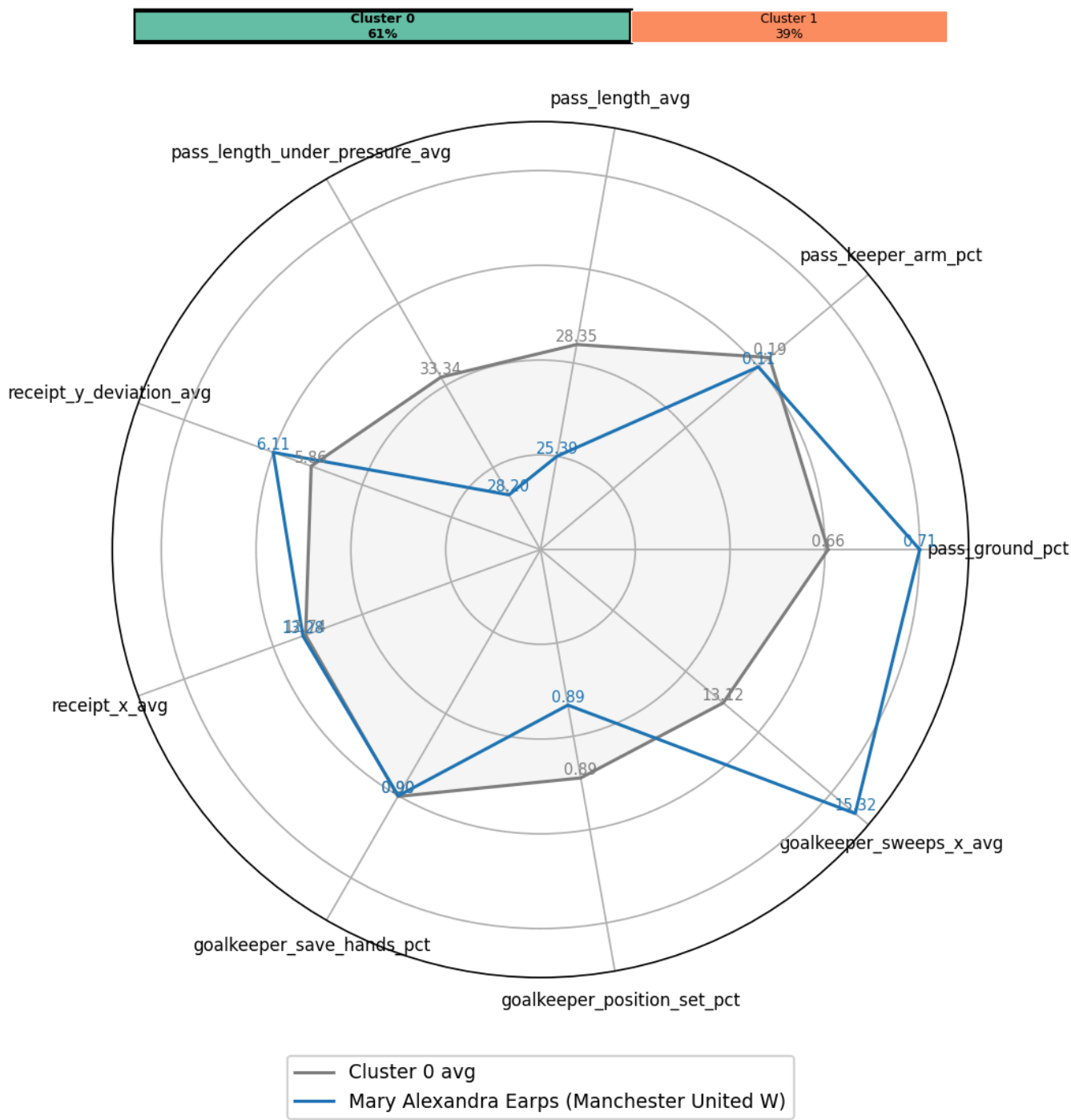


Figure 53: Mary Earps — Raw Statistics Comparison, 2 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) vs FA Women's Super League peers (raw)

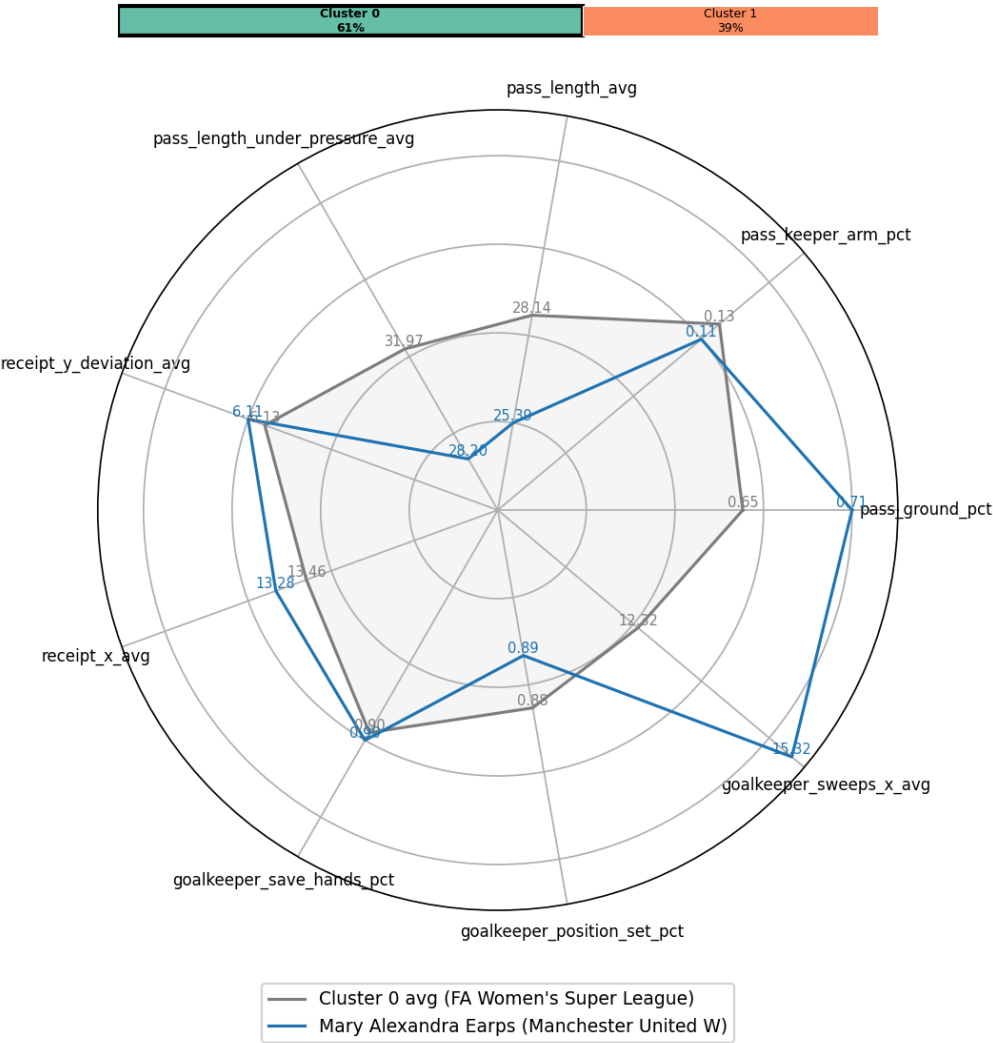


Figure 54: Mary Earps — League-Aware Raw Statistics Comparison, 2 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) (adj)

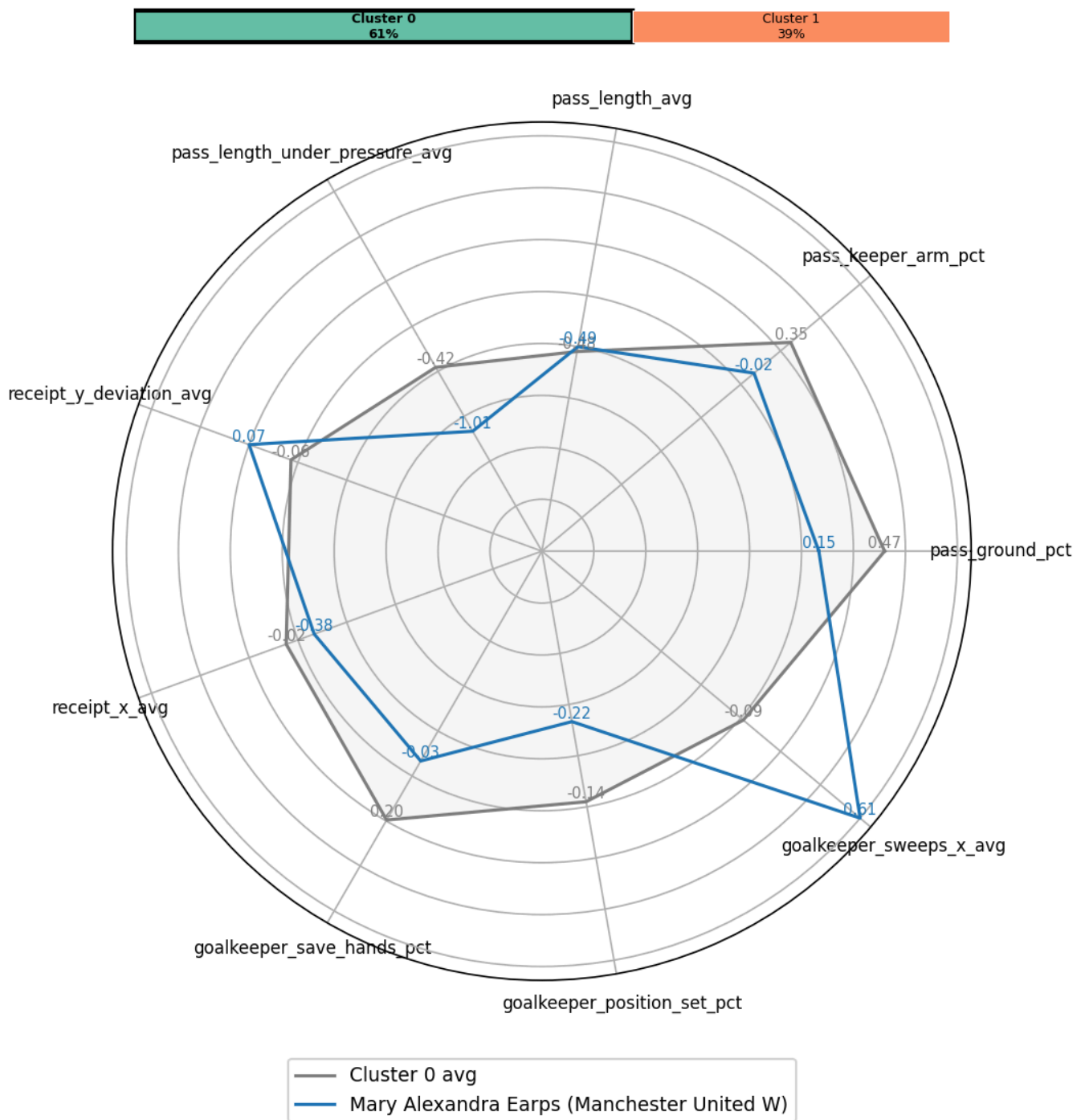


Figure 55: Mary Earps — Adjusted Statistics Comparison, 2 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) vs FA Women's Super League peers (adj)

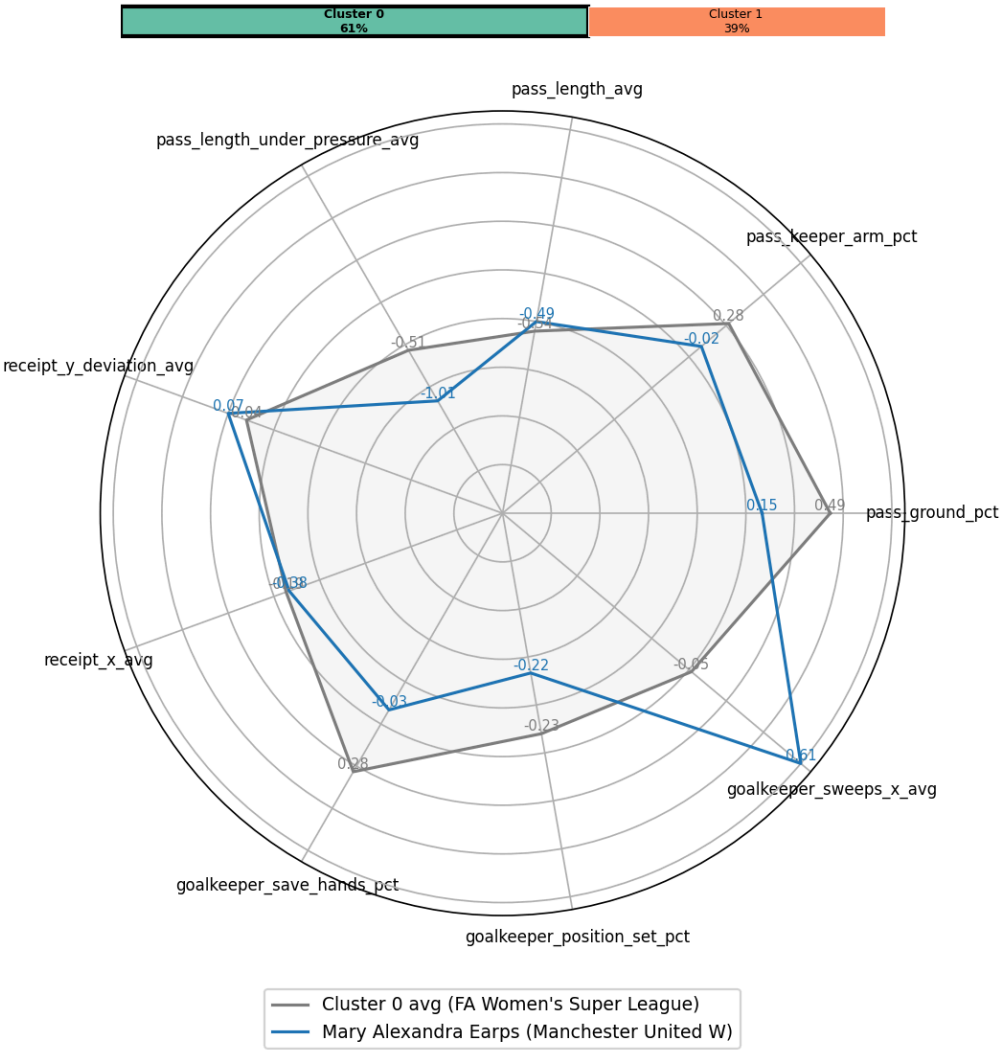


Figure 56: Mary Earps — League-Aware Adjusted Statistics Comparison, 2 Clusters

3 cluster plots

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All_Features — Mary Alexandra Earps (Manchester United W) (raw)

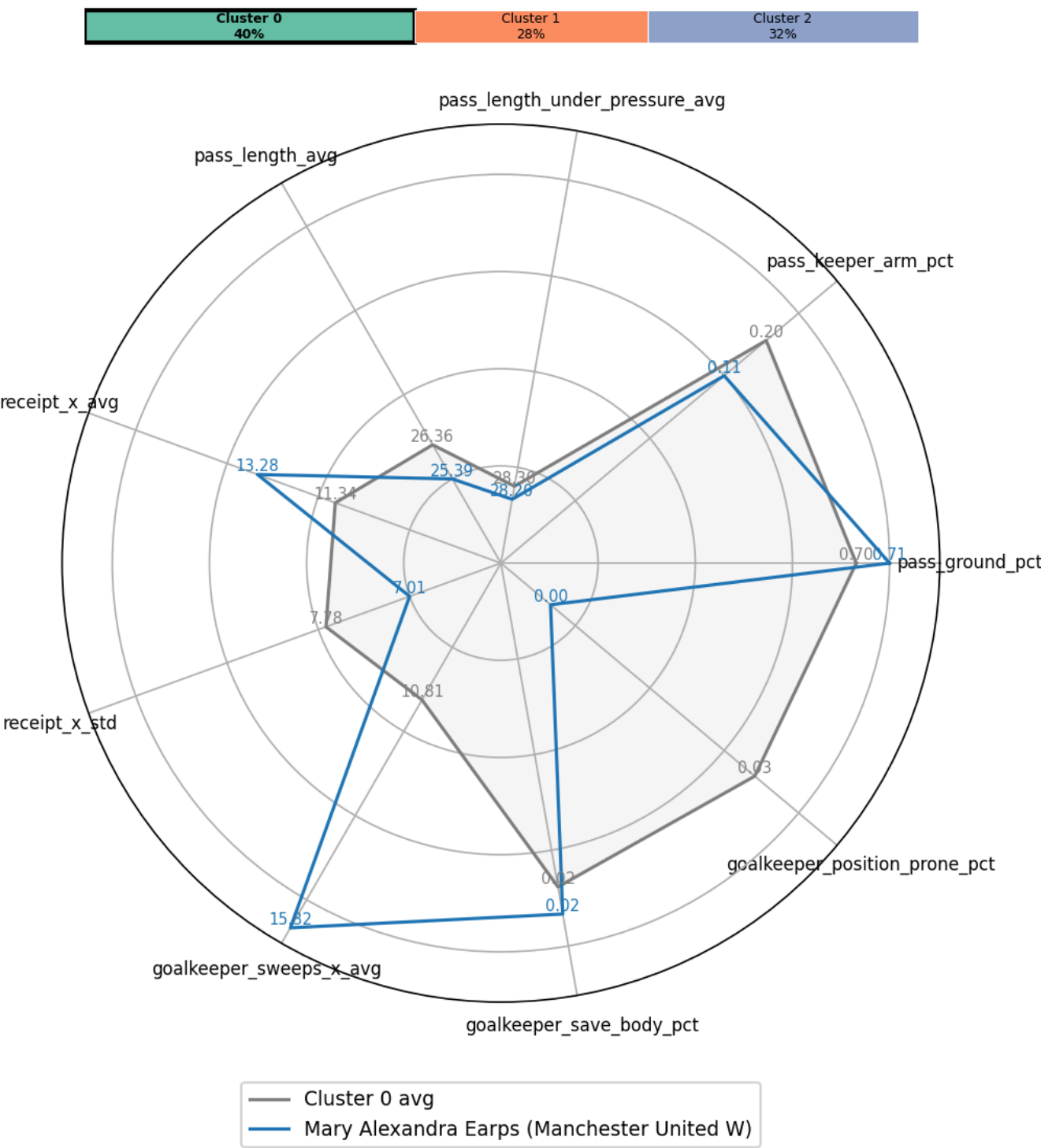


Figure 57: Mary Earps — Raw Statistics Comparison, 3 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) vs FA Women's Super League peers (raw)

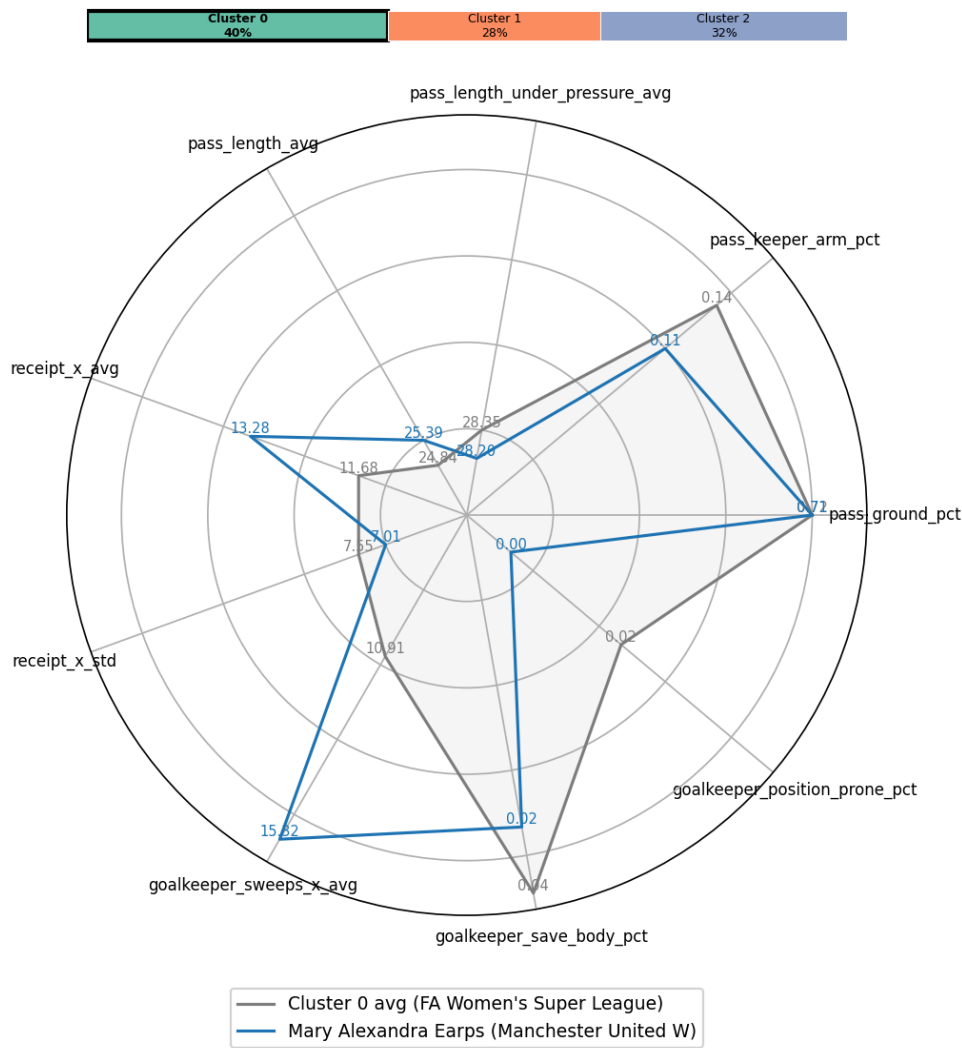


Figure 58: Mary Earps — League-Aware Raw Statistics Comparison, 3 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) (adj)

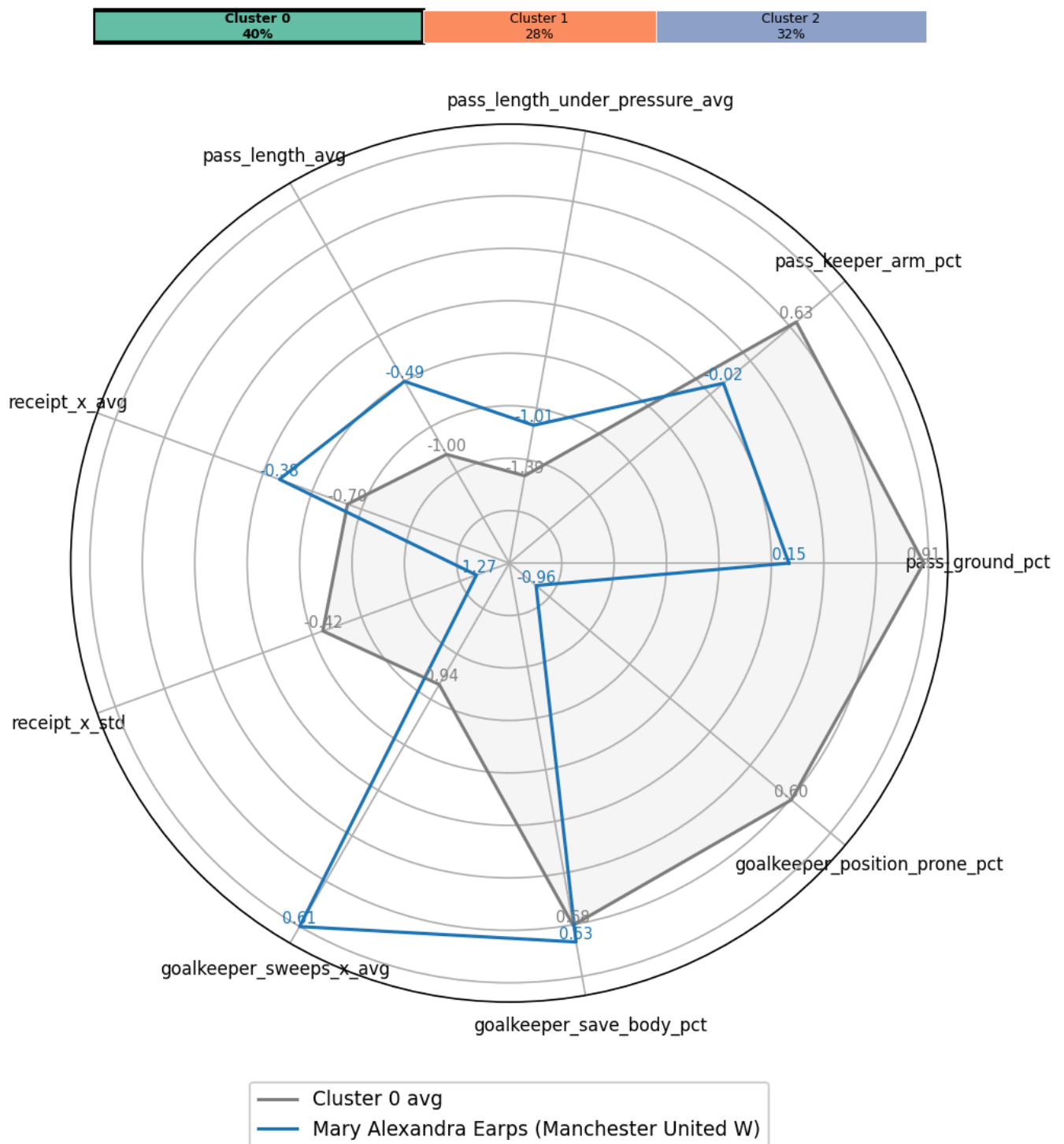


Figure 59: Mary Earps — Adjusted Statistics Comparison, 3 Clusters

All_Features — Mary Alexandra Earps (Manchester United W) vs FA Women's Super League peers (adj)

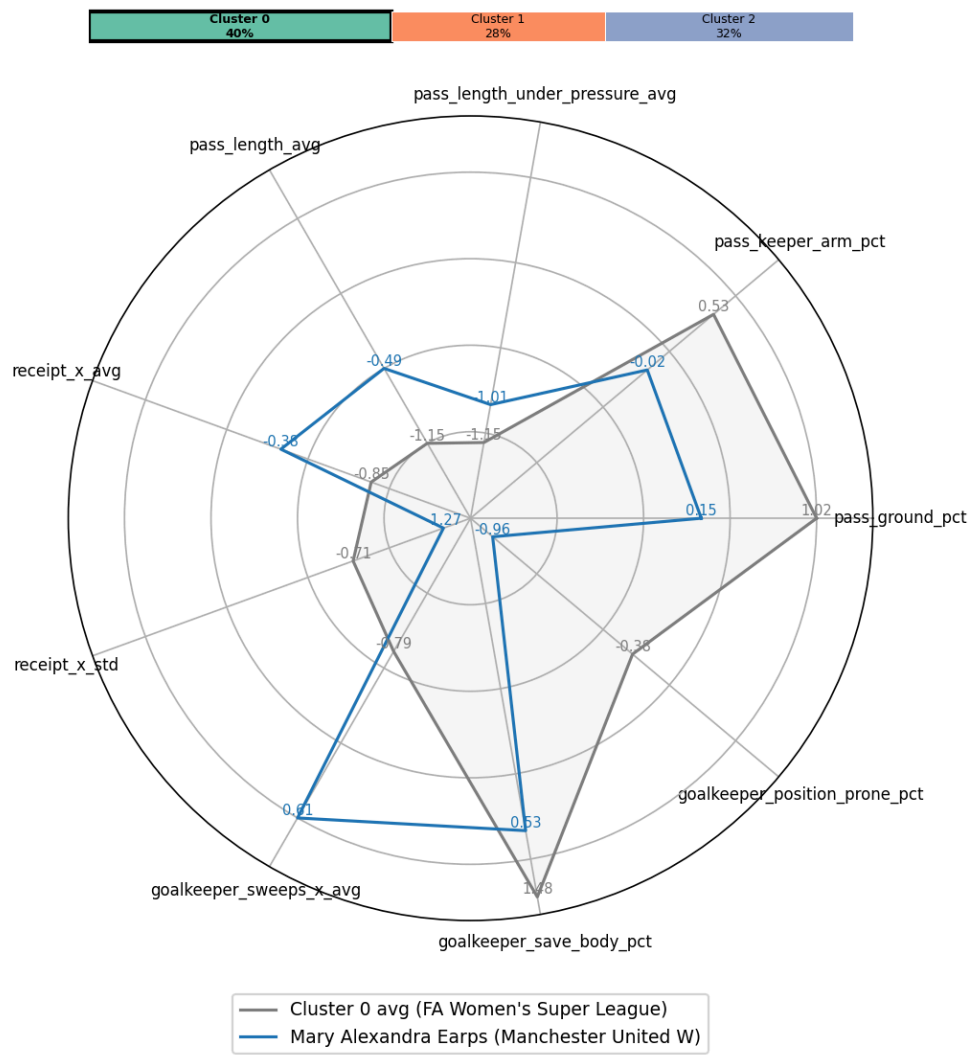


Figure 60: Mary Earps — League-Aware Adjusted Statistics Comparison, 3 Clusters